RiskBid: Supplementary Proofs, Protocols, and Diagnostics

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This supplement contains technical proofs, experimental protocols, additional figures, and diagnostic analyses supporting the main manuscript. Section labels A–G correspond to the references in the main text.

A Technical Assumptions and Proofs

This appendix collects the technical conditions and proofs supporting the main text. We first collect the notation, regularity assumptions, and metric identities used in the analysis of RiskBid. We then verify the feasibility of affine-hull smoothing, derive the finite-difference error bound, and prove the descent, convergence, and oracle-complexity results.

A.1 Additional Notation, Oracle Semantics, Regularity Conditions, and Metric Identities

We use the notation of the main text and distinguish the scalar target from its oracle implementation. Write F for the target on the budget affine hull. In *smoothed-oracle mode*, $F = f_\delta$ is the affine-hull ball smoothing of the original locally Lipschitz objective. In *direct empirical-oracle mode*, denoted by $M_{\text{smooth}} = 0$, the target is $F = \widehat{f}_n$ and is evaluated directly, without a Monte Carlo averaging layer. The direct target is smooth on each regular active cell, where the order statistic and clipping regime are fixed; Appendix A.4 gives the cell decomposition, uniform branch constants, and a stratified trajectory certificate that separately accounts for interface crossings. The main-text theorems use the common notation F, Λ_F, and $\mathcal{G}_F$. The proofs below are written for f_δ; the same arguments apply to a smooth target after replacing the target-specific constants, while Proposition 1 supplies the direct empirical specialization without requiring the entire finite trajectory to remain in one cell.

The feasible set and the tangent space of the budget hyperplane are

$$\begin{aligned} \mathcal{X} &= \left\{ x \in \mathbb{R}^d : \mathbf{1}^\top x = 1,\ l_i \le x_i \le u_i,\ i \in [d] \right\}, \\ \mathsf{T} &= \{ v \in \mathbb{R}^d : \mathbf{1}^\top v = 0 \}. \end{aligned} \tag{1}$$

Let $r = d - 1$, and let $Q \in \mathbb{R}^{d \times r}$ be an orthonormal basis of T:

$$Q^\top Q = I_r, \qquad Q^\top \mathbf{1} = 0. \tag{2}$$

For a fixed reference point $x_{\text{ref}} \in \operatorname{aff}(\mathcal{X})$, every point on the budget affine hull has the unique reduced-coordinate representation

$$x = x_{\text{ref}} + Qz, \qquad z = Q^\top (x - x_{\text{ref}}) \in \mathbb{R}^r. \tag{3}$$

Uniqueness follows from $Q^\top Q = I_r$: if $x_{\text{ref}} + Qz_1 = x_{\text{ref}} + Qz_2$, then $z_1 - z_2 = Q^\top Q(z_1 - z_2) = 0$.

For the smoothed-oracle mode, define

$$\begin{aligned} \phi(z) &= f(x_{\text{ref}} + Qz), \\ \phi_\delta(z) &= \mathbb{E}_{v \sim \nu_r} \left[\phi(z + \delta v) \right], \end{aligned} \tag{4}$$

where ν_r is the uniform distribution over

$$\mathbb{B}_r = \{ v \in \mathbb{R}^r : \|v\|_2 \le 1 \}. \tag{5}$$

The corresponding objective in the allocation space is

$$f_\delta(x_{\text{ref}} + Qz) = \phi_\delta(z). \tag{6}$$

All gradients of f_δ are affine-hull gradients:

$$\nabla_{\text{aff}} f_\delta(x) = Q \nabla \phi_\delta(z), \qquad x = x_{\text{ref}} + Qz. \tag{7}$$

Because $Q \nabla \phi_\delta(z) \in \operatorname{range}(Q) = \mathsf{T}$, this gradient contains no artificial component in the forbidden total-budget direction. For readability, we occasionally write $\nabla f_\delta(x)$ when the affine-hull meaning is unambiguous.

For a smoothing radius $\delta > 0$ and a finite-difference radius $h > 0$, set

$$\mathcal{X}_\delta = \left\{ x \in \operatorname{ri}(\mathcal{X}) : l_i + \delta \le x_i \le u_i - \delta,\ i \in [d] \right\}, \tag{8}$$

and

$$\mathcal{X}_{\delta,h} = \left\{ x \in \operatorname{ri}(\mathcal{X}) : l_i + \delta + h \le x_i \le u_i - \delta - h,\ i \in [d] \right\}. \tag{9}$$

For the finite-horizon algorithm in the main text, h is fixed. For the asymptotic convergence theorem, it is useful to separate the geometric safety buffer from the vanishing differentiation radius. We therefore fix $\bar{h} > 0$, require $0 < h_t \le \bar{h}$, and maintain iterates in the common tube $\mathcal{X}_{\delta,\bar{h}}$. The finite-difference query at iteration t uses h_t, whereas the feasible-step safeguard may use the fixed inner bounds $l_i + \delta + \bar{h}$ and $u_i - \delta - \bar{h}$. The main pseudocode is recovered by the special case $h_t \equiv \bar{h} \equiv h$.

The following assumptions are the exact hypotheses used in the deterministic proofs.

Assumption 1 (Common compact query tube). *There exist $\bar{h} > 0$ and a nonempty compact set*

$$K \subset \mathcal{X}_{\delta,\bar{h}} \subset \operatorname{ri}(\mathcal{X}) \tag{10}$$

such that every iterate belongs to K, every update segment $[x_t, x_{t+1}]$ lies in $\mathcal{X}_\delta$, and every reduced-coordinate finite-difference segment

$$\{ x_t + sQe_j : 0 \le s \le h_t \}, \qquad j = 1, \dots, r, \tag{11}$$

lies in $\mathcal{X}_\delta$. Consequently,

$$f^\star_{\delta,K} = \inf_{x \in K} f_\delta(x) > -\infty \tag{12}$$

is well defined. In the fixed-h implementation, the feasible-step rule in Appendix A.10 guarantees membership in $\mathcal{X}_{\delta,h}$; the additional inclusion in a compact K is the standard bounded-interior hypothesis needed for uniform metric constants.

Assumption 2 (Affine-hull smoothness). *There exists $\Lambda_\delta > 0$ such that, for all x, y in an open neighborhood of the segments specified in Eqs. (10)–(11), with $x - y \in \mathsf{T}$,*

$$\|\nabla_{\text{aff}} f_\delta(x) - \nabla_{\text{aff}} f_\delta(y)\|_2 \le \Lambda_\delta \|x - y\|_2. \tag{13}$$

If the unsmoothed reduced objective ϕ is L-Lipschitz on the relevant δ-neighborhood, Appendix A.6 establishes the bound

$$\Lambda_\delta \le \frac{Lr}{\delta}\mathbb{E}_{u\sim\mathrm{Unif}(\mathbb{S}^{r-1})}|u_1| \le \frac{L\sqrt{r}}{\delta} \le \frac{L\sqrt{d}}{\delta}. \tag{14}$$

Assumption 3 (Finite-difference accuracy). *At iteration t, the lifted estimator satisfies*

$$\|\widehat{g}_t - \nabla_{\mathrm{aff}} f_\delta(x_t)\|_2 \le \zeta_{h_t}. \tag{15}$$

For the forward differences of the main algorithm, Appendix A.7 proves, without suppressing the reduced dimension,

$$\zeta_{h_t} \le \frac{\sqrt{r}\,\Lambda_\delta h_t}{2} \le \frac{\sqrt{d}\,\Lambda_\delta h_t}{2}. \tag{16}$$

Thus $h_t \to 0$ implies $\zeta_{h_t} \to 0$. A stochastic oracle is treated separately in Appendix A.18; there the deterministic bias and the zero-mean oracle error are handled separately.

On the compact set K, the diagonal entropy–barrier Hessian and its inverse are continuous and positive definite. Hence their extreme eigenvalues attain strictly positive finite bounds. In particular, there exist constants $0 < \underline{w} \le \overline{w} < \infty$ such that

$$\underline{w} I \preceq H_X(x)^{-1} \preceq \overline{w} I, \qquad x \in K. \tag{17}$$

The constants $\underline{w}$ and $\overline{w}$ are not algorithmic tuning parameters. They enter only in the norm comparisons required by the descent and convergence proofs.

We next record the entropy–barrier metric identities repeatedly used in the proofs [1, 5, 8]. Fix $x \in \mathrm{ri}(X)$ and write

$$H = H_X(x), \qquad W = H^{-1}. \tag{18}$$

Since H is symmetric positive definite, so is W, and

$$\mathbf{1}^\top W \mathbf{1} > 0. \tag{19}$$

For any vector $y \in \mathbb{R}^d$, define the W-weighted centering operator

$$\begin{aligned} \Pi_W y &= y - \lambda_W(y)\mathbf{1}, \\ \lambda_W(y) &= \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}}. \end{aligned} \tag{20}$$

Then

$$\begin{aligned} \mathbf{1}^\top W \Pi_W y &= \mathbf{1}^\top W\,(y - \lambda_W(y)\mathbf{1}) \\ &= \mathbf{1}^\top W y - \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}} \mathbf{1}^\top W \mathbf{1} \\ &= 0. \end{aligned} \tag{21}$$

Thus $\Pi_W y$ is centered with respect to the W-weighted pairing. Note that $\Pi_W y$ need not lie in T; rather, $W\Pi_W y$ is tangent. The entropy–barrier Riemannian gradient associated with the covector y is

$$\mathfrak{g}_W(y) = W\Pi_W y = W\,(y - \lambda_W(y)\mathbf{1})\,. \tag{22}$$

By Eq. (21),

$$\mathbf{1}^\top \mathfrak{g}_W(y) = \mathbf{1}^\top W \Pi_W y = 0. \tag{23}$$

Therefore $\mathfrak{g}_W(y)$ lies in the tangent space of the budget hyperplane. The squared H-metric norm of $\mathfrak{g}_W(y)$ is

$$\begin{aligned} \|\mathfrak{g}_W(y)\|_H^2 &= \mathfrak{g}_W(y)^\top H \mathfrak{g}_W(y) \\ &= (W\Pi_W y)^\top H (W\Pi_W y) \\ &= (\Pi_W y)^\top W^\top H W (\Pi_W y) \\ &= (\Pi_W y)^\top W H W (\Pi_W y) \\ &= (\Pi_W y)^\top W (\Pi_W y), \end{aligned} \tag{24}$$

where we used that W is symmetric and $WHW = W$. In particular, for $y = \nabla_{\mathrm{aff}} F(x)$, define

$$\lambda_F(x) = \frac{\mathbf{1}^\top H_X(x)^{-1} \nabla_{\mathrm{aff}} F(x)}{\mathbf{1}^\top H_X(x)^{-1}\mathbf{1}}. \tag{25}$$

Then the main-text stationarity measure is

$$\begin{aligned} \mathcal{G}_F(x)^2 = (\nabla_{\mathrm{aff}} F(x) - \lambda_F(x)\mathbf{1})^\top H_X(x)^{-1} \\ \times (\nabla_{\mathrm{aff}} F(x) - \lambda_F(x)\mathbf{1})\,. \end{aligned} \tag{26}$$

Equivalently,

$$\mathcal{G}_F(x) = \left\| H_X(x)^{-1} (\nabla_{\mathrm{aff}} F(x) - \lambda_F(x)\mathbf{1}) \right\|_{H_X(x)}. \tag{27}$$

For the smoothed target $F = f_\delta$, we write $\lambda_F = \lambda_\delta$ and $\mathcal{G}_F = \mathcal{G}_\delta$.

A.2 VaR-Adjusted Objective and Static-Panel Specialization

The convergence analysis is stated for the scalar target F on the budget affine hull and therefore does not require a particular decomposition of the black-box objective. For the static allocation problem studied in the main text, let ξ index a return vector $\mathbf{R}(\xi)$ and set

$$\begin{aligned} R(x,\xi) &= x^\top \mathbf{R}(\xi), \\ \mu(x) &= \mathbb{E}[R(x,\xi)], \\ q_\alpha(x) &= q_\alpha(R(x,\xi)). \end{aligned} \tag{28}$$

This is the specialization used throughout the empirical study. The smoothed-oracle analysis acts on the resulting scalar objective, while the direct empirical formulas below additionally use the linear representation of the static mixture.

The optimized and reported population criterion is

$$S_{\mathrm{VaR}}(x) = \frac{\mu(x) - \rho}{\max\{\rho - q_\alpha(x), \kappa_\rho\}}, \qquad f(x) = -S_{\mathrm{VaR}}(x). \tag{29}$$

The fixed scale $\kappa_\rho > 0$ prevents a vanishing denominator without altering the numerator or introducing a mean-dependent branch. The resulting score is finite on the whole feasible set, but its lower-tail interpretation is sign-dependent. The qualification can be stated as

$$D_\kappa(x) = \max\{\rho - q_\alpha(x), \kappa_\rho\}.$$

Away from the clipping interface,

$$\frac{\partial S_{\mathrm{VaR}}}{\partial \mu} = \frac{1}{D_\kappa(x)} > 0.$$

On the unclipped branch $D_\kappa(x) = \rho - q_\alpha(x)$,

$$\frac{\partial S_{\mathrm{VaR}}}{\partial q_\alpha} = \frac{\mu(x) - \rho}{(\rho - q_\alpha(x))^2},$$

whereas on the clipped branch the score is locally independent of q_α. Consequently, a higher lower-tail quantile improves the score

when $\mu(x) > \rho$, is neutral at $\mu(x) = \rho$, and lowers the score when $\mu(x) < \rho$ on the unclipped branch. Thus Eq. (29) defines one scalar optimization criterion on both sides of the target floor, but it is a monotone lower-tail-protection criterion only in the super-target region. Below the target floor, negative values should be read as normalized recovery or mean-shortfall scores, not as a globally tail-monotone risk ranking. This qualification affects the interpretation of the objective, not the target-agnostic convergence arguments below, which apply to the scalar function F under their stated regularity conditions.

For empirical samples $\{\mathbf{R}^{(j)}\}_{j=1}^{n}$, define

$$\widehat{\mu}_n(x) = \frac{1}{n}\sum_{j=1}^{n} x^\top \mathbf{R}^{(j)}, \qquad \widehat{q}_{\alpha,n}(x) = R_{(\lceil n\alpha \rceil)}(x), \tag{30}$$

and set

$$\begin{aligned} \widehat{S}_{\mathrm{VaR},n}(x) &= \frac{\widehat{\mu}_n(x) - \rho}{\max\{\rho - \widehat{q}_{\alpha,n}(x), \kappa_\rho\}}, \\ \widehat{f}_n(x) &= -\widehat{S}_{\mathrm{VaR},n}(x). \end{aligned} \tag{31}$$

This is the scalar objective evaluated in direct empirical-oracle mode and the metric used for validation model selection. The same sign-aware qualification applies empirically: when $\widehat{\mu}_n(x) < \rho$, a higher negative score need not correspond to a higher empirical lower-tail quantile. For that reason, the empirical tables report mean return and reference-floor violation rates alongside $\widehat{S}_{\mathrm{VaR},n}$ rather than interpreting the scalar score alone as a uniform downside-risk ordering.

A.3 Lipschitz Continuity of the Empirical VaR-Adjusted Objective

Fix a compact convex queried domain $\mathcal{D} \subseteq \mathcal{X}$ and assume $\|\mathbf{R}^{(j)}\|_2 \le M_R$. Each sample return, the empirical mean, and the empirical order statistic are then M_R-Lipschitz. Let

$$A(x) = \rho - \widehat{\mu}_n(x), \qquad B_\kappa(x) = \max\{\rho - \widehat{q}_{\alpha,n}(x), \kappa_\rho\}. \tag{32}$$

Then $\widehat{f}_n = A/B_\kappa$, $B_\kappa \ge \kappa_\rho$, and B_κ is M_R-Lipschitz. If $|A(x)| \le M_A$ on $\mathcal{D}$, then

$$\begin{aligned} |\widehat{f}_n(x) - \widehat{f}_n(y)| &\le \frac{|A(x) - A(y)|}{B_\kappa(x)} + \frac{|A(y)|\,|B_\kappa(x) - B_\kappa(y)|}{B_\kappa(x) B_\kappa(y)} \\ &\le \left(\frac{M_R}{\kappa_\rho} + \frac{M_A M_R}{\kappa_\rho^2}\right)\|x - y\|_2. \end{aligned} \tag{33}$$

Thus $\widehat{f}_n$ is globally Lipschitz on $\mathcal{D}$ with

$$L_{\mathrm{emp}} = \frac{M_R}{\kappa_\rho} + \frac{M_A M_R}{\kappa_\rho^2}. \tag{34}$$

The only nonsmoothness in the empirical criterion arises from order-statistic changes and the denominator-clipping interface.

A.4 Direct Empirical Oracle and Stratified Active-Cell Trajectories

The smooth-target theorem applies globally to the affine-hull smoothed objective on its common query tube. The main experiments instead use the direct empirical target, denoted by $M_{\mathrm{smooth}} = 0$ in the implementation. The direct target is continuous and globally Lipschitz, but only piecewise smooth. We therefore separate two issues that should not be conflated. First, on every regular active cell, the empirical objective admits an exact smooth branch and the usual finite-difference descent argument applies. Second, an implemented trajectory may cross order-statistic or clipping interfaces. Proposition 1 assembles the cellwise inequalities into a trajectory-level certificate: regular iterations contribute genuine stationarity descent, while crossing iterations enter through a separate interface debit. This yields a finite-horizon result for every direct trajectory on the compact query tube, without pretending that a crossing iterate belongs to one globally smooth model.

Let

$$\bar{R} = \frac{1}{n}\sum_{j=1}^{n} \mathbf{R}^{(j)}, \qquad k = \lceil n\alpha \rceil, \tag{35}$$

and recall

$$\widehat{\mu}_n(x) = \bar{R}^\top x, \qquad \widehat{q}_{\alpha,n}(x) = R_{(k)}(x). \tag{36}$$

Consider the finite collection of affine hyperplanes in $\mathrm{aff}(\mathcal{X})$. In the unions below, permanently identical affine return functions are merged, and sets that coincide with the entire affine hull are omitted; this avoids treating a structural identity as a kink.

$$\mathcal{H}_{\mathrm{ord}} = \bigcup_{1 \le a < b \le n} \left\{x : (\mathbf{R}^{(a)} - \mathbf{R}^{(b)})^\top x = 0\right\}, \tag{37}$$

and

$$\mathcal{H}_{\mathrm{clip}} = \bigcup_{j=1}^{n} \left\{x : \rho - (\mathbf{R}^{(j)})^\top x = \kappa_\rho\right\}. \tag{38}$$

The complement of $\mathcal{H}_{\mathrm{ord}} \cup \mathcal{H}_{\mathrm{clip}}$ has finitely many connected components. We call each nonempty component intersected with $\mathrm{ri}(\mathcal{X})$ a *regular active cell* and denote it by C.

Fix one regular active cell C. No pair of sample returns ties on C, so there is one permutation π_C satisfying

$$(\mathbf{R}^{(\pi_C(1))})^\top x < \cdots < (\mathbf{R}^{(\pi_C(n))})^\top x, \qquad x \in C. \tag{39}$$

Consequently, the empirical lower-tail quantile is represented throughout the cell by one fixed sample vector

$$q_C = \mathbf{R}^{(\pi_C(k))}, \qquad \widehat{q}_{\alpha,n}(x) = q_C^\top x. \tag{40}$$

Moreover, because C does not meet $\mathcal{H}_{\mathrm{clip}}$, exactly one of the following regimes holds on the whole cell:

$$\rho - q_C^\top x > \kappa_\rho \quad \text{for every } x \in C, \tag{41}$$

or

$$\rho - q_C^\top x < \kappa_\rho \quad \text{for every } x \in C. \tag{42}$$

Define the affine numerator

$$A(x) = \rho - \bar{R}^\top x \tag{43}$$

and, in the unclipped regime, the affine denominator

$$B_C(x) = \rho - q_C^\top x. \tag{44}$$

The direct empirical minimization objective $\widehat{f}_n = -\widehat{S}_{\mathrm{VaR},n}$ is therefore

$$\widehat{f}_n(x) = \begin{cases} A(x)/B_C(x), & \text{if Eq. (41) holds,} \\ A(x)/\kappa_\rho, & \text{if Eq. (42) holds.} \end{cases} \tag{45}$$

The clipped branch is affine, whereas the unclipped branch is linear-fractional; the sign of $A(x)$ does not alter this partition. On the unclipped branch, the quotient rule gives

$$\nabla \widehat{f}_n(x) = -\frac{\bar{R}}{B_C(x)} + \frac{A(x)}{B_C(x)^2} q_C. \tag{46}$$

This formula is used for regularity proofs and offline residual checks; the reported optimization runs receive only scalar target-oracle values. For completeness, write $B = B_C(x)$, $A = A(x)$, and $q = q_C$. Since

$$\begin{aligned} \nabla A &= -\bar{R}, & \nabla B &= -q, \\ \nabla\left(B^{-1}\right) &= B^{-2} q, & \nabla\left(B^{-2}\right) &= 2B^{-3} q, \end{aligned} \tag{47}$$

we have

$$\begin{aligned} \nabla^2 \widehat{f}_n(x) &= \nabla\left(-B^{-1}\bar{R} + AB^{-2} q\right) \\ &= -\bar{R}\,\nabla(B^{-1})^\top + q\,\nabla(AB^{-2})^\top \\ &= -\frac{\bar{R} q^\top}{B^2} - \frac{q\bar{R}^\top}{B^2} + \frac{2A}{B^3} q q^\top. \end{aligned} \tag{48}$$

The expression is symmetric, as required for a twice continuously differentiable scalar branch.

Let $K \subset \mathrm{ri}(\mathcal{X})$ be a compact set containing the direct trajectory and all of its query and update segments. Set $M_A = \sup_{x\in K} |A(x)|$ and $M_R = \max_j \|\mathbf{R}^{(j)}\|_2$. On the closure of every regular cell that meets K,

$$|A(x)| \le M_A, \qquad B_C(x) \ge \kappa_\rho, \qquad x \in K \cap \overline{C}. \tag{49}$$

Using $\|ab^\top\|_{2\to 2} = \|a\|_2 \|b\|_2$ in Eq. (48) yields the cellwise and uniform bounds

$$\begin{aligned} \|\nabla^2 \widehat{f}_{n,C}(x)\|_{2\to 2} &\le \frac{2\|\bar{R}\|_2 \|q_C\|_2}{\kappa_\rho^2} + \frac{2M_A \|q_C\|_2^2}{\kappa_\rho^3} \\ &=: \Lambda_C \le \frac{2\|\bar{R}\|_2 M_R}{\kappa_\rho^2} + \frac{2M_A M_R^2}{\kappa_\rho^3} =: \overline{\Lambda}_n. \end{aligned} \tag{50}$$

Here $\widehat{f}_{n,C}$ denotes the smooth affine or linear-fractional branch associated with C, extended to a neighborhood of $K \cap \overline{C}$. On a clipped branch its Hessian is zero. The mean-value formula therefore gives, whenever the segment $[x, y]$ is contained in one cell closure,

$$\begin{aligned} &\nabla \widehat{f}_{n,C}(y) - \nabla \widehat{f}_{n,C}(x) \\ &\quad = \int_0^1 \nabla^2 \widehat{f}_{n,C}(x + s(y-x))(y-x)\, ds, \\ &\|\nabla \widehat{f}_{n,C}(y) - \nabla \widehat{f}_{n,C}(x)\|_2 \\ &\quad \le \overline{\Lambda}_n \|y - x\|_2. \end{aligned} \tag{51}$$

For $x \in \overline{C}$, define the lifted affine-hull branch gradient

$$g_C(x) = QQ^\top \nabla \widehat{f}_{n,C}(x). \tag{52}$$

At a differentiable point this is the ordinary affine-hull gradient of $\widehat{f}_n$. At an ordering or clipping interface it is a limiting branch gradient and hence belongs to the Clarke affine-hull subdifferential $\partial^{\mathrm{C}}_{\mathrm{aff}} \widehat{f}_n(x)$. The branchwise stationarity measure is

$$\mathcal{G}_{n,C}(x) = \left\| H_{\mathcal{X}}(x)^{-1} \left(g_C(x) - \lambda_{n,C}(x)\mathbf{1}\right) \right\|_{H_{\mathcal{X}}(x)}, \tag{53}$$

where $\lambda_{n,C}(x) = \frac{\mathbf{1}^\top H_{\mathcal{X}}(x)^{-1} g_C(x)}{\mathbf{1}^\top H_{\mathcal{X}}(x)^{-1}\mathbf{1}}$. For reduced coordinates, write $\widehat{\phi}_n(z) = \widehat{f}_n(x_{\mathrm{ref}} + Qz)$ and $\widehat{\phi}_{n,C}(z) = \widehat{f}_{n,C}(x_{\mathrm{ref}} + Qz)$.

For a horizon T, call iteration t *regular* if there exists one regular active cell C_t whose closure contains x_t, every finite-difference segment $\{x_t + sQe_j : 0 \le s \le h_t\}$, and the update segment $[x_t, x_{t+1}]$. Let $\mathcal{R}_T$ be the set of regular iterations and let $\mathcal{B}_T = \{0, \dots, T-1\} \setminus \mathcal{R}_T$ be the interface set. Write $\Gamma_T^{\mathrm{reg}} = \sum_{t\in\mathcal{R}_T} \gamma_t$ and define the a posteriori interface debit $\mathfrak{D}_T = \sum_{t\in\mathcal{B}_T} [\widehat{f}_n(x_{t+1}) - \widehat{f}_n(x_t)]_+$. All center values except possibly the terminal one are already available from the reused baseline queries. If a terminal evaluation is not recorded, the last increment—and, more conservatively, the entire debit—can be replaced without an additional oracle call by the Lipschitz envelope $\mathfrak{D}_T \le L_{\mathrm{emp}} \sum_{t\in\mathcal{B}_T} \gamma_t \|d_t\|_2$.

Proposition 1 (Direct-oracle stratified descent, convergence, and complexity)**.** *Suppose that every iterate, finite-difference segment, and update segment lies in the compact set $K \subset \mathrm{ri}(\mathcal{X})$. Run Algorithm 1 with the direct empirical scalar oracle $\widehat{f}_n$, with no Monte Carlo smoothing layer. Then, for every $t \in \mathcal{R}_T$, the estimator approximates the branch gradient selected by the actual query envelope:*

$$\|\widehat{g}_t - g_{C_t}(x_t)\|_2 \le \zeta_{h_t}^{\mathrm{dir}} := \frac{\sqrt{r}\,\overline{\Lambda}_n h_t}{2}. \tag{54}$$

There exist the same geometry-dependent constants $c_1, c_2, c_3 > 0$ as in the smooth descent argument such that, whenever $\gamma_t \le c_1/\overline{\Lambda}_n$ for $t \in \mathcal{R}_T$,

$$\begin{aligned} c_2 \sum_{t\in\mathcal{R}_T} \gamma_t \mathcal{G}_{n,C_t}(x_t)^2 &\le \Delta_{n,K} + c_3 \sum_{t\in\mathcal{R}_T} \gamma_t (\zeta_{h_t}^{\mathrm{dir}})^2 + \mathfrak{D}_T, \\ \min_{t\in\mathcal{R}_T} \mathcal{G}_{n,C_t}(x_t)^2 &\le \frac{\Delta_{n,K} + c_3 \sum_{t\in\mathcal{R}_T} \gamma_t (\zeta_{h_t}^{\mathrm{dir}})^2 + \mathfrak{D}_T}{c_2 \Gamma_T^{\mathrm{reg}}}, \end{aligned} \tag{55}$$

with the second line understood when $\Gamma_T^{\mathrm{reg}} > 0$ and $\Delta_{n,K} = \widehat{f}_n(x_0) - \inf_{x\in K} \widehat{f}_n(x)$. Thus active-cell crossings do not invalidate the direct run; they are isolated from the smooth descent contribution and charged exactly through $\mathfrak{D}_T$.

If, along an infinite trajectory,

$$\begin{aligned} \Gamma_T^{\mathrm{reg}} &\to \infty, \\ \frac{\mathfrak{D}_T + \sum_{t\in\mathcal{R}_T} \gamma_t (\zeta_{h_t}^{\mathrm{dir}})^2}{\Gamma_T^{\mathrm{reg}}} &\to 0, \end{aligned}$$

then the trajectory has a regular-iterate accumulation point $\bar{x}$ satisfying the Clarke–KKT condition

$$0 \in \partial^{\mathrm{C}}_{\mathrm{aff}} \widehat{f}_n(\bar{x}) + \mathcal{N}_{\mathcal{X}}(\bar{x}). \tag{56}$$

If, more strongly, there exist t_0 and one cell C_∞ such that every iterate, probe, and update segment after t_0 lies in $\overline{C_\infty}$, then under $h_t \to 0$, $\sum_t \gamma_t = \infty$, $\sum_t \gamma_t^2 < \infty$, and $\sum_t \gamma_t (\zeta_{h_t}^{\mathrm{dir}})^2 < \infty$, every accumulation point satisfies Eq. (56). Each iteration uses exactly d direct empirical-objective evaluations.

Proof. Fix $t \in \mathcal{R}_T$. By definition, all coordinate probes and the update segment are represented by one smooth branch $\widehat{f}_{n,C_t}$. Applying the fundamental theorem of calculus to each reduced coordinate and using Eq. (51) gives

$$\left| \frac{\widehat{\phi}_n(z + h_t e_j) - \widehat{\phi}_n(z)}{h_t} - \partial_j \widehat{\phi}_{n,C_t}(z) \right| \le \frac{\overline{\Lambda}_n h_t}{2}. \tag{57}$$

Stacking the r coordinates and using $Q^\top Q = I_r$ proves Eq. (54). The barrier projection, tangent identity, and feasible-step proof depend only on the estimator vector and the geometry of $\mathcal{X}$. Hence the smooth one-step argument gives, on every regular iteration, $\widehat{f}_n(x_{t+1}) - \widehat{f}_n(x_t) \le -c_2\gamma_t \mathcal{G}_{n,C_t}(x_t)^2 + c_3\gamma_t(\zeta^{\mathrm{dir}}_{h_t})^2$. For $t \in \mathcal{B}_T$, no smooth-branch descent assertion is made; by the definition of the positive part, $\widehat{f}_n(x_{t+1}) - \widehat{f}_n(x_t) \le [\widehat{f}_n(x_{t+1}) - \widehat{f}_n(x_t)]_+$. Summing both kinds of increments, telescoping, and using $\widehat{f}_n(x_T) \ge \inf_{x\in K} \widehat{f}_n(x)$ proves the first line of Eq. (55); division by Γ^{reg}_T proves the second. The Lipschitz upper bound on $\mathfrak{D}_T$ follows from Appendix A.3 and $x_{t+1} - x_t = \gamma_t d_t$.

Under the stated ratio condition, the weighted regular-iterate average of $\mathcal{G}_{n,C_t}(x_t)^2$ tends to zero. Consequently there is a sequence of regular indices $t_k \to \infty$ with $\mathcal{G}_{n,C_{t_k}}(x_{t_k}) \to 0$. Compactness gives a convergent subsequence, and finiteness of the active-cell stratification gives a further subsequence associated with one cell C. The branch gradients and the barrier metric are continuous on $K \cap \overline{C}$; therefore the limiting branch gradient satisfies the metric KKT relation. Because every limiting branch gradient of a continuous piecewise-C^1 function belongs to its Clarke subdifferential, Eq. (56) follows. Under eventual active-cell identification, the original quasi-Fejér and accumulation-point proof applies to the smooth extension $\widehat{f}_{n,C_\infty}$ on the entire tail. Its limiting gradient is again a Clarke branch gradient, so every accumulation point satisfies the same inclusion. Finally, one shared baseline value and $r = d - 1$ forward values are queried, giving $1 + r = d$ direct evaluations per iteration. □

The proposition deliberately distinguishes a trajectory certificate from an unconditional nonsmooth KKT theorem. A crossing iteration is never passed off as a smooth step: its entire adverse contribution is visible in $\mathfrak{D}_T$. The finite-horizon inequality therefore applies to the same direct empirical mode used in the main tables, including trajectories with nonzero crossing rates. The asymptotic Clarke conclusion requires either a vanishing debit relative to accumulated regular step or eventual active-cell identification. Affine-hull smoothing remains the mode that supplies an unconditional globally differentiable target and the corresponding global smooth-target KKT certificate.

A.5 Feasibility of Affine-Hull Smoothing

We show that affine-hull smoothing preserves the budget equality and the box constraints. Recall that $Q^\top Q = I_r$ and $Q^\top \mathbf{1} = 0$. Hence $\|Q\|_{2\to 2} = 1$.

Figure 1 illustrates the geometric role of the safe region $\mathcal{X}_{\delta,h}$. Smoothing perturbations are taken only along directions in the budget tangent space, while finite-difference query points are kept away from the box boundary so that subsequent smoothing queries remain feasible.

Let $x \in \mathcal{X}_\delta$ and $v \in \mathbb{B}_r$. Since $Qv \in \mathsf{T}$, the affine equality is preserved:

$$\begin{aligned}\mathbf{1}^\top(x + \delta Qv) &= \mathbf{1}^\top x + \delta \mathbf{1}^\top Qv \\ &= 1 + \delta(Q^\top \mathbf{1})^\top v \\ &= 1. \end{aligned} \tag{58}$$

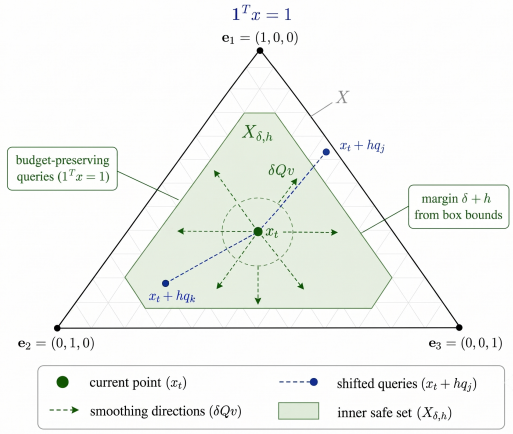


Figure 1: Safe affine-hull queries. The inner region $\mathcal{X}_{\delta,h}$ leaves a margin from the box boundary, so smoothing and finite-difference queries remain on the budget hyperplane and stay feasible.

Therefore $x + \delta Qv \in \mathrm{aff}(\mathcal{X})$. It remains to check the box constraints. For each $i \in [d]$,

$$\begin{aligned}|(Qv)_i| &= |e_i^\top Qv| \\ &\le \|e_i\|_2 \|Qv\|_2 \\ &\le \|Q\|_{2\to 2}\|v\|_2 \\ &\le 1. \end{aligned} \tag{59}$$

Since $x \in \mathcal{X}_\delta$, we have $l_i + \delta \le x_i \le u_i - \delta$. Hence

$$\begin{aligned}x_i + \delta(Qv)_i &\ge x_i - \delta|(Qv)_i| \\ &\ge x_i - \delta \\ &\ge l_i, \end{aligned} \tag{60}$$

and

$$\begin{aligned}x_i + \delta(Qv)_i &\le x_i + \delta|(Qv)_i| \\ &\le x_i + \delta \\ &\le u_i. \end{aligned} \tag{61}$$

Thus $l_i \le x_i + \delta(Qv)_i \le u_i$ for every $i \in [d]$. Combining this with Eq. (58), we obtain

$$x + \delta Qv \in \mathcal{X}. \tag{62}$$

Therefore every affine-hull smoothing query used to evaluate f_δ is feasible.

When finite differences are also used, the algorithm starts from $x \in \mathcal{X}_{\delta,h}$. The additional margin h ensures that the reduced-coordinate queries $z + he_j$ remain inside the region where subsequent δ-smoothing queries are feasible, as illustrated in Figure 1. This is the reason for using $\mathcal{X}_{\delta,h}$ in Algorithm 1.

A.6 Gradient Formula for Uniform Smoothing

This subsection records the standard gradient identity for uniform ball smoothing in the reduced coordinates. Let

$$\begin{aligned}\phi_\delta(z) &= \mathbb{E}_{v\sim\nu_r}\left[\phi(z+\delta v)\right] \\ &= \frac{1}{\mathrm{Vol}(\mathbb{B}_r)}\int_{\mathbb{B}_r}\phi(z+\delta v)\,dv,\end{aligned} \tag{63}$$

where ν_r is the uniform distribution over $\mathbb{B}_r$. Assume that ϕ is locally Lipschitz on an open set containing $z+\delta\mathbb{B}_r$. Then ϕ_δ is differentiable, and

$$\nabla\phi_\delta(z) = \frac{r}{\delta}\mathbb{E}_{u\sim\mathrm{Unif}(\mathbb{S}^{r-1})}\left[\phi(z+\delta u)u\right]. \tag{64}$$

The identity follows from differentiating the ball average in the weak sense and applying the divergence theorem to the boundary $\partial\mathbb{B}_r = \mathbb{S}^{r-1}$:

$$\begin{aligned}\nabla\phi_\delta(z) &= \frac{1}{\delta\,\mathrm{Vol}(\mathbb{B}_r)}\int_{\partial\mathbb{B}_r}\phi(z+\delta u)u\,dS(u) \\ &= \frac{\mathrm{Area}(\mathbb{S}^{r-1})}{\delta\,\mathrm{Vol}(\mathbb{B}_r)}\mathbb{E}_{u\sim\mathrm{Unif}(\mathbb{S}^{r-1})}\left[\phi(z+\delta u)u\right] \\ &= \frac{r}{\delta}\mathbb{E}_{u\sim\mathrm{Unif}(\mathbb{S}^{r-1})}\left[\phi(z+\delta u)u\right],\end{aligned} \tag{65}$$

where $\mathrm{Area}(\mathbb{S}^{r-1}) = r\,\mathrm{Vol}(\mathbb{B}_r)$.

We next justify the $O(L\sqrt{r}/\delta)$ gradient-Lipschitz constant used in the main text. Let z, y be two reduced points for which all translated balls are contained in the Lipschitz neighborhood, and let $a\in\mathbb{R}^r$ satisfy $\|a\|_2 = 1$. From Eq. (64),

$$\begin{aligned}&a^\top\left(\nabla\phi_\delta(z) - \nabla\phi_\delta(y)\right) \\ &\quad= \frac{r}{\delta}\mathbb{E}_u\left[\left(\phi(z+\delta u) - \phi(y+\delta u)\right)a^\top u\right].\end{aligned} \tag{66}$$

Taking absolute values, applying the L-Lipschitz property pointwise, and using rotational invariance of the uniform sphere distribution gives

$$\begin{aligned}\left|a^\top\left(\nabla\phi_\delta(z) - \nabla\phi_\delta(y)\right)\right| &\le \frac{rL}{\delta}\|z-y\|_2\mathbb{E}_u|a^\top u| \\ &= \frac{rL}{\delta}\|z-y\|_2\mathbb{E}_u|u_1|.\end{aligned} \tag{67}$$

Since $\mathbb{E}[u_1^2] = 1/r$, Cauchy–Schwarz yields

$$\mathbb{E}|u_1| \le \sqrt{\mathbb{E}[u_1^2]} = \frac{1}{\sqrt{r}}. \tag{68}$$

Taking the supremum over all unit vectors a in Eq. (67), we obtain

$$\|\nabla\phi_\delta(z) - \nabla\phi_\delta(y)\|_2 \le \frac{L\sqrt{r}}{\delta}\|z-y\|_2. \tag{69}$$

Because Q is an isometry from $\mathbb{R}^r$ onto T, the identical bound holds for the lifted affine-hull gradient. This proves Eq. (14) with a specified constant rather than an order statement alone.

The smoothing bias is controlled by Lipschitz continuity. If ϕ is L-Lipschitz on $z+\delta\mathbb{B}_r$, then

$$\begin{aligned}|\phi_\delta(z) - \phi(z)| &= \left|\mathbb{E}_{v\sim\nu_r}\left[\phi(z+\delta v) - \phi(z)\right]\right| \\ &\le \mathbb{E}_{v\sim\nu_r}\left[|\phi(z+\delta v) - \phi(z)|\right] \\ &\le L\delta\,\mathbb{E}_{v\sim\nu_r}\left[\|v\|_2\right] \\ &\le L\delta.\end{aligned} \tag{70}$$

Since $f_\delta(x_{\mathrm{ref}}+Qz) = \phi_\delta(z)$ and $\nabla_{\mathrm{aff}}f_\delta(x_{\mathrm{ref}}+Qz) = Q\nabla\phi_\delta(z)$, Eq. (64) gives the affine-hull gradient used by RiskBid.

A.7 Finite-Difference Bias

This subsection verifies Assumption 3 for the forward-difference estimator used by RiskBid. Let $x = x_{\mathrm{ref}}+Qz$, and suppose that the line segments $\{z+se_j : 0\le s\le h,\ j=1,\dots,r\}$ lie in the reduced region corresponding to $\mathcal{X}_\delta$. By Assumption 2 and the orthonormality of Q, ϕ_δ has Λ_δ-Lipschitz gradient on these segments. For $j=1,\dots,r$, define

$$\widehat{U}_j(z) = \frac{\phi_\delta(z+he_j) - \phi_\delta(z)}{h}. \tag{71}$$

By the fundamental theorem of calculus,

$$\widehat{U}_j(z) - \partial_j\phi_\delta(z) = \frac{1}{h}\int_0^h\left[\partial_j\phi_\delta(z+se_j) - \partial_j\phi_\delta(z)\right]ds. \tag{72}$$

Therefore,

$$\begin{aligned}\left|\widehat{U}_j(z) - \partial_j\phi_\delta(z)\right| &\le \frac{1}{h}\int_0^h\Big|e_j^\top\left[\nabla\phi_\delta(z+se_j)\right. \\ &\qquad\left. - \nabla\phi_\delta(z)\right]\Big|ds \\ &\le \frac{1}{h}\int_0^h\left\|\nabla\phi_\delta(z+se_j)\right. \\ &\qquad\left. - \nabla\phi_\delta(z)\right\|_2 ds \\ &\le \frac{1}{h}\int_0^h\Lambda_\delta s\,ds \\ &= \frac{\Lambda_\delta h}{2}.\end{aligned} \tag{73}$$

Let $\widehat{U}(z) = \left(\widehat{U}_1(z),\dots,\widehat{U}_r(z)\right)^\top$. Stacking the coordinate-wise bounds gives

$$\begin{aligned}\left\|\widehat{U}(z) - \nabla\phi_\delta(z)\right\|_2 &= \left(\sum_{j=1}^r\left|\widehat{U}_j(z) - \partial_j\phi_\delta(z)\right|^2\right)^{1/2} \\ &\le \frac{\sqrt{r}\,\Lambda_\delta h}{2} \\ &\le \frac{\sqrt{d}\,\Lambda_\delta h}{2},\end{aligned} \tag{74}$$

where $r = d-1$. The lifted estimator is

$$\widehat{g} = Q\widehat{U}(z). \tag{75}$$

Using $\nabla_{\mathrm{aff}}f_\delta(x) = Q\nabla\phi_\delta(z)$, and $Q^\top Q = I_r$, we obtain

$$\begin{aligned}\|\widehat{g} - \nabla_{\mathrm{aff}}f_\delta(x)\|_2 &= \left\|Q\left(\widehat{U}(z) - \nabla\phi_\delta(z)\right)\right\|_2 \\ &= \left\|\widehat{U}(z) - \nabla\phi_\delta(z)\right\|_2 \\ &\le \frac{\sqrt{d}\,\Lambda_\delta h}{2}.\end{aligned} \tag{76}$$

Thus Assumption 3 holds with

$$\zeta_h \le \frac{\sqrt{r}\,\Lambda_\delta h}{2} \le \frac{\sqrt{d}\,\Lambda_\delta h}{2}. \tag{77}$$

A.8 Pure-Simplex Limit of RiskBid

This subsection derives the pure-simplex form of RiskBid from the same entropy–barrier potential used in the box-constrained method. When $l_i = 0$ and $u_i = 1$ for all i, the upper-barrier sum in Eq. (93) is empty, so $\theta_{\mathcal{X}}(x) = \sum_i x_i \log x_i$. Consider

$$\Delta = \left\{x \in \mathbb{R}^d : x_i \geq 0,\ i \in [d],\ \mathbf{1}^\top x = 1\right\}, \tag{78}$$

and its relative interior

$$\mathrm{ri}(\Delta) = \left\{x \in \mathbb{R}^d : x_i > 0,\ i \in [d],\ \mathbf{1}^\top x = 1\right\}. \tag{79}$$

On $\mathrm{ri}(\Delta)$, the negative entropy potential is

$$\psi_\Delta(x) = \sum_{i=1}^{d} x_i \log x_i, \tag{80}$$

with Hessian

$$H_\Delta(x) = \nabla^2 \psi_\Delta(x) = \mathrm{diag}\left(\frac{1}{x_1}, \ldots, \frac{1}{x_d}\right). \tag{81}$$

Since $x_i > 0$ for all i, $H_\Delta(x)$ is positive definite. For a differentiable function g, the Shahshahani Riemannian gradient is the solution of

$$\mathrm{grad}_\Delta g(x) = \arg\min_{v \in \mathsf{T}} \left\{\frac{1}{2} v^\top H_\Delta(x) v - \nabla g(x)^\top v\right\}, \tag{82}$$

where $\mathsf{T} = \{v \in \mathbb{R}^d : \mathbf{1}^\top v = 0\}$. The Lagrangian for the constraint $\mathbf{1}^\top v = 0$ is

$$\mathcal{L}(v, \lambda) = \frac{1}{2} v^\top H_\Delta(x) v - \nabla g(x)^\top v + \lambda \mathbf{1}^\top v. \tag{83}$$

The first-order condition gives

$$H_\Delta(x) v = \nabla g(x) - \lambda \mathbf{1},\ v = H_\Delta(x)^{-1} \left(\nabla g(x) - \lambda \mathbf{1}\right). \tag{84}$$

Using $\mathbf{1}^\top v = 0$, we obtain

$$\lambda = \frac{\mathbf{1}^\top H_\Delta(x)^{-1} \nabla g(x)}{\mathbf{1}^\top H_\Delta(x)^{-1} \mathbf{1}}. \tag{85}$$

Since $H_\Delta(x)^{-1} = \mathrm{diag}(x_1, \ldots, x_d)$ and $\mathbf{1}^\top x = 1$, this reduces to

$$\lambda = \sum_{s=1}^{d} x_s\, \partial_{x_s} g(x). \tag{86}$$

Therefore, for every $i \in [d]$,

$$[\mathrm{grad}_\Delta g(x)]_i = x_i \left(\partial_{x_i} g(x) - \sum_{s=1}^{d} x_s\, \partial_{x_s} g(x)\right). \tag{87}$$

Replacing $\nabla g(x_t)$ by the lifted finite-difference estimator $\widehat{g}_t$, the pure-simplex descent direction is

$$d_{t,i}^{\Delta} = -x_{t,i} \left(\widehat{g}_{t,i} - \sum_{s=1}^{d} x_{t,s} \widehat{g}_{t,s}\right), \qquad i \in [d]. \tag{88}$$

It is tangent to the simplex because

$$\begin{aligned} \sum_{i=1}^{d} d_{t,i}^{\Delta} &= -\sum_{i=1}^{d} x_{t,i} \left(\widehat{g}_{t,i} - \sum_{s=1}^{d} x_{t,s} \widehat{g}_{t,s}\right) \\ &= -\sum_{i=1}^{d} x_{t,i} \widehat{g}_{t,i} + \left(\sum_{i=1}^{d} x_{t,i}\right) \left(\sum_{s=1}^{d} x_{t,s} \widehat{g}_{t,s}\right) \\ &= 0. \end{aligned} \tag{89}$$

Define

$$\tau_t^{\Delta} = \min_{i : d_{t,i}^{\Delta} < 0} \frac{x_{t,i}}{-d_{t,i}^{\Delta}}, \qquad \gamma_t^{\Delta} = \min\{\eta_t, \beta \tau_t^{\Delta}\}, \quad \beta \in (0, 1), \tag{90}$$

where the minimum over an empty set is $+\infty$. The resulting update is

$$x_{t+1,i} = x_{t,i} + \gamma_t^{\Delta} d_{t,i}^{\Delta} \tag{91}$$

preserves $\mathbf{1}^\top x = 1$ and satisfies $x_{t+1} \in \mathrm{ri}(\Delta)$ whenever $x_t \in \mathrm{ri}(\Delta)$. This is the feasible pure-simplex limit used in the main text.

A.9 Derivation of the Entropy–Barrier Projection

Consider the box-constrained budget polytope

$$\mathcal{X} = \{x \in \mathbb{R}^d : \mathbf{1}^\top x = 1,\ l_i \leq x_i \leq u_i,\ i \in [d]\}. \tag{92}$$

On $\mathrm{ri}(\mathcal{X})$, use the same separable entropy–barrier potential as in the main text:

$$\theta_{\mathcal{X}}(x) = \sum_{i=1}^{d} (x_i - l_i) \log(x_i - l_i) - \sum_{i : u_i < 1} \log(u_i - x_i). \tag{93}$$

Its Hessian is

$$H_{\mathcal{X}}(x) = \nabla^2 \theta_{\mathcal{X}}(x) = \mathrm{diag}(h_1(x), \ldots, h_d(x)), \tag{94}$$

where

$$h_i(x) = \frac{1}{x_i - l_i} + \frac{\mathbf{1}_{\{u_i < 1\}}}{(u_i - x_i)^2}. \tag{95}$$

Thus $H_{\mathcal{X}}(x)$ is positive definite. Write $W = H_{\mathcal{X}}(x)^{-1}$.

For a differentiable function g, the equality-constrained Riemannian gradient solves

$$\mathrm{grad}_{\mathcal{X}}\, g(x) = \arg\min_{v : \mathbf{1}^\top v = 0} \left\{\frac{1}{2} v^\top H_{\mathcal{X}}(x) v - \nabla g(x)^\top v\right\}. \tag{96}$$

The first-order condition gives

$$v = W\{\nabla g(x) - \lambda \mathbf{1}\}. \tag{97}$$

Imposing $\mathbf{1}^\top v = 0$ yields

$$\lambda_{\mathcal{X}}(x) = \frac{\mathbf{1}^\top W \nabla g(x)}{\mathbf{1}^\top W \mathbf{1}}. \tag{98}$$

Therefore

$$\mathrm{grad}_{\mathcal{X}}\, g(x) = W\{\nabla g(x) - \lambda_{\mathcal{X}}(x) \mathbf{1}\}. \tag{99}$$

Equivalently, with $A = \mathbf{1}^\top$, the $H_{\mathcal{X}}(x)$-orthogonal tangent projection is

$$P_{\mathcal{X}}(x) = I - W A^\top (A W A^\top)^{-1} A = I - W \mathbf{1} \frac{\mathbf{1}^\top}{\mathbf{1}^\top W \mathbf{1}}. \tag{100}$$

Applying it to the unconstrained Riemannian gradient gives

$$P_{\mathcal{X}}(x) W \nabla g(x) = W\{\nabla g(x) - \lambda_{\mathcal{X}}(x) \mathbf{1}\} = \mathrm{grad}_{\mathcal{X}}\, g(x). \tag{101}$$

Hence the descent direction used by RiskBid is

$$d_g(x) = -W\{\nabla g(x) - \lambda_{\mathcal{X}}(x) \mathbf{1}\}. \tag{102}$$

Componentwise,

$$W_{ii} = \left[\frac{1}{x_i - l_i} + \frac{\mathbf{1}_{\{u_i < 1\}}}{(u_i - x_i)^2}\right]^{-1}. \tag{103}$$

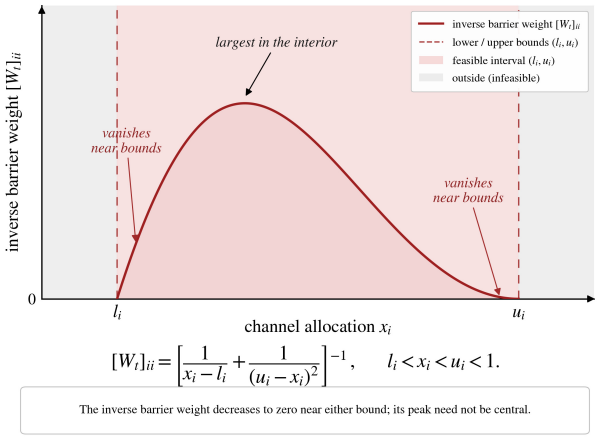


Figure 2: Inverse entropy–barrier weight for a nonredundant upper cap $u_i < 1$. The inverse metric weight vanishes at both bounds, but its profile need not be symmetric because the lower and upper barrier terms differ.

The asymmetry follows directly from the two barrier terms: $W_{ii}(x) \sim x_i - l_i$ as $x_i \downarrow l_i$, whereas $W_{ii}(x) \sim (u_i - x_i)^2$ as $x_i \uparrow u_i < 1$. The boundary limits are

$$\lim_{x_i \downarrow l_i} W_{ii} = 0, \qquad \lim_{x_i \uparrow u_i} W_{ii} = 0 \quad \text{when } u_i < 1. \tag{104}$$

When $u_i = 1$, the upper constraint is redundant under the simplex equality and is intentionally omitted from the potential. This is the key property that makes the general metric reduce exactly to the Shahshahani metric on the pure simplex.

A.10 Feasibility of the Discrete Step

At iteration t, RiskBid uses

$$d_t = -W_t(\widehat{g}_t - \widehat{\lambda}_t \mathbf{1}), \qquad \widehat{\lambda}_t = \frac{\mathbf{1}^\top W_t \widehat{g}_t}{\mathbf{1}^\top W_t \mathbf{1}}, \tag{105}$$

where $W_t = H_{\mathcal{X}}(x_t)^{-1}$. Therefore

$$\mathbf{1}^\top d_t = -\mathbf{1}^\top W_t \widehat{g}_t + \widehat{\lambda}_t \mathbf{1}^\top W_t \mathbf{1} = 0. \tag{106}$$

Hence the update $x_{t+1} = x_t + \gamma_t d_t$ preserves $\mathbf{1}^\top x_{t+1} = 1$.

To match the algorithm in the main text, define the safe coordinate bounds

$$\underline{l}_i = l_i + \delta + h, \qquad \overline{u}_i = u_i - \delta - h. \tag{107}$$

For $x_t \in \mathcal{X}_{\delta,h}$, let

$$\tau_t = \min\left\{ \min_{i: d_{t,i} > 0} \frac{\overline{u}_i - x_{t,i}}{d_{t,i}}, \quad \min_{i: d_{t,i} < 0} \frac{x_{t,i} - \underline{l}_i}{-d_{t,i}} \right\}, \tag{108}$$

with the minimum over an empty set interpreted as $+\infty$, and choose

$$\gamma_t = \min\{\eta_t, \beta\tau_t\}, \qquad \beta \in (0,1). \tag{109}$$

If $d_{t,i} > 0$, then

$$x_{t+1,i} \le x_{t,i} + \beta(\overline{u}_i - x_{t,i}) \le \overline{u}_i. \tag{110}$$

If $d_{t,i} < 0$, then

$$x_{t+1,i} \ge x_{t,i} - \beta(x_{t,i} - \underline{l}_i) \ge \underline{l}_i. \tag{111}$$

The opposite bound is automatic in each case, and coordinates with $d_{t,i} = 0$ are unchanged. Thus

$$x_{t+1} \in \mathcal{X}_{\delta,h} \qquad \text{whenever} \qquad x_t \in \mathcal{X}_{\delta,h}. \tag{112}$$

This proves the exact feasibility statement used by Algorithm 1.

A.11 Projection Identities Used in the Descent Proof

Fix $x \in \mathrm{ri}(\mathcal{X})$ and write $H = H_{\mathcal{X}}(x)$ and $W = H^{-1}$. For any $y \in \mathbb{R}^d$, recall the W-weighted centering operator

$$\Pi_W y = y - \lambda_W(y)\mathbf{1}, \tag{113}$$

$$\lambda_W(y) = \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}}. \tag{114}$$

Since W is positive definite, $\mathbf{1}^\top W \mathbf{1} > 0$, and therefore Π_W is well defined.

Let $p = \Pi_W \nabla f_\delta(x), \qquad e = \widehat{g} - \nabla f_\delta(x),$ $\widetilde{e} = \Pi_W e$. The operator Π_W is linear. Indeed, for any $y_1, y_2 \in \mathbb{R}^d$ and $a, b \in \mathbb{R}$,

$$\begin{aligned}
\Pi_W(ay_1 + by_2) &= ay_1 + by_2 - \frac{\mathbf{1}^\top W(ay_1 + by_2)}{\mathbf{1}^\top W \mathbf{1}}\mathbf{1} \\
&= ay_1 + by_2 - \left\{ a\frac{\mathbf{1}^\top W y_1}{\mathbf{1}^\top W \mathbf{1}} \right. \\
&\qquad \left. + b\frac{\mathbf{1}^\top W y_2}{\mathbf{1}^\top W \mathbf{1}} \right\}\mathbf{1} \\
&= a\left(y_1 - \frac{\mathbf{1}^\top W y_1}{\mathbf{1}^\top W \mathbf{1}}\mathbf{1}\right) \\
&\quad + b\left(y_2 - \frac{\mathbf{1}^\top W y_2}{\mathbf{1}^\top W \mathbf{1}}\mathbf{1}\right) \\
&= a\Pi_W y_1 + b\Pi_W y_2.
\end{aligned} \tag{115}$$

Therefore

$$\begin{aligned}
\Pi_W \widehat{g} &= \Pi_W\{\nabla f_\delta(x) + e\} \\
&= \Pi_W \nabla f_\delta(x) + \Pi_W e \\
&= p + \widetilde{e}.
\end{aligned} \tag{116}$$

The true stationarity norm is

$$\mathcal{G}_\delta(x)^2 = p^\top W p. \tag{117}$$

The centering construction gives the following orthogonality identities. First,

$$\begin{aligned}
\mathbf{1}^\top W p &= \mathbf{1}^\top W\{\nabla f_\delta(x) - \lambda_W(\nabla f_\delta(x))\mathbf{1}\} \\
&= \mathbf{1}^\top W \nabla f_\delta(x) - \lambda_W(\nabla f_\delta(x))\mathbf{1}^\top W \mathbf{1} \\
&= \mathbf{1}^\top W \nabla f_\delta(x) - \frac{\mathbf{1}^\top W \nabla f_\delta(x)}{\mathbf{1}^\top W \mathbf{1}}\mathbf{1}^\top W \mathbf{1} \\
&= 0.
\end{aligned} \tag{118}$$

Second, applying the same calculation to e gives

$$\begin{aligned}\mathbf{1}^\top W\widetilde{e} &= \mathbf{1}^\top W\Pi_W e\\ &= \mathbf{1}^\top W\left\{e - \frac{\mathbf{1}^\top We}{\mathbf{1}^\top W\mathbf{1}}\mathbf{1}\right\}\\ &= \mathbf{1}^\top We - \frac{\mathbf{1}^\top We}{\mathbf{1}^\top W\mathbf{1}}\mathbf{1}^\top W\mathbf{1}\\ &= 0. \end{aligned} \tag{119}$$

Since

$$p = \nabla f_\delta(x) - \lambda_W(\nabla f_\delta(x))\mathbf{1}, \tag{120}$$

we also have

$$\nabla f_\delta(x) = p + \lambda_W(\nabla f_\delta(x))\mathbf{1}. \tag{121}$$

Hence

$$\begin{aligned}\nabla f_\delta(x)^\top Wp &= \{p + \lambda_W(\nabla f_\delta(x))\mathbf{1}\}^\top Wp\\ &= p^\top Wp + \lambda_W(\nabla f_\delta(x))\mathbf{1}^\top Wp\\ &= p^\top Wp + \lambda_W(\nabla f_\delta(x))\cdot 0\\ &= p^\top Wp. \end{aligned} \tag{122}$$

Similarly,

$$\begin{aligned}\nabla f_\delta(x)^\top W\widetilde{e} &= \{p + \lambda_W(\nabla f_\delta(x))\mathbf{1}\}^\top W\widetilde{e}\\ &= p^\top W\widetilde{e} + \lambda_W(\nabla f_\delta(x))\mathbf{1}^\top W\widetilde{e}\\ &= p^\top W\widetilde{e} + \lambda_W(\nabla f_\delta(x))\cdot 0\\ &= p^\top W\widetilde{e}. \end{aligned} \tag{123}$$

Finally, since W is positive definite, the bilinear form

$$\langle a, b\rangle_W = a^\top Wb \tag{124}$$

is an inner product. Therefore Cauchy's inequality and Young's inequality give

$$\begin{aligned}\left|p^\top W\widetilde{e}\right| &\le (p^\top Wp)^{1/2}(\widetilde{e}^\top W\widetilde{e})^{1/2}\\ &\le \frac{1}{2}p^\top Wp + \frac{1}{2}\widetilde{e}^\top W\widetilde{e}. \end{aligned} \tag{125}$$

A.12 Feasibility of the Pure-Simplex Update

We prove the feasibility statement used in the pure-simplex limit of RiskBid. Let $x_t \in \mathrm{ri}(\Delta)$, where

$$\Delta = \left\{x \in \mathbb{R}^d : x_i \ge 0,\ i \in [d],\ \mathbf{1}^\top x = 1\right\}. \tag{126}$$

Given the lifted finite-difference estimator $\widehat{g}_t$, define the pure-simplex direction

$$d^\Delta_{t,i} = -x_{t,i}\left(\widehat{g}_{t,i} - \sum_{s=1}^d x_{t,s}\widehat{g}_{t,s}\right), \qquad i \in [d]. \tag{127}$$

By Eq. (87), this is the negative Shahshahani Riemannian gradient with $\nabla g(x_t)$ replaced by $\widehat{g}_t$.

First, the direction is tangent to the simplex:

$$\begin{aligned}\sum_{i=1}^d d^\Delta_{t,i} &= -\sum_{i=1}^d x_{t,i}\left(\widehat{g}_{t,i} - \sum_{s=1}^d x_{t,s}\widehat{g}_{t,s}\right)\\ &= -\sum_{i=1}^d x_{t,i}\widehat{g}_{t,i} + \left(\sum_{i=1}^d x_{t,i}\right)\left(\sum_{s=1}^d x_{t,s}\widehat{g}_{t,s}\right)\\ &= 0. \end{aligned} \tag{128}$$

Therefore, for the update

$$x_{t+1} = x_t + \gamma^\Delta_t d^\Delta_t, \tag{129}$$

we have

$$\mathbf{1}^\top x_{t+1} = \mathbf{1}^\top x_t + \gamma^\Delta_t \mathbf{1}^\top d^\Delta_t = 1. \tag{130}$$

It remains to ensure nonnegativity. Define the maximal safe step by

$$\tau^\Delta_t = \min_{i: d^\Delta_{t,i} < 0} \frac{x_{t,i}}{-d^\Delta_{t,i}}, \tag{131}$$

where the minimum over an empty set is interpreted as $+\infty$. Choose

$$0 \le \gamma^\Delta_t \le \beta\tau^\Delta_t, \qquad \beta \in (0,1). \tag{132}$$

For any coordinate with $d^\Delta_{t,i} < 0$, we have

$$\gamma^\Delta_t \le \beta\frac{x_{t,i}}{-d^\Delta_{t,i}}, \tag{133}$$

and hence

$$\begin{aligned}x_{t+1,i} &= x_{t,i} + \gamma^\Delta_t d^\Delta_{t,i}\\ &\ge x_{t,i} - \beta x_{t,i}\\ &= (1-\beta)x_{t,i}\\ &> 0. \end{aligned} \tag{134}$$

For any coordinate with $d^\Delta_{t,i} \ge 0$,

$$x_{t+1,i} = x_{t,i} + \gamma^\Delta_t d^\Delta_{t,i} \ge x_{t,i} > 0. \tag{135}$$

Thus $x_{t+1,i} > 0$ for every $i \in [d]$. Together with Eq. (130), this proves

$$x_{t+1} \in \mathrm{ri}(\Delta). \tag{136}$$

Therefore the pure-simplex update of RiskBid is feasible under the safe step in Eq. (132).

A.13 Proof of Theorem 1

Proof. Fix an iteration t and write

$$x = x_t, \qquad \gamma = \gamma_t, \qquad W = W_t, \qquad \widehat{g} = \widehat{g}_t. \tag{137}$$

Let

$$\begin{aligned}p &= \Pi_W \nabla_{\mathrm{aff}} f_\delta(x),\\ e &= \widehat{g} - \nabla_{\mathrm{aff}} f_\delta(x),\\ \widetilde{e} &= \Pi_W e. \end{aligned} \tag{138}$$

Since Π_W is linear for fixed W,

$$\Pi_W \widehat{g} = p + \widetilde{e}. \tag{139}$$

Thus the update direction can be written as

$$d_t = -W\Pi_W\widehat{g} = -W(p + \widetilde{e}). \tag{140}$$

By Eq. (23), $d_t \in \mathsf{T}$.

The safe-step feasibility result in Appendix A.10 implies

$$x_{t+1} = x + \gamma d_t \in \mathcal{X}_{\delta,h}. \tag{141}$$

Hence the affine-hull smoothness condition in Assumption 2 applies along this step. Therefore,

$$f_\delta(x + \gamma d_t) \le f_\delta(x) + \gamma\nabla_{\mathrm{aff}} f_\delta(x)^\top d_t + \frac{\Lambda_\delta\gamma^2}{2}\|d_t\|_2^2. \tag{142}$$

We first bound the linear term. By the definition of Π_W, both p and $\widetilde{e}$ satisfy

$$\mathbf{1}^\top Wp = 0, \qquad \mathbf{1}^\top W\widetilde{e} = 0. \tag{143}$$

Therefore, we have

$$\begin{cases} \nabla_{\mathrm{aff}} f_\delta(x)^\top W p = p^\top W p, \\ \nabla_{\mathrm{aff}} f_\delta(x)^\top W \widetilde{e} = p^\top W \widetilde{e}. \end{cases} \tag{144}$$

Using Eq. (140), we obtain

$$\begin{aligned} \nabla_{\mathrm{aff}} f_\delta(x)^\top d_t &= -\nabla_{\mathrm{aff}} f_\delta(x)^\top W(p+\widetilde{e}) \\ &= -p^\top W p - p^\top W \widetilde{e} \\ &\le -p^\top W p + \left| p^\top W \widetilde{e} \right| \\ &\le -\frac{1}{2} p^\top W p + \frac{1}{2} \widetilde{e}^\top W \widetilde{e}. \end{aligned} \tag{145}$$

The last step uses Young's inequality under the W-inner product.

We next control the perturbation term. On the compact set $K \subset \mathcal{X}_{\delta,\bar{h}}$ in Assumption 1, Eq. (17) gives

$$\underline{w} I \preceq W \preceq \overline{w} I. \tag{146}$$

For any $u \in \mathbb{R}^d$,

$$\begin{aligned} \|\Pi_W u\|_2 &= \left\| u - \frac{\mathbf{1}^\top W u}{\mathbf{1}^\top W \mathbf{1}} \mathbf{1} \right\|_2 \\ &\le \|u\|_2 + \frac{\|W\mathbf{1}\|_2 \|u\|_2}{\mathbf{1}^\top W \mathbf{1}} \|\mathbf{1}\|_2 \\ &\le \left(1 + \frac{\overline{w}}{\underline{w}}\right) \|u\|_2. \end{aligned} \tag{147}$$

Assumption 3 gives $\|e\|_2 \le \zeta_h$. Hence

$$\begin{aligned} \widetilde{e}^\top W \widetilde{e} &\le \overline{w} \|\widetilde{e}\|_2^2 \\ &= \overline{w} \|\Pi_W e\|_2^2 \\ &\le \overline{w} \left(1 + \frac{\overline{w}}{\underline{w}}\right)^2 \|e\|_2^2 \\ &\le C_\Pi \zeta_h^2, \end{aligned} \tag{148}$$

where

$$C_\Pi = \overline{w} \left(1 + \frac{\overline{w}}{\underline{w}}\right)^2. \tag{149}$$

We also need a Euclidean bound on the update direction. From Eq. (140),

$$\begin{aligned} \|d_t\|_2^2 &= \|W(p+\widetilde{e})\|_2^2 \\ &\le 2\|Wp\|_2^2 + 2\|W\widetilde{e}\|_2^2 \\ &\le 2\overline{w}\, p^\top W p + 2\overline{w}\, \widetilde{e}^\top W \widetilde{e} \\ &\le 2\overline{w}\, p^\top W p + 2\overline{w} C_\Pi \zeta_h^2 \\ &\le C_d \left(p^\top W p + \zeta_h^2 \right), \end{aligned} \tag{150}$$

with

$$C_d = 2\overline{w} \max\{1, C_\Pi\}. \tag{151}$$

Substituting Eqs. (145), (148), and (150) into Eq. (142) yields

$$\begin{aligned} f_\delta(x_{t+1}) &\le f_\delta(x_t) - \frac{\gamma_t}{2} p^\top W p \\ &\quad + \frac{\gamma_t}{2} \widetilde{e}^\top W \widetilde{e} + \frac{\Lambda_\delta \gamma_t^2}{2} \|d_t\|_2^2 \\ &\le f_\delta(x_t) - \frac{\gamma_t}{2} p^\top W p + \frac{\gamma_t C_\Pi}{2} \zeta_h^2 \\ &\quad + \frac{\Lambda_\delta \gamma_t^2 C_d}{2} \left(p^\top W p + \zeta_h^2 \right) \\ &= f_\delta(x_t) - \left(\frac{1}{2} - \frac{\Lambda_\delta \gamma_t C_d}{2} \right) \gamma_t p^\top W p \\ &\quad + \left(\frac{C_\Pi}{2} + \frac{\Lambda_\delta \gamma_t C_d}{2} \right) \gamma_t \zeta_h^2. \end{aligned} \tag{152}$$

Choose

$$c_1 \le \frac{1}{2 C_d}. \tag{153}$$

Under the theorem condition $\gamma_t \le c_1 / \Lambda_\delta$,

$$\frac{1}{2} - \frac{\Lambda_\delta \gamma_t C_d}{2} \ge \frac{1}{4}, \qquad \frac{C_\Pi}{2} + \frac{\Lambda_\delta \gamma_t C_d}{2} \le \frac{C_\Pi}{2} + \frac{1}{4}. \tag{154}$$

Finally, by Eq. (26),

$$p^\top W p = \mathcal{G}_\delta(x_t)^2. \tag{155}$$

Therefore,

$$\begin{aligned} f_\delta(x_{t+1}) &\le f_\delta(x_t) - \frac{1}{4} \gamma_t \mathcal{G}_\delta(x_t)^2 \\ &\quad + \left(\frac{C_\Pi}{2} + \frac{1}{4} \right) \gamma_t \zeta_h^2. \end{aligned} \tag{156}$$

Thus Theorem 1 holds with

$$c_2 = \frac{1}{4}, \qquad c_3 = \frac{C_\Pi}{2} + \frac{1}{4}. \tag{157}$$

□

Asymptotic radius convention. The finite-horizon pseudocode uses one fixed radius h. The convergence statement in the main text uses a standard diminishing-radius variant. In that variant, the estimator uses $h_t \downarrow 0$, while feasibility is maintained with one fixed buffer $\bar{h} \ge \sup_t h_t$ and the common tube $\mathcal{X}_{\delta,\bar{h}}$. This convention is necessary: if the safe tube itself changed with h_t, compactness and the metric bounds would not follow from one fixed set. No experimental value is altered by this analytical convention.

A.14 Proof of Theorem 2

Proof. Let

$$\zeta_t = \zeta_{h_t}. \tag{158}$$

Under the assumptions of the theorem, the iterates remain in the compact set $K \subset \mathcal{X}_{\delta,\bar{h}}$. Hence $K \subset \mathrm{ri}(\mathcal{X})$, the metric bounds in Eq. (17) hold uniformly on K, and $\mathcal{G}_\delta$ is continuous on K. By Theorem 1, for every t,

$$c_2 \gamma_t \mathcal{G}_\delta(x_t)^2 \le f_\delta(x_t) - f_\delta(x_{t+1}) + c_3 \gamma_t \zeta_t^2. \tag{159}$$

Summing Eq. (159) from $t = 0$ to $T - 1$ gives

$$c_2 \sum_{t=0}^{T-1} \gamma_t \mathcal{G}_\delta(x_t)^2 \le \sum_{t=0}^{T-1} \left[f_\delta(x_t) - f_\delta(x_{t+1}) \right] + c_3 \sum_{t=0}^{T-1} \gamma_t \zeta_t^2 = f_\delta(x_0) - f_\delta(x_T) + c_3 \sum_{t=0}^{T-1} \gamma_t \zeta_t^2. \tag{160}$$

Since K is compact and f_δ is continuous, define

$$f_\star = \inf_{x \in K} f_\delta(x) > -\infty. \tag{161}$$

Using $x_T \in K$ and $\sum_{t=0}^{\infty} \gamma_t \zeta_t^2 < \infty$, we obtain

$$\sum_{t=0}^{\infty} \gamma_t \mathcal{G}_\delta(x_t)^2 < \infty. \tag{162}$$

Since $\sum_{t=0}^{\infty} \gamma_t = \infty$, Eq. (162) implies

$$\liminf_{t \to \infty} \mathcal{G}_\delta(x_t) = 0. \tag{163}$$

Indeed, if there were $\varepsilon > 0$ and T_0 such that $\mathcal{G}_\delta(x_t) \ge \varepsilon$ for all $t \ge T_0$, then

$$\sum_{t=T_0}^{\infty} \gamma_t \mathcal{G}_\delta(x_t)^2 \ge \varepsilon^2 \sum_{t=T_0}^{\infty} \gamma_t = \infty, \tag{164}$$

contradicting Eq. (162).

We next prove the full convergence $\mathcal{G}_\delta(x_t) \to 0$. From Eq. (150), there exists a constant $C_x > 0$ such that

$$\|x_{t+1} - x_t\|_2 \le C_x \gamma_t \left(\mathcal{G}_\delta(x_t) + \zeta_t \right). \tag{165}$$

Because K is compact and $\mathcal{G}_\delta$ is continuous on K, $\mathcal{G}_\delta$ is uniformly continuous on K. Suppose, for contradiction, that $\mathcal{G}_\delta(x_t) \not\to 0$. Then there exist $\varepsilon > 0$ and infinitely many indices t_j such that

$$\mathcal{G}_\delta(x_{t_j}) \ge \varepsilon. \tag{166}$$

By uniform continuity, there exists $\rho_\varepsilon > 0$ such that, for any $x, y \in K$,

$$\|x - y\|_2 \le \rho_\varepsilon \quad \Longrightarrow \quad |\mathcal{G}_\delta(x) - \mathcal{G}_\delta(y)| \le \frac{\varepsilon}{2}. \tag{167}$$

Fix a small constant $\alpha_\varepsilon > 0$. Since $\sum_t \gamma_t = \infty$, for each sufficiently large t_j we can choose $m_j > t_j$ as the first index satisfying

$$\sum_{t=t_j}^{m_j - 1} \gamma_t \ge \alpha_\varepsilon. \tag{168}$$

Because $\sum_{t=0}^{\infty} \gamma_t^2 < \infty$, necessarily $\gamma_t \to 0$. Therefore, for all sufficiently large j,

$$\sum_{t=t_j}^{m_j - 1} \gamma_t \le 2\alpha_\varepsilon. \tag{169}$$

For any $k \in \{t_j, \dots, m_j\}$, Eq. (165) gives

$$\begin{aligned} \|x_k - x_{t_j}\|_2 &\le C_x \sum_{t=t_j}^{k-1} \gamma_t \left(\mathcal{G}_\delta(x_t) + \zeta_t \right) \\ &\le C_x \left[\left(\sum_{t=t_j}^{k-1} \gamma_t \right)^{1/2} \left(\sum_{t=t_j}^{k-1} \gamma_t \mathcal{G}_\delta(x_t)^2 \right)^{1/2} + \left(\sum_{t=t_j}^{k-1} \gamma_t \right)^{1/2} \left(\sum_{t=t_j}^{k-1} \gamma_t \zeta_t^2 \right)^{1/2} \right]. \end{aligned} \tag{170}$$

The two tail sums

$$\sum_{t=t_j}^{\infty} \gamma_t \mathcal{G}_\delta(x_t)^2, \qquad \sum_{t=t_j}^{\infty} \gamma_t \zeta_t^2 \tag{171}$$

converge to zero as $j \to \infty$. Therefore, using Eq. (169), we have

$$\max_{t_j \le k \le m_j} \|x_k - x_{t_j}\|_2 \le \rho_\varepsilon \tag{172}$$

for all sufficiently large j. Combining Eqs. (166), (167), and (172), we get

$$\mathcal{G}_\delta(x_k) \ge \frac{\varepsilon}{2}, \qquad k = t_j, \dots, m_j. \tag{173}$$

Hence

$$\sum_{t=t_j}^{m_j - 1} \gamma_t \mathcal{G}_\delta(x_t)^2 \ge \sum_{t=t_j}^{m_j - 1} \gamma_t \left(\frac{\varepsilon}{2} \right)^2 \ge \frac{\varepsilon^2}{4} \alpha_\varepsilon. \tag{174}$$

Choosing the indices t_j recursively so that the blocks are disjoint gives infinitely many disjoint blocks, each contributing at least $\varepsilon^2 \alpha_\varepsilon / 4$ to the finite sum in Eq. (162). This contradiction proves

$$\mathcal{G}_\delta(x_t) \to 0. \tag{175}$$

Let $\bar{x}$ be any accumulation point of $\{x_t\}$. Since K is compact, there exists a subsequence $x_{t_j} \to \bar{x}$, and $\bar{x} \in K \subset \mathrm{ri}(\mathcal{X})$. By continuity of $\mathcal{G}_\delta$ and Eq. (175),

$$\mathcal{G}_\delta(\bar{x}) = \lim_{j \to \infty} \mathcal{G}_\delta(x_{t_j}) = 0. \tag{176}$$

Let

$$W(\bar{x}) = H_{\mathcal{X}}(\bar{x})^{-1}. \tag{177}$$

By Eq. (26),

$$\mathcal{G}_\delta(\bar{x})^2 = p(\bar{x})^\top W(\bar{x}) p(\bar{x}), \tag{178}$$

where

$$p(\bar{x}) = \nabla_{\mathrm{aff}} f_\delta(\bar{x}) - \lambda_\delta(\bar{x}) \mathbf{1}. \tag{179}$$

Since $\bar{x} \in \mathrm{ri}(\mathcal{X})$, $W(\bar{x})$ is positive definite. Thus Eq. (176) implies

$$p(\bar{x}) = 0, \qquad \text{or equivalently,} \qquad \nabla_{\mathrm{aff}} f_\delta(\bar{x}) - \lambda_\delta(\bar{x}) \mathbf{1} = 0. \tag{180}$$

Because $\bar{x} \in \mathrm{ri}(\mathcal{X})$, no box constraint is active. Hence the upper and lower box multipliers can be chosen as

$$\nu^+ = 0, \qquad \nu^- = 0. \tag{181}$$

Together with

$$\mathbf{1}^\top \bar{x} = 1, \qquad l_i < \bar{x}_i < u_i, \quad i \in [d], \tag{182}$$

Eq. (180) gives the KKT condition for the smoothed constrained problem:

$$0 \in \nabla_{\mathrm{aff}} f_\delta(\bar{x}) + \mathcal{N}_{\mathcal{X}}(\bar{x}). \tag{183}$$

Therefore, under Assumption 1, every accumulation point is an interior KKT point of the fixed-δ smoothed problem. This is a conditional statement: it assumes that the entire trajectory remains in a common compact subset of $\mathrm{ri}(\mathcal{X})$ and does not establish that such a subset contains a full-domain optimizer. Boundary KKT points and trajectories approaching active box constraints are not claimed by this theorem. □

A.15 Inner Approximation of Boundary Optima

Let $x^\circ \in \mathrm{ri}(\mathcal{X})$ and define

$$m_\circ = \min_i\{x_i^\circ - l_i,\ u_i - x_i^\circ\} > 0, \qquad D_{\mathcal{X}} = \sup_{x,y\in\mathcal{X}} \|x-y\|_2. \quad (184)$$

For $0 < s < m_\circ$, let $\mathcal{X}_s = \{x \in \mathcal{X} : l_i + s \le x_i \le u_i - s\}$.

Proposition 2 (Boundary-value approximation). *If g is L_g-Lipschitz on $\mathcal{X}$, then*

$$0 \le \inf_{x\in\mathcal{X}_s} g(x) - \inf_{x\in\mathcal{X}} g(x) \le \frac{L_g D_{\mathcal{X}}}{m_\circ} s. \quad (185)$$

Proof. For any $x \in \mathcal{X}$, set $\theta = s/m_\circ$ and $x_s = (1-\theta)x + \theta x^\circ$. Convexity preserves the budget equality and box constraints, while $x_{s,i} - l_i \ge \theta m_\circ = s$ and $u_i - x_{s,i} \ge \theta m_\circ = s$. Hence $x_s \in \mathcal{X}_s$ and $\|x_s - x\|_2 \le (s/m_\circ)D_{\mathcal{X}}$. Applying Lipschitz continuity to a minimizer, or to an infimizing sequence, proves Eq. (185). □

This proposition is only an objective-value statement. It explains how a sequence of shrinking inner domains can approximate the optimal value of a boundary solution. For a fixed positive tube width, the displayed upper bound may be nonzero; the proposition does not show recovery of a boundary optimizer, convergence of box multipliers, or stationarity of a point excluded by the fixed tube. In particular, it does not convert the interior Riemannian stationarity theorem into a boundary-KKT theorem.

A.16 Proof of Theorem 3

The safe-step rule can shorten a nominal step. We therefore first prove a bound in terms of the *realized cumulative step length*. The usual $O(\epsilon^{-2})$ iteration statement then follows under either the nonbinding-safeguard condition or, more generally, a uniform positive lower bound on the realized steps. This ordering specifies each condition used by the proof.

Proof. Let

$$f_{\delta,K}^\star = \inf_{x\in K} f_\delta(x), \qquad \Delta_{\delta,K} = f_\delta(x_0) - f_{\delta,K}^\star. \quad (186)$$

Compactness of K and continuity of f_δ imply $0 \le \Delta_{\delta,K} < \infty$. In the main theorem the corresponding gap is

$$\Delta_{F,K} = F(x_0) - \inf_{x\in K} F(x). \quad (187)$$

For $F = f_\delta$, this is precisely $\Delta_{\delta,K}$. Restricting the infimum to the compact query region avoids evaluating f_δ outside its stated smoothing domain; no extension to all of $\mathcal{X}$ is assumed.

For a horizon $T \ge 1$, define

$$\Gamma_T = \sum_{t=0}^{T-1} \gamma_t. \quad (188)$$

Assume $\Gamma_T > 0$, $\gamma_t \le c_1/\Lambda_\delta$, and use a fixed finite-difference radius h over this finite horizon. The descent theorem gives

$$f_\delta(x_{t+1}) \le f_\delta(x_t) - c_2\gamma_t \mathcal{G}_\delta(x_t)^2 + c_3\gamma_t\zeta_h^2. \quad (189)$$

Rearranging,

$$c_2\gamma_t \mathcal{G}_\delta(x_t)^2 \le f_\delta(x_t) - f_\delta(x_{t+1}) + c_3\gamma_t\zeta_h^2. \quad (190)$$

Summing from $t = 0$ to $T-1$ and telescoping the objective terms yields

$$c_2 \sum_{t=0}^{T-1} \gamma_t \mathcal{G}_\delta(x_t)^2 \le f_\delta(x_0) - f_\delta(x_T) + c_3\zeta_h^2 \sum_{t=0}^{T-1} \gamma_t \le \Delta_{\delta,K} + c_3\zeta_h^2\Gamma_T. \quad (191)$$

Let τ_T minimize $\mathcal{G}_\delta(x_t)$ over $t = 0,\dots,T-1$. Since every weight γ_t is nonnegative,

$$\mathcal{G}_\delta(x_{\tau_T})^2\Gamma_T \le \sum_{t=0}^{T-1} \gamma_t \mathcal{G}_\delta(x_t)^2. \quad (192)$$

Combining Eqs. (191) and (192) gives the safeguard-robust certificate

$$\mathcal{G}_\delta(x_{\tau_T})^2 \le \frac{\Delta_{\delta,K}}{c_2\Gamma_T} + \frac{c_3}{c_2}\zeta_h^2. \quad (193)$$

This inequality remains true even when the safeguard binds; the only quantity that changes is Γ_T.

Choose h so that

$$\zeta_h \le \frac{\epsilon}{2}\sqrt{\frac{c_2}{c_3}}. \quad (194)$$

Then

$$\frac{c_3}{c_2}\zeta_h^2 \le \frac{\epsilon^2}{4}. \quad (195)$$

If, in addition,

$$\Gamma_T \ge \frac{4\Delta_{\delta,K}}{c_2\epsilon^2}, \quad (196)$$

then

$$\frac{\Delta_{\delta,K}}{c_2\Gamma_T} \le \frac{\epsilon^2}{4}. \quad (197)$$

Substitution into Eq. (193) gives

$$\mathcal{G}_\delta(x_{\tau_T})^2 \le \frac{\epsilon^2}{2} \le \epsilon^2, \qquad \mathcal{G}_\delta(x_{\tau_T}) \le \epsilon. \quad (198)$$

We now recover an iteration bound. Suppose the realized steps satisfy

$$\gamma_t \ge \underline{\gamma} > 0, \qquad t = 0,\dots,T-1. \quad (199)$$

Then $\Gamma_T \ge T\underline{\gamma}$, so Eq. (196) follows whenever

$$T \ge \frac{4\Delta_{\delta,K}}{c_2\underline{\gamma}\epsilon^2}. \quad (200)$$

In particular, if the nominal step is constant, $\eta_t \equiv \gamma$, and the finite-horizon safeguard is nonbinding,

$$\beta\tau_t \ge \gamma, \qquad t = 0,\dots,T-1, \quad (201)$$

then the realized step is $\gamma_t = \gamma$. Taking

$$\gamma = \theta/\Lambda_\delta, \qquad 0 < \theta \le c_1, \quad (202)$$

gives

$$T = O\left(\frac{\Lambda_\delta\Delta_{\delta,K}}{\epsilon^2}\right). \quad (203)$$

Thus the constant-step rate in the main text is a corollary of the safeguard-robust cumulative-step inequality Eq. (193).

One iteration reuses one baseline value and evaluates one forward point in each of the $r = d - 1$ reduced directions. Therefore

$$1 + r = d \tag{204}$$

scalar target-oracle evaluations are required per iteration. Under Eq. (203), the total number of exact smoothed oracle evaluations is

$$N_{\text{smooth}} = dT = O\left(\frac{d\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right). \tag{205}$$

Using the affine-hull smoothing bound $\Lambda_\delta \le L\sqrt{r}/\delta \le L\sqrt{d}/\delta$ gives

$$N_{\text{smooth}} = O\left(\frac{d^{3/2} L \Delta_{\delta,K}}{\delta \epsilon^2}\right). \tag{206}$$

A valid optional simplification follows from a diameter bound, not from an inverse-δ assumption on the objective gap. Ball smoothing preserves the Lipschitz constant: for $x, y \in \mathcal{X}_\delta$,

$$\begin{aligned} |f_\delta(x) - f_\delta(y)| &= |\mathbb{E}_v[f(x + \delta Q v) - f(y + \delta Q v)]| \\ &\le L\|x - y\|_2. \end{aligned} \tag{207}$$

Let $D_K = \sup_{x,y \in K} \|x - y\|_2$. Then

$$\Delta_{\delta,K} \le L D_K. \tag{208}$$

Indeed, for any minimizer $x_K^\star$ of f_δ on compact K,

$$\Delta_{\delta,K} = f_\delta(x_0) - f_\delta(x_K^\star) \le L\|x_0 - x_K^\star\|_2 \le L D_K. \tag{209}$$

Consequently,

$$N_{\text{smooth}} = O\left(\frac{d^{3/2} L^2 D_K}{\delta \epsilon^2}\right). \tag{210}$$

No general $\Delta_{\delta,K} = O(L/\delta)$ claim is needed or used.

If one exact smoothed-oracle value is approximated by an average of $M \ge 1$ primitive objective evaluations, the deterministic accounting is

$$N_{\text{sim}} = M N_{\text{smooth}}. \tag{211}$$

The stochastic accuracy required of M is analyzed in Appendix A.18. In direct empirical-oracle mode, there is no Monte Carlo smoothing layer. Proposition 1 replaces the single-cell gap by the regular-step mass and the interface debit. In particular, suppose that $\mathfrak{D}_T \le \mathfrak{D}_\infty$, the regular realized-step mass satisfies $\Gamma_T^{\text{reg}} \ge \pi T\theta/\overline{\Lambda}_n$ for constants $\pi, \theta > 0$, and h is chosen so that the finite-difference term is at most $\epsilon^2/4$. Then the same best-iterate calculation gives

$$N_{\text{direct}} = O\left(\frac{d\,\overline{\Lambda}_n(\Delta_{n,K} + \mathfrak{D}_\infty)}{\pi \epsilon^2}\right). \tag{212}$$

Under eventual active-cell identification, for every fixed $\pi < 1$ the regular-mass condition holds for all sufficiently large horizons, and $\mathfrak{D}_\infty$ is the finite debit of the initial prefix. In this mode, $M_{\text{smooth}} = 0$ denotes direct evaluation and contributes no Monte Carlo multiplicative factor to the oracle count. □

Observable output certificate. The index τ_T is used to state a best-iterate existence guarantee and does not require an extra black-box query. The algorithm may return

$$\widehat{\tau} \in \arg\min_{0 \le t < T} \|d_t\|_{H_t}. \tag{213}$$

Let $e_t = \widehat{g}_t - \nabla_{\text{aff}} f_\delta(x_t)$. Since

$$d_t = -W_t \Pi_{W_t} \widehat{g}_t, \qquad \operatorname{grad}_{\mathcal{X}} f_\delta(x_t) = W_t \Pi_{W_t} \nabla_{\text{aff}} f_\delta(x_t), \tag{214}$$

the reverse triangle inequality in the H_t norm gives

$$\begin{aligned} \left|\|d_t\|_{H_t} - \mathcal{G}_\delta(x_t)\right| &\le \|W_t \Pi_{W_t} e_t\|_{H_t} \\ &= \left((\Pi_{W_t} e_t)^\top W_t (\Pi_{W_t} e_t)\right)^{1/2} \\ &\le \sqrt{C_\Pi}\, \zeta_h. \end{aligned} \tag{215}$$

Therefore

$$\begin{aligned} \mathcal{G}_\delta(x_{\widehat{\tau}}) &\le \|d_{\widehat{\tau}}\|_{H_{\widehat{\tau}}} + \sqrt{C_\Pi}\zeta_h \\ &\le \min_{0 \le t < T} \|d_t\|_{H_t} + \sqrt{C_\Pi}\zeta_h \\ &\le \min_{0 \le t < T} \mathcal{G}_\delta(x_t) + 2\sqrt{C_\Pi}\zeta_h. \end{aligned} \tag{216}$$

Thus the computable return rule has the same asymptotic complexity. For a computable ϵ certificate, it suffices to strengthen the bias condition by a constant factor, for example

$$\zeta_h \le \min\left\{\frac{\epsilon}{4}\sqrt{\frac{c_2}{c_3}}, \frac{\epsilon}{4\sqrt{C_\Pi}}\right\}, \tag{217}$$

and to enlarge the constant in the lower bound on Γ_T or T. The orders in Eqs. (205) and (206) are unchanged.

A.17 Unsmoothing and Clarke–KKT Consistency

The preceding convergence theorem concerns one fixed smoothing radius $\delta > 0$. A stationary point from a single fixed-δ run is therefore a KKT point of the smoothed problem only. Consistency with the original nonsmooth objective requires the vanishing-smoothing sequence stated below.

We first state the precise consistency result proved in this subsection. This also provides the label used by cross-references in the appendix.

For $x = x_{\text{ref}} + Qz$, define the affine-hull Clarke subdifferential of f by

$$\partial_C^{\text{aff}} f(x) = Q\, \partial_C \phi(z), \qquad \phi(z) = f(x_{\text{ref}} + Qz). \tag{218}$$

Thus $\partial_C^{\text{aff}} f(x)$ is the Clarke subdifferential of the restriction of f to the affine hull, lifted back to the original allocation space.

Corollary 1 (Unsmoothing and Clarke–KKT consistency)**.** *Let $\delta_j \downarrow 0$, and let $x_{\delta_j} \in \mathcal{X}_{\delta_j}$ satisfy the following normal-cone residual condition*

$$\operatorname{dist}\left(0, \nabla_{\text{aff}} f_{\delta_j}(x_{\delta_j}) + \mathcal{N}_{\mathcal{X}}(x_{\delta_j})\right) \le \epsilon_{\delta_j}, \qquad \epsilon_{\delta_j} \to 0. \tag{219}$$

Assume that $x_{\delta_j} \to \bar{x}$ and that f is locally Lipschitz on an open affine-hull neighborhood containing every sufficiently small smoothing ball around x_{δ_j}. Then $\bar{x}$ satisfies the affine-hull Clarke–KKT condition

$$0 \in \partial_C^{\text{aff}} f(\bar{x}) + \mathcal{N}_{\mathcal{X}}(\bar{x}). \tag{220}$$

Proof of Corollary 1. We use the affine-hull formulation from the main text. For each smoothing radius $\delta > 0$, write

$$x_\delta = x_{\text{ref}} + Qz_\delta, \qquad f_\delta(x_\delta) = \phi_\delta(z_\delta), \tag{221}$$

where

$$\phi_\delta(z) = \mathbb{E}_{v\sim\nu_r}\left[\phi(z+\delta v)\right]. \tag{222}$$

Since x_δ is an ϵ_δ-stationary point of the smoothed problem in the normal-cone sense, there exist $r_\delta \in \mathbb{R}^d$ and $n_\delta \in \mathcal{N}_{\mathcal{X}}(x_\delta)$ such that

$$\|r_\delta\|_2 \le \epsilon_\delta, \qquad \epsilon_\delta \to 0, \tag{223}$$

and

$$r_\delta = \nabla_{\text{aff}} f_\delta(x_\delta) + n_\delta. \tag{224}$$

We first relate the smoothed gradient to nearby Clarke subgradients. Since ϕ is locally Lipschitz, the gradient of the uniformly smoothed function belongs to the convex hull of nearby Clarke subgradients:

$$\nabla\phi_\delta(z_\delta) \in \operatorname{conv}\left(\bigcup_{\|v\|_2\le 1} \partial_{\mathrm{C}}\phi(z_\delta+\delta v)\right). \tag{225}$$

Multiplying by Q and using Eq. (218), we obtain

$$\begin{aligned}
\nabla_{\text{aff}} f_\delta(x_\delta) &= Q\nabla\phi_\delta(z_\delta) \\
&\in Q\operatorname{conv}\left(\bigcup_{\|v\|_2\le 1} \partial_{\mathrm{C}}\phi(z_\delta+\delta v)\right) \\
&= \operatorname{conv}\left(\bigcup_{\|v\|_2\le 1} Q\,\partial_{\mathrm{C}}\phi(z_\delta+\delta v)\right) \\
&= \operatorname{conv}\left(\bigcup_{\|v\|_2\le 1} \partial_{\mathrm{C}}^{\text{aff}} f(x_\delta+\delta Qv)\right).
\end{aligned} \tag{226}$$

Now take any sequence $\delta_j \downarrow 0$ such that

$$x_{\delta_j} \to \bar{x}. \tag{227}$$

Define $\xi_j = \nabla_{\text{aff}} f_{\delta_j}(x_{\delta_j})$. By Eq. (226),

$$\xi_j \in \operatorname{conv}\left(\bigcup_{\|v\|_2\le 1} \partial_{\mathrm{C}}^{\text{aff}} f(x_{\delta_j}+\delta_j Qv)\right). \tag{228}$$

Since f is locally Lipschitz near $\bar{x}$, the affine-hull Clarke subgradients are locally bounded. Hence $\{\xi_j\}$ is bounded. Passing to a subsequence if necessary, assume

$$\xi_j \to \bar{\xi}. \tag{229}$$

We claim that

$$\bar{\xi} \in \partial_{\mathrm{C}}^{\text{aff}} f(\bar{x}). \tag{230}$$

To prove the claim, apply Carathéodory's theorem to Eq. (228). For each j, there exist weights $\theta_{j,\ell} \ge 0$, directions $v_{j,\ell} \in \mathbb{R}^r$, and subgradients $\xi_{j,\ell}$, with $\ell = 0,\dots,d$, such that

$$\|v_{j,\ell}\|_2 \le 1, \qquad \xi_{j,\ell} \in \partial_{\mathrm{C}}^{\text{aff}} f(y_{j,\ell}), \tag{231}$$

where

$$y_{j,\ell} = x_{\delta_j} + \delta_j Q v_{j,\ell}, \tag{232}$$

and

$$\sum_{\ell=0}^{d} \theta_{j,\ell} = 1, \qquad \theta_{j,\ell} \ge 0, \tag{233}$$

$$\xi_j = \sum_{\ell=0}^{d} \theta_{j,\ell}\xi_{j,\ell}. \tag{234}$$

For every fixed ℓ, the points $y_{j,\ell}$ converge to $\bar{x}$ because

$$\begin{aligned}
\|y_{j,\ell} - \bar{x}\|_2 &= \|x_{\delta_j} + \delta_j Q v_{j,\ell} - \bar{x}\|_2 \\
&\le \|x_{\delta_j} - \bar{x}\|_2 + \delta_j \|Q v_{j,\ell}\|_2 \\
&\le \|x_{\delta_j} - \bar{x}\|_2 + \delta_j \|v_{j,\ell}\|_2 \\
&\le \|x_{\delta_j} - \bar{x}\|_2 + \delta_j \\
&\to 0.
\end{aligned} \tag{235}$$

By local boundedness of Clarke subgradients, after passing to a further subsequence we may assume, for every $\ell = 0,\dots,d$, that

$$\theta_{j,\ell} \to \bar{\theta}_\ell, \qquad \xi_{j,\ell} \to \bar{\xi}_\ell. \tag{236}$$

The affine-hull Clarke subdifferential has a closed graph for locally Lipschitz functions. Hence

$$y_{j,\ell} \to \bar{x}, \qquad \xi_{j,\ell} \to \bar{\xi}_\ell, \qquad \xi_{j,\ell} \in \partial_{\mathrm{C}}^{\text{aff}} f(y_{j,\ell}) \tag{237}$$

imply

$$\bar{\xi}_\ell \in \partial_{\mathrm{C}}^{\text{aff}} f(\bar{x}), \qquad \ell = 0,\dots,d. \tag{238}$$

Taking limits in Eqs. (233) and (234), we get

$$\sum_{\ell=0}^{d} \bar{\theta}_\ell = 1, \qquad \bar{\theta}_\ell \ge 0, \tag{239}$$

$$\bar{\xi} = \sum_{\ell=0}^{d} \bar{\theta}_\ell \bar{\xi}_\ell. \tag{240}$$

Since $\partial_{\mathrm{C}}^{\text{aff}} f(\bar{x})$ is convex, Eqs. (238)–(240) prove

$$\bar{\xi} \in \partial_{\mathrm{C}}^{\text{aff}} f(\bar{x}). \tag{241}$$

It remains to pass the normal-cone term to the limit. From Eq. (224),

$$n_{\delta_j} = r_{\delta_j} - \xi_j. \tag{242}$$

Since

$$\|r_{\delta_j}\|_2 \le \epsilon_{\delta_j}, \qquad \epsilon_{\delta_j} \to 0, \qquad \xi_j \to \bar{\xi}, \tag{243}$$

we have

$$n_{\delta_j} = r_{\delta_j} - \xi_j \to 0 - \bar{\xi} = -\bar{\xi}. \tag{244}$$

Moreover,

$$n_{\delta_j} \in \mathcal{N}_{\mathcal{X}}(x_{\delta_j}), \qquad x_{\delta_j} \to \bar{x}. \tag{245}$$

Because $\mathcal{X}$ is closed and convex, the graph of $\mathcal{N}_{\mathcal{X}}$ is closed. Therefore

$$-\bar{\xi} \in \mathcal{N}_{\mathcal{X}}(\bar{x}). \tag{246}$$

Combining Eqs. (241) and (246), we obtain

$$0 \in \partial_{\mathrm{C}}^{\text{aff}} f(\bar{x}) + \mathcal{N}_{\mathcal{X}}(\bar{x}). \tag{247}$$

This is the affine-hull Clarke–KKT condition for the original nonsmooth constrained problem.

Finally, the affine-hull form is equivalent to the usual constrained Clarke–KKT condition on $\mathcal{X}$. The equality-normal direction $\operatorname{span}\{\mathbf{1}\}$ is contained in $\mathcal{N}_{\mathcal{X}}(\bar{x})$, while $\partial_{\mathrm{C}}^{\text{aff}} f(\bar{x})$ records the tangential Clarke subgradients along the budget hyperplane.

Hence Eq. (247) proves the consistency statement claimed in Corollary 1. In particular, this corollary does not turn a single fixed-δ run into a Clarke–KKT guarantee for the original objective. □

A.18 Stochastic Smoothing Oracle

The deterministic analysis above uses exact evaluations of the smoothed objective f_δ. With a stochastic black-box evaluator, these values can be approximated by Monte Carlo averaging. For $x \in \mathcal{X}_\delta$ and an integer $M \geq 1$, define

$$\widehat{f}_{\delta,M}(x) = \frac{1}{M}\sum_{m=1}^{M} f(x + \delta Q v_m), \qquad v_m \overset{\text{i.i.d.}}{\sim} \nu_r. \tag{248}$$

Here ν_r is the uniform distribution on $\mathbb{B}_r$. This is the scalar-oracle route used by the finite-difference estimator. A direct sphere-gradient route, introduced in Appendix A.28, avoids the additional finite-difference layer and fits the same stochastic decomposition with zero deterministic difference bias.

For the stochastic descent statement, we formulate the assumption directly at the lifted gradient-estimator level. This isolates the variance contribution and avoids hiding the finite-difference amplification. Let $\mathcal{F}_t$ be the filtration generated by the algorithm up to the beginning of iteration t. Assume that the lifted estimator admits the decomposition

$$\widehat{g}_t = \nabla_{\text{aff}} f_\delta(x_t) + b_t + \xi_t, \tag{249}$$

where

$$\|b_t\|_2 \leq \zeta_h, \tag{250}$$

$$\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0, \tag{251}$$

$$\mathbb{E}\left[\|\xi_t\|_2^2 \mid \mathcal{F}_t\right] \leq \frac{\sigma_g^2}{M}. \tag{252}$$

The vector b_t is the deterministic finite-difference bias, while ξ_t is the stochastic oracle noise. If the noise is specified at the scalar function-oracle level, then the variance proxy σ_g^2 includes the finite-difference factor and the dimension-dependent aggregation factor.

The stochastic update direction is

$$d_t = -W_t \Pi_{W_t} \widehat{g}_t. \tag{253}$$

Since W_t and Π_{W_t} are $\mathcal{F}_t$-measurable and Π_{W_t} is linear, Eq. (249) gives

$$\begin{aligned}\Pi_{W_t}\widehat{g}_t &= \Pi_{W_t}\left(\nabla_{\text{aff}} f_\delta(x_t) + b_t + \xi_t\right)\\ &= \Pi_{W_t}\nabla_{\text{aff}} f_\delta(x_t) + \Pi_{W_t} b_t + \Pi_{W_t}\xi_t.\end{aligned} \tag{254}$$

The proof of Theorem 1 applies pathwise with the total estimation error

$$b_t + \xi_t = \widehat{g}_t - \nabla_{\text{aff}} f_\delta(x_t). \tag{255}$$

Thus, for the same constants $c_2, c_3 > 0$ and for $\gamma_t \leq c_1/\Lambda_\delta$,

$$\begin{aligned}f_\delta(x_{t+1}) \leq\ & f_\delta(x_t) - c_2\gamma_t \mathcal{G}_\delta(x_t)^2\\ &+ c_3\gamma_t\|b_t + \xi_t\|_2^2.\end{aligned} \tag{256}$$

Taking conditional expectation with respect to $\mathcal{F}_t$ yields

$$\begin{aligned}\mathbb{E}\left[f_\delta(x_{t+1}) \mid \mathcal{F}_t\right] \leq\ & f_\delta(x_t) - c_2\gamma_t \mathcal{G}_\delta(x_t)^2\\ &+ c_3\gamma_t \mathbb{E}\left[\|b_t + \xi_t\|_2^2 \mid \mathcal{F}_t\right].\end{aligned} \tag{257}$$

Moreover,

$$\begin{aligned}\mathbb{E}\left[\|b_t + \xi_t\|_2^2 \mid \mathcal{F}_t\right] &= \mathbb{E}\left[\|b_t\|_2^2 + 2b_t^\top \xi_t + \|\xi_t\|_2^2 \mid \mathcal{F}_t\right]\\ &= \|b_t\|_2^2 + 2b_t^\top \mathbb{E}[\xi_t \mid \mathcal{F}_t]\\ &\quad + \mathbb{E}\left[\|\xi_t\|_2^2 \mid \mathcal{F}_t\right]\\ &\leq \zeta_h^2 + \frac{\sigma_g^2}{M}.\end{aligned} \tag{258}$$

Substituting Eq. (258) into Eq. (257) gives

$$\begin{aligned}\mathbb{E}\left[f_\delta(x_{t+1}) \mid \mathcal{F}_t\right] \leq\ & f_\delta(x_t) - c_2\gamma_t \mathcal{G}_\delta(x_t)^2\\ &+ c_3\gamma_t\left(\zeta_h^2 + \frac{\sigma_g^2}{M}\right).\end{aligned} \tag{259}$$

Taking total expectation and summing from $t = 0$ to $T-1$, we obtain

$$\begin{aligned}c_2\sum_{t=0}^{T-1}\gamma_t \mathbb{E}\left[\mathcal{G}_\delta(x_t)^2\right] &\leq \mathbb{E}\left[f_\delta(x_0) - f_\delta(x_T)\right]\\ &\quad + c_3\sum_{t=0}^{T-1}\gamma_t\left(\zeta_h^2 + \frac{\sigma_g^2}{M}\right)\\ &\leq \Delta_{\delta,K} + c_3\sum_{t=0}^{T-1}\gamma_t\left(\zeta_h^2 + \frac{\sigma_g^2}{M}\right).\end{aligned} \tag{260}$$

Under the same nonbinding-safeguard condition used in Appendix A.16, a constant realized step $\gamma = \Theta(1/\Lambda_\delta)$ gives

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\mathcal{G}_\delta(x_t)^2\right] \leq \frac{\Delta_{\delta,K}}{c_2\gamma T} + \frac{c_3}{c_2}\left(\zeta_h^2 + \frac{\sigma_g^2}{M}\right). \tag{261}$$

Therefore, choosing

$$\begin{cases} h = O\left(\dfrac{\epsilon}{\sqrt{d}\,\Lambda_\delta}\right),\\[2ex] M = \Theta\left(\max\left\{1, \dfrac{\sigma_g^2}{\epsilon^2}\right\}\right),\\[2ex] T = \Theta\left(\max\left\{1, \dfrac{\Lambda_\delta\Lambda_{\delta,K}}{\epsilon^2}\right\}\right)\end{cases} \tag{262}$$

ensures

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}\left[\mathcal{G}_\delta(x_t)^2\right] = O(\epsilon^2). \tag{263}$$

Equivalently, if τ is drawn uniformly from $\{0, \ldots, T-1\}$, then

$$\mathbb{E}\left[\mathcal{G}_\delta(x_\tau)^2\right] = O(\epsilon^2). \tag{264}$$

The deterministic oracle-complexity result is recovered by setting $\sigma_g^2 = 0$.

A.19 Why CVaR Optimization Is Not Equivalent

This subsection shows that replacing VaR by CVaR changes the induced preference order. The two criteria use different lower-tail summaries, so CVaR optimization is not a smooth reformulation of the VaR-adjusted objective. Let

$$Q_\alpha(x) = q_\alpha(R(x)), \qquad C_\alpha(x) = \text{CVaR}_\alpha(R(x)), \tag{265}$$

where C_α denotes the lower-tail CVaR of the return distribution. The VaR-adjusted score ranks x above y if

$$\frac{\mu(x)-\rho}{\max\{\rho-Q_\alpha(x),\kappa_\rho\}} > \frac{\mu(y)-\rho}{\max\{\rho-Q_\alpha(y),\kappa_\rho\}}. \tag{266}$$

The corresponding CVaR-adjusted score ranks x above y if

$$\frac{\mu(x)-\rho}{\max\{\rho-C_\alpha(x),\kappa_\rho\}} > \frac{\mu(y)-\rho}{\max\{\rho-C_\alpha(y),\kappa_\rho\}}. \tag{267}$$

The two rankings need not agree.

Consider $\alpha = 0.1$ and $\rho = 5$. Let two candidate allocations induce the following return distributions:

$$R_x = \begin{cases} 0, & \text{with probability } 0.1, \\ 100/9, & \text{with probability } 0.9, \end{cases} \tag{268}$$

and

$$R_y = \begin{cases} -100, & \text{with probability } 0.05, \\ 1, & \text{with probability } 0.05, \\ 299/18, & \text{with probability } 0.9. \end{cases} \tag{269}$$

Their means are equal:

$$\mu(x) = 0.1\cdot 0 + 0.9\cdot\frac{100}{9} = 10, \tag{270}$$

$$\mu(y) = 0.05\cdot(-100) + 0.05\cdot 1 + 0.9\cdot\frac{299}{18} = -5 + \frac{1}{20} + \frac{299}{20} = 10. \tag{271}$$

Thus

$$\mu(x)-\rho = \mu(y)-\rho = 5. \tag{272}$$

For the lower 10% VaR, the quantiles are

$$Q_\alpha(x) = 0, \qquad Q_\alpha(y) = 1. \tag{273}$$

Choose any $\kappa_\rho < 4$, so clipping is inactive for both VaR denominators in this example. Therefore,

$$\frac{\mu(x)-\rho}{\max\{\rho-Q_\alpha(x),\kappa_\rho\}} = \frac{5}{5-0} = 1, \tag{274}$$

$$\frac{\mu(y)-\rho}{\max\{\rho-Q_\alpha(y),\kappa_\rho\}} = \frac{5}{5-1} = \frac{5}{4}. \tag{275}$$

Hence the VaR-adjusted score ranks y above x. For the lower-tail CVaR, the worst 10% of R_x equals 0. For R_y, the worst 10% consists of the two atoms -100 and 1, each with probability 0.05. Hence

$$C_\alpha(x) = 0, \tag{276}$$

$$C_\alpha(y) = \frac{0.05\cdot(-100) + 0.05\cdot 1}{0.1} = -49.5. \tag{277}$$

The CVaR-adjusted scores are therefore

$$\frac{\mu(x)-\rho}{\max\{\rho-C_\alpha(x),\kappa_\rho\}} = \frac{5}{5-0} = 1, \tag{278}$$

$$\frac{\mu(y)-\rho}{\max\{\rho-C_\alpha(y),\kappa_\rho\}} = \frac{5}{5-(-49.5)} = \frac{5}{54.5} < 1. \tag{279}$$

Thus the CVaR-adjusted score ranks x above y, while the VaR-adjusted score ranks y above x. Both allocations satisfy $\mu > \rho$, so the reversal is not caused by a target-floor sign change. Replacing VaR by CVaR therefore changes the optimization criterion rather than merely smoothing it.

A.20 KKT Conditions in Component Form

For the smoothed constrained problem

$$\min_{x\in\mathcal{X}} f_\delta(x), \tag{280}$$

write the constraints as

$$\begin{cases} c_0(x) = \mathbf{1}^\top x - 1 = 0, \\ c_i^+(x) = x_i - u_i \le 0, \quad i = 1,\dots,d, \\ c_i^-(x) = l_i - x_i \le 0, \quad i = 1,\dots,d. \end{cases} \tag{281}$$

Using the sign convention in the main text, define the Lagrangian by

$$\mathcal{L}(x,\lambda,\nu^+,\nu^-) = f_\delta(x) - \lambda\left(\mathbf{1}^\top x - 1\right) + \sum_{i=1}^d \nu_i^+(x_i-u_i) + \sum_{i=1}^d \nu_i^-(l_i-x_i). \tag{282}$$

Here $\lambda\in\mathbb{R}$ is the equality multiplier, and $\nu_i^+, \nu_i^- \ge 0$ are the upper- and lower-box multipliers.

The stationarity condition is

$$\begin{aligned} 0 &= \nabla_x\mathcal{L}(x,\lambda,\nu^+,\nu^-) \\ &= \nabla_{\text{aff}} f_\delta(x) - \lambda\nabla_x\left(\mathbf{1}^\top x - 1\right) \\ &\quad + \sum_{i=1}^d \nu_i^+\nabla_x(x_i-u_i) + \sum_{i=1}^d \nu_i^-\nabla_x(l_i-x_i) \\ &= \nabla_{\text{aff}} f_\delta(x) - \lambda\mathbf{1} + \sum_{i=1}^d \nu_i^+ e_i - \sum_{i=1}^d \nu_i^- e_i \\ &= \nabla_{\text{aff}} f_\delta(x) - \lambda\mathbf{1} + \nu^+ - \nu^-. \end{aligned} \tag{283}$$

The remaining KKT conditions are

$$\begin{cases} \nu_i^+ \ge 0, \quad \nu_i^- \ge 0, & i = 1,\dots,d, \\ \nu_i^+(x_i-u_i) = 0, \quad \nu_i^-(l_i-x_i) = 0, & i = 1,\dots,d, \\ l_i \le x_i \le u_i, & i = 1,\dots,d, \\ \mathbf{1}^\top x = 1. \end{cases} \tag{284}$$

Equivalently, the KKT conditions can be written in normal-cone form as

$$0 \in \nabla_{\text{aff}} f_\delta(x) + \mathcal{N}_{\mathcal{X}}(x). \tag{285}$$

Indeed, the normal cone to $\mathcal{X}$ at x consists of all vectors of the form

$$-\lambda\mathbf{1} + \nu^+ - \nu^-, \tag{286}$$

where $\lambda\in\mathbb{R}$, $\nu_i^+ \ge 0$, $\nu_i^- \ge 0$, and

$$\nu_i^+ = 0 \quad \text{whenever } x_i < u_i, \tag{287}$$

$$\nu_i^- = 0 \quad \text{whenever } x_i > l_i. \tag{288}$$

Thus

$$\begin{aligned} &0 = \nabla_{\text{aff}} f_\delta(x) - \lambda\mathbf{1} + \nu^+ - \nu^- \\ &\iff -\nabla_{\text{aff}} f_\delta(x) = -\lambda\mathbf{1} + \nu^+ - \nu^- \in \mathcal{N}_{\mathcal{X}}(x) \\ &\iff 0 \in \nabla_{\text{aff}} f_\delta(x) + \mathcal{N}_{\mathcal{X}}(x). \end{aligned} \tag{289}$$

Therefore the component-wise KKT system in Eqs. (283) and (284) is equivalent to the normal-cone condition in Eq. (285).

A.21 Auxiliary Definitions and Facts for Cleaner Proofs

We collect several elementary facts used in the appendix. These statements are included only to specify the perturbation, norm, and projection steps.

Definition 1 (Feasible perturbation radius)**.** For $x \in \mathrm{ri}(\mathcal{X})$, define its box-margin radius by

$$r_{\mathcal{X}}(x) = \min_{i\in[d]} \{x_i - l_i,\ u_i - x_i\}\,. \tag{290}$$

For a reduced perturbation direction $a \in \mathbb{R}^r$, define its lifted coordinate size by

$$\omega_Q(a) = \|Qa\|_\infty. \tag{291}$$

Fact 1 (A useful norm chain)**.** *Let $Q \in \mathbb{R}^{d\times r}$ satisfy*

$$Q^\top Q = I_r, \qquad Q^\top \mathbf{1} = 0. \tag{292}$$

Then, for every $a \in \mathbb{R}^r$,

$$\|Qa\|_\infty \le \|Qa\|_2 = \|a\|_2. \tag{293}$$

Proof. The first inequality follows from the comparison between the sup-norm and the Euclidean norm:

$$\|Qa\|_\infty \le \|Qa\|_2. \tag{294}$$

The equality follows from the orthonormality of the columns of Q:

$$\begin{aligned}\|Qa\|_2^2 &= (Qa)^\top (Qa)\\ &= a^\top Q^\top Qa\\ &= a^\top I_r a\\ &= \|a\|_2^2.\end{aligned} \tag{295}$$

Taking square roots and combining with Eq. (294) proves Eq. (293). □

Fact 2 (Feasibility of reduced perturbations)**.** *Let $x \in \mathrm{ri}(\mathcal{X})$ and $a \in \mathbb{R}^r$. If*

$$\eta\, \omega_Q(a) \le r_{\mathcal{X}}(x) \tag{296}$$

for some $\eta \ge 0$, then

$$x + \eta Qa \in \mathcal{X}. \tag{297}$$

If the inequality in Eq. (296) *is strict, then*

$$x + \eta Qa \in \mathrm{ri}(\mathcal{X}). \tag{298}$$

Proof. First, since $Q^\top \mathbf{1} = 0$,

$$\begin{aligned}\mathbf{1}^\top (x + \eta Qa) &= \mathbf{1}^\top x + \eta \mathbf{1}^\top Qa\\ &= 1 + \eta (Q^\top \mathbf{1})^\top a\\ &= 1.\end{aligned} \tag{299}$$

Thus the affine budget equality is preserved.

For each coordinate $i \in [d]$, the lower bound satisfies

$$\begin{aligned}x_i + \eta (Qa)_i &\ge x_i - \eta |(Qa)_i|\\ &\ge x_i - \eta \|Qa\|_\infty\\ &= x_i - \eta \omega_Q(a)\\ &\ge x_i - r_{\mathcal{X}}(x)\\ &\ge l_i.\end{aligned} \tag{300}$$

Similarly, the upper bound satisfies

$$\begin{aligned}x_i + \eta (Qa)_i &\le x_i + \eta |(Qa)_i|\\ &\le x_i + \eta \|Qa\|_\infty\\ &= x_i + \eta \omega_Q(a)\\ &\le x_i + r_{\mathcal{X}}(x)\\ &\le u_i.\end{aligned} \tag{301}$$

Combining Eq. (299) with Eqs. (300)–(301) gives

$$x + \eta Qa \in \mathcal{X}. \tag{302}$$

If $\eta\omega_Q(a) < r_{\mathcal{X}}(x)$, then the inequalities in Eqs. (300) and (301) are strict, so

$$x + \eta Qa \in \mathrm{ri}(\mathcal{X}). \tag{303}$$

□

Fact 3 (Weighted centering is an oblique projection)**.** *Let $W \succ 0$ and define*

$$\Pi_W y = y - \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}} \mathbf{1}. \tag{304}$$

Then $\Pi_W y$ is centered with respect to the W-weighted pairing, $W\Pi_W y$ lies in the tangent space T*, and Π_W is idempotent:*

$$\begin{cases} \mathbf{1}^\top W \Pi_W y = 0, \\ \mathbf{1}^\top (W \Pi_W y) = 0, \\ \Pi_W (\Pi_W y) = \Pi_W y. \end{cases} \tag{305}$$

Proof. By the definition of Π_W,

$$\begin{aligned}\mathbf{1}^\top W \Pi_W y &= \mathbf{1}^\top W \left(y - \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}} \mathbf{1} \right)\\ &= \mathbf{1}^\top W y - \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}} \mathbf{1}^\top W \mathbf{1}\\ &= \mathbf{1}^\top W y - \mathbf{1}^\top W y\\ &= 0.\end{aligned} \tag{306}$$

This proves the first identity in Eq. (305). Since

$$\mathbf{1}^\top (W \Pi_W y) = \mathbf{1}^\top W \Pi_W y, \tag{307}$$

The second identity in Eq. (305) also follows. Hence $W\Pi_W y \in$ T. For idempotence, define

$$\lambda_W(y) = \frac{\mathbf{1}^\top W y}{\mathbf{1}^\top W \mathbf{1}}. \tag{308}$$

Then

$$\Pi_W y = y - \lambda_W(y) \mathbf{1}. \tag{309}$$

Using Eq. (306),

$$\begin{aligned}\lambda_W(\Pi_W y) &= \frac{\mathbf{1}^\top W (\Pi_W y)}{\mathbf{1}^\top W \mathbf{1}}\\ &= \frac{0}{\mathbf{1}^\top W \mathbf{1}}\\ &= 0.\end{aligned} \tag{310}$$

Therefore,

$$\begin{aligned}\Pi_W (\Pi_W y) &= \Pi_W y - \lambda_W(\Pi_W y) \mathbf{1}\\ &= \Pi_W y - 0 \cdot \mathbf{1}\\ &= \Pi_W y.\end{aligned} \tag{311}$$

□

Fact 4 (Order-statistic stability). *For $a, b \in \mathbb{R}^n$, let $a_{(k)}$ and $b_{(k)}$ denote their k-th order statistics. Then*

$$|a_{(k)} - b_{(k)}| \le \|a - b\|_\infty. \tag{312}$$

Proof. Let

$$\eta = \|a - b\|_\infty. \tag{313}$$

Then, for every coordinate j,

$$a_j - \eta \le b_j \le a_j + \eta. \tag{314}$$

Sorting preserves coordinatewise domination, so

$$a_{(k)} - \eta \le b_{(k)} \le a_{(k)} + \eta. \tag{315}$$

Equivalently,

$$-\eta \le b_{(k)} - a_{(k)} \le \eta. \tag{316}$$

Therefore,

$$|a_{(k)} - b_{(k)}| \le \eta = \|a - b\|_\infty. \tag{317}$$

This proves Eq. (312). □

A.22 Construction of the Tangent Basis

This subsection gives one construction of the tangent basis used in the affine-hull formulation. Recall that

$$\mathsf{T} = \{y \in \mathbb{R}^d : \mathbf{1}^\top y = 0\}, \qquad r = d - 1. \tag{318}$$

The algorithm only requires a matrix $Q \in \mathbb{R}^{d \times r}$ whose columns form an orthonormal basis of T, namely

$$Q^\top Q = I_r, \qquad Q^\top \mathbf{1} = 0. \tag{319}$$

A convenient construction starts from the centering matrix

$$C = I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top. \tag{320}$$

First, C is symmetric:

$$\begin{aligned} C^\top &= \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right)^\top \\ &= I_d^\top - \frac{1}{d}(\mathbf{1}\mathbf{1}^\top)^\top \\ &= I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top \\ &= C. \end{aligned} \tag{321}$$

Second, C is idempotent:

$$\begin{aligned} C^2 &= \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right)\left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right) \\ &= I_d - \frac{2}{d}\mathbf{1}\mathbf{1}^\top + \frac{1}{d^2}\mathbf{1}(\mathbf{1}^\top\mathbf{1})\mathbf{1}^\top \\ &= I_d - \frac{2}{d}\mathbf{1}\mathbf{1}^\top + \frac{1}{d}\mathbf{1}\mathbf{1}^\top \\ &= I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top \\ &= C. \end{aligned} \tag{322}$$

Moreover,

$$\begin{aligned} C\mathbf{1} &= \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right)\mathbf{1} \\ &= \mathbf{1} - \frac{1}{d}\mathbf{1}(\mathbf{1}^\top\mathbf{1}) \\ &= \mathbf{1} - \frac{1}{d}\mathbf{1}\cdot d \\ &= 0. \end{aligned} \tag{323}$$

Thus $\mathrm{range}(C) \subseteq \mathsf{T}$. Conversely, if $y \in \mathsf{T}$, then $\mathbf{1}^\top y = 0$, and hence

$$\begin{aligned} Cy &= \left(I_d - \frac{1}{d}\mathbf{1}\mathbf{1}^\top\right)y \\ &= y - \frac{1}{d}\mathbf{1}(\mathbf{1}^\top y) \\ &= y - \frac{1}{d}\mathbf{1}\cdot 0 \\ &= y. \end{aligned} \tag{324}$$

Therefore $\mathsf{T} \subseteq \mathrm{range}(C)$, and we conclude that

$$\mathrm{range}(C) = \mathsf{T}. \tag{325}$$

Since C is symmetric and idempotent, it is the Euclidean orthogonal projector onto T. Let

$$C = U\Sigma U^\top \tag{326}$$

be an eigenvalue decomposition with

$$\Sigma = \mathrm{diag}\left(\underbrace{1, \ldots, 1}_{r \text{ times}}, 0\right). \tag{327}$$

Write $U = [Q\ q_0]$, where $Q \in \mathbb{R}^{d \times r}$ contains the eigenvectors associated with the eigenvalue 1, and $q_0 \in \mathbb{R}^d$ is an eigenvector associated with the eigenvalue 0. Because U is orthogonal,

$$\begin{aligned} I_d &= U^\top U \\ &= \begin{pmatrix} Q^\top \\ q_0^\top \end{pmatrix}[Q\ q_0] \\ &= \begin{pmatrix} Q^\top Q & Q^\top q_0 \\ q_0^\top Q & q_0^\top q_0 \end{pmatrix}. \end{aligned} \tag{328}$$

Hence $Q^\top Q = I_r$. It follows we obtain

$$\begin{aligned} QQ^\top &= [Q\ q_0]\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}[Q\ q_0]^\top \\ &= U\Sigma U^\top \\ &= C. \end{aligned} \tag{329}$$

Therefore,

$$\begin{aligned} \|Q^\top \mathbf{1}\|_2^2 &= (Q^\top \mathbf{1})^\top (Q^\top \mathbf{1}) \\ &= \mathbf{1}^\top QQ^\top \mathbf{1} \\ &= \mathbf{1}^\top C\mathbf{1} \\ &= \mathbf{1}^\top 0 \\ &= 0. \end{aligned} \tag{330}$$

Thus $Q^\top \mathbf{1} = 0$. The columns of Q form an orthonormal basis of T.

A.23 Finite-Difference Queries in a Strictly Feasible Tube

The main text uses $\mathcal{X}_\delta$ to ensure that affine-hull smoothing queries remain feasible. Forward finite differences require a slightly stronger condition: after moving from z to $z + he_j$, the entire δ-smoothing ball must still be feasible. This is why Algorithm 1 keeps the iterates in the stricter tube $\mathcal{X}_{\delta,h}$. Recall that

$$\mathcal{X}_{\delta,h} = \left\{x \in \mathrm{aff}(\mathcal{X}) : l_i + \delta + h \le x_i \le u_i - \delta - h, i \in [d]\right\}, \quad (331)$$

and

$$\mathcal{X}_\delta = \{x \in \mathrm{aff}(\mathcal{X}) : l_i + \delta \le x_i \le u_i - \delta,\ i \in [d]\}. \quad (332)$$

Let

$$x = x_{\mathrm{ref}} + Qz \in \mathcal{X}_{\delta,h}. \quad (333)$$

For the j-th forward finite-difference query, where $j = 1, \dots, r$, define

$$x^{(j)} = x_{\mathrm{ref}} + Q(z + he_j) = x + hQe_j. \quad (334)$$

We first show that $x^{(j)} \in \mathcal{X}_\delta$. Since $Q^\top Q = I_r$, for every coordinate $i \in [d]$,

$$\begin{aligned} |(Qe_j)_i| &\le \|Qe_j\|_2 \\ &= \sqrt{(Qe_j)^\top (Qe_j)} \\ &= \sqrt{e_j^\top Q^\top Q e_j} \\ &= \sqrt{e_j^\top e_j} \\ &= 1. \end{aligned} \quad (335)$$

Using Eq. (335), the lower box margin of $x^{(j)}$ satisfies

$$\begin{aligned} x_i^{(j)} &= x_i + h(Qe_j)_i \\ &\ge x_i - h|(Qe_j)_i| \\ &\ge x_i - h \\ &\ge l_i + \delta + h - h \\ &= l_i + \delta. \end{aligned} \quad (336)$$

Similarly, the upper box margin satisfies

$$\begin{aligned} x_i^{(j)} &= x_i + h(Qe_j)_i \\ &\le x_i + h|(Qe_j)_i| \\ &\le x_i + h \\ &\le u_i - \delta - h + h \\ &= u_i - \delta. \end{aligned} \quad (337)$$

The affine equality is also preserved:

$$\begin{aligned} \mathbf{1}^\top x^{(j)} &= \mathbf{1}^\top (x + hQe_j) \\ &= \mathbf{1}^\top x + h\mathbf{1}^\top Qe_j \\ &= 1 + h(Q^\top \mathbf{1})^\top e_j \\ &= 1 + h\,0^\top e_j \\ &= 1. \end{aligned} \quad (338)$$

Combining Eqs. (336), (337), and (338), we obtain

$$x^{(j)} \in \mathcal{X}_\delta, \qquad j = 1, \dots, r. \quad (339)$$

It remains to check that the smoothing queries around $x^{(j)}$ are feasible. Let $v \in \mathbb{B}_r$, where

$$\mathbb{B}_r = \{v \in \mathbb{R}^r : \|v\|_2 \le 1\}. \quad (340)$$

Since $x^{(j)} \in \mathcal{X}_\delta$, we have

$$l_i + \delta \le x_i^{(j)} \le u_i - \delta, \qquad i \in [d]. \quad (341)$$

The equality constraint of the smoothing query is preserved because

$$\begin{aligned} \mathbf{1}^\top \left(x^{(j)} + \delta Qv\right) &= \mathbf{1}^\top x^{(j)} + \delta \mathbf{1}^\top Qv \\ &= 1 + \delta(Q^\top \mathbf{1})^\top v \\ &= 1. \end{aligned} \quad (342)$$

For the coordinate size of the lifted smoothing direction, we have

$$\begin{aligned} |(Qv)_i| &\le \|Qv\|_2 \\ &= \sqrt{(Qv)^\top (Qv)} \\ &= \sqrt{v^\top Q^\top Qv} \\ &= \sqrt{v^\top v} \\ &= \|v\|_2 \\ &\le 1. \end{aligned} \quad (343)$$

Therefore, for every $i \in [d]$, the lower box constraint satisfies

$$\begin{aligned} x_i^{(j)} + \delta(Qv)_i &\ge x_i^{(j)} - \delta|(Qv)_i| \\ &\ge x_i^{(j)} - \delta \\ &\ge l_i + \delta - \delta \\ &= l_i. \end{aligned} \quad (344)$$

The upper box constraint satisfies

$$\begin{aligned} x_i^{(j)} + \delta(Qv)_i &\le x_i^{(j)} + \delta|(Qv)_i| \\ &\le x_i^{(j)} + \delta \\ &\le u_i - \delta + \delta \\ &= u_i. \end{aligned} \quad (345)$$

Combining Eqs. (342), (344), and (345), we conclude that

$$x^{(j)} + \delta Qv \in \mathcal{X}, \qquad \forall v \in \mathbb{B}_r, \qquad j = 1, \dots, r. \quad (346)$$

Thus every forward finite-difference query and every smoothing query used to evaluate it stays on the affine budget hyperplane and remains box-feasible.

A.24 Euclidean and Barrier-Riemannian Stationarity

The main text measures stationarity using the barrier-Riemannian quantity $\mathcal{G}_\delta(x)$. We show that, on compact subsets of the relative interior, this measure is equivalent to the usual Euclidean projected-gradient norm.

Let $K \subset \mathrm{ri}(\mathcal{X})$ be compact. Since $H_\mathcal{X}(x)$ is continuous and positive definite on K, there exist constants $0 < m_K \le M_K < \infty$ such that

$$m_K I_d \preceq H_\mathcal{X}(x) \preceq M_K I_d, \qquad x \in K. \quad (347)$$

Equivalently, with

$$W(x) = H_\mathcal{X}(x)^{-1}, \quad (348)$$

we have

$$\frac{1}{M_K} I_d \preceq W(x) \preceq \frac{1}{m_K} I_d, \qquad x \in K. \tag{349}$$

For compactness of notation, write

$$g_\delta(x) = \nabla_{\mathrm{aff}} f_\delta(x). \tag{350}$$

Define the Euclidean centered gradient by

$$\begin{aligned} \lambda_E(x) &= \frac{\mathbf{1}^\top g_\delta(x)}{\mathbf{1}^\top \mathbf{1}} = \frac{1}{d} \mathbf{1}^\top g_\delta(x), \\ p_E(x) &= g_\delta(x) - \lambda_E(x) \mathbf{1}. \end{aligned} \tag{351}$$

Then $p_E(x)$ is the Euclidean projection of $g_\delta(x)$ onto the orthogonal complement of $\mathrm{span}\{\mathbf{1}\}$. Indeed, for any $\lambda \in \mathbb{R}$,

$$\begin{aligned} \|g_\delta(x) - \lambda \mathbf{1}\|_2^2 &= \|p_E(x) + (\lambda_E(x) - \lambda)\mathbf{1}\|_2^2 \\ &= \|p_E(x)\|_2^2 + 2(\lambda_E(x) - \lambda)\mathbf{1}^\top p_E(x) \\ &\quad + (\lambda_E(x) - \lambda)^2 \|\mathbf{1}\|_2^2. \end{aligned} \tag{352}$$

Moreover,

$$\begin{aligned} \mathbf{1}^\top p_E(x) &= \mathbf{1}^\top g_\delta(x) - \lambda_E(x) \mathbf{1}^\top \mathbf{1} \\ &= \mathbf{1}^\top g_\delta(x) - \frac{\mathbf{1}^\top g_\delta(x)}{d} d \\ &= 0. \end{aligned} \tag{353}$$

Substituting Eq. (353) into Eq. (352) gives

$$\begin{aligned} \|g_\delta(x) - \lambda \mathbf{1}\|_2^2 &= \|p_E(x)\|_2^2 + d(\lambda_E(x) - \lambda)^2 \\ &\geq \|p_E(x)\|_2^2. \end{aligned} \tag{354}$$

Hence

$$\|p_E(x)\|_2^2 = \min_{\lambda \in \mathbb{R}} \|g_\delta(x) - \lambda \mathbf{1}\|_2^2. \tag{355}$$

Next define the barrier-weighted centered gradient by

$$\lambda_\delta(x) = \frac{\mathbf{1}^\top W(x) g_\delta(x)}{\mathbf{1}^\top W(x) \mathbf{1}}, \qquad p_H(x) = g_\delta(x) - \lambda_\delta(x) \mathbf{1}. \tag{356}$$

Then

$$\begin{aligned} \mathbf{1}^\top W(x) p_H(x) &= \mathbf{1}^\top W(x) \left(g_\delta(x) - \lambda_\delta(x) \mathbf{1}\right) \\ &= \mathbf{1}^\top W(x) g_\delta(x) - \lambda_\delta(x) \mathbf{1}^\top W(x) \mathbf{1} \\ &= \mathbf{1}^\top W(x) g_\delta(x) \\ &\quad - \frac{\mathbf{1}^\top W(x) g_\delta(x)}{\mathbf{1}^\top W(x) \mathbf{1}} \mathbf{1}^\top W(x) \mathbf{1} \\ &= 0. \end{aligned} \tag{357}$$

Thus $p_H(x)$ is centered with respect to the $W(x)$-weighted pairing. We now show that $p_H(x)$ minimizes the weighted quadratic over scalar shifts. For any $\lambda \in \mathbb{R}$, define

$$q_\lambda(x) = g_\delta(x) - \lambda \mathbf{1}. \tag{358}$$

Since

$$q_\lambda(x) = p_H(x) + (\lambda_\delta(x) - \lambda)\mathbf{1}, \tag{359}$$

we have

$$\begin{aligned} q_\lambda(x)^\top W(x) q_\lambda(x) &= \left[p_H(x) + (\lambda_\delta(x) - \lambda)\mathbf{1}\right]^\top W(x) \\ &\quad \times \left[p_H(x) + (\lambda_\delta(x) - \lambda)\mathbf{1}\right] \\ &= p_H(x)^\top W(x) p_H(x) \\ &\quad + 2(\lambda_\delta(x) - \lambda)\mathbf{1}^\top W(x) p_H(x) \\ &\quad + (\lambda_\delta(x) - \lambda)^2 \mathbf{1}^\top W(x) \mathbf{1} \\ &= p_H(x)^\top W(x) p_H(x) \\ &\quad + (\lambda_\delta(x) - \lambda)^2 \mathbf{1}^\top W(x) \mathbf{1} \\ &\geq p_H(x)^\top W(x) p_H(x), \end{aligned} \tag{360}$$

where the fourth equality uses Eq. (357). Therefore,

$$p_H(x)^\top W(x) p_H(x) = \min_{\lambda \in \mathbb{R}} q_\lambda(x)^\top W(x) q_\lambda(x). \tag{361}$$

By the definition of the barrier-Riemannian stationarity measure,

$$\mathcal{G}_\delta(x)^2 = p_H(x)^\top W(x) p_H(x). \tag{362}$$

Using Eq. (361) and then Eq. (349), we obtain

$$\begin{aligned} \mathcal{G}_\delta(x)^2 &= p_H(x)^\top W(x) p_H(x) \\ &= \min_{\lambda \in \mathbb{R}} q_\lambda(x)^\top W(x) q_\lambda(x) \\ &\leq p_E(x)^\top W(x) p_E(x) \\ &\leq \frac{1}{m_K} \|p_E(x)\|_2^2. \end{aligned} \tag{363}$$

This proves the upper bound

$$\mathcal{G}_\delta(x) \leq \frac{1}{\sqrt{m_K}} \|p_E(x)\|_2. \tag{364}$$

For the converse direction, use the Euclidean minimizing property in Eq. (355) and the lower metric bound in Eq. (349). We have

$$\begin{aligned} \|p_E(x)\|_2^2 &= \min_{\lambda \in \mathbb{R}} \|g_\delta(x) - \lambda \mathbf{1}\|_2^2 \\ &\leq \|p_H(x)\|_2^2 \\ &\leq M_K p_H(x)^\top W(x) p_H(x) \\ &= M_K \mathcal{G}_\delta(x)^2. \end{aligned} \tag{365}$$

Thus

$$\frac{1}{\sqrt{M_K}} \|p_E(x)\|_2 \leq \mathcal{G}_\delta(x). \tag{366}$$

Combining the two inequalities gives

$$\begin{aligned} \frac{1}{\sqrt{M_K}} \|p_E(x)\|_2 &\leq \mathcal{G}_\delta(x) \\ &\leq \frac{1}{\sqrt{m_K}} \|p_E(x)\|_2, \quad x \in K. \end{aligned} \tag{367}$$

Therefore, on compact subsets of $\mathrm{ri}(\mathcal{X})$, the barrier-Riemannian stationarity measure is equivalent to the Euclidean projected-gradient norm. In this sense, $\mathcal{G}_\delta(x)$ is not a weaker stationarity notion in the relative interior; it is a geometrically scaled version of Euclidean projected stationarity.

A.25 Armijo Feasible Line Search

Algorithm 1 uses the safeguarded step

$$\gamma_t = \min\{\eta_t, \beta\tau_t\}. \tag{368}$$

A practical alternative is to combine feasibility backtracking with an Armijo decrease test. Since the method is zeroth-order, the line search should be stated in terms of the finite-difference estimator rather than the inaccessible true gradient. Fix an iteration t and write

$$\begin{aligned} H_t &= H_{\mathcal{X}}(x_t), \\ W_t &= H_t^{-1}, \\ \widehat{g}_t &= \nabla_{\mathrm{aff}} f_\delta(x_t) + e_t. \end{aligned} \tag{369}$$

Define the weighted centered quantities

$$p_t = \Pi_{W_t} \nabla_{\mathrm{aff}} f_\delta(x_t), \tag{370}$$

$$\widehat{p}_t = \Pi_{W_t} \widehat{g}_t, \tag{371}$$

$$\widetilde{e}_t = \Pi_{W_t} e_t. \tag{372}$$

By linearity of Π_{W_t},

$$\begin{aligned} \widehat{p}_t &= \Pi_{W_t} \widehat{g}_t \\ &= \Pi_{W_t} \left(\nabla_{\mathrm{aff}} f_\delta(x_t) + e_t\right) \\ &= \Pi_{W_t} \nabla_{\mathrm{aff}} f_\delta(x_t) + \Pi_{W_t} e_t \\ &= p_t + \widetilde{e}_t. \end{aligned} \tag{373}$$

The inexact Riemannian descent direction is

$$d_t = -W_t \widehat{p}_t. \tag{374}$$

Because $\widehat{p}_t$ is W_t-centered,

$$\mathbf{1}^\top d_t = -\mathbf{1}^\top W_t \widehat{p}_t = 0. \tag{375}$$

Thus the direction is tangent to the affine budget hyperplane. Assume

$$x_t \in \mathcal{X}_{\delta,h}, \qquad d_t \neq 0. \tag{376}$$

Define the lower and upper tube boundaries by

$$a_i = l_i + \delta + h, \qquad b_i = u_i - \delta - h, \qquad i \in [d]. \tag{377}$$

The maximal feasible step that keeps the next iterate in $\mathcal{X}_{\delta,h}$ is

$$\tau_t = \min\left\{ \min_{i:d_{t,i}>0} \frac{b_i - x_{t,i}}{d_{t,i}}, \min_{i:d_{t,i}<0} \frac{x_{t,i} - a_i}{-d_{t,i}} \right\}, \tag{378}$$

where the minimum over an empty index set is interpreted as $+\infty$. Since $d_t \neq 0$ and $\mathbf{1}^\top d_t = 0$, the direction has at least one positive and one negative component. Because $x_t \in \mathcal{X}_{\delta,h}$, we have

$$a_i \le x_{t,i} \le b_i, \qquad i \in [d], \tag{379}$$

and hence $\tau_t \in [0, \infty]$. If $\tau_t = 0$, the feasible line-search step is zero and the algorithm should reduce the nominal step or declare that the current iterate lies on the boundary of the safe tube. In the usual interior case $x_t \in \mathrm{ri}(\mathcal{X}_{\delta,h})$, we have $\tau_t > 0$. Choose a feasibility safety factor

$$\beta_{\mathrm{feas}} \in (0,1), \tag{380}$$

a backtracking contraction factor

$$\theta \in (0,1), \tag{381}$$

and an Armijo parameter

$$\sigma \in (0,1). \tag{382}$$

When $\tau_t > 0$, the line search selects the largest step size in the grid

$$\left\{\beta_{\mathrm{feas}}\tau_t,\ \beta_{\mathrm{feas}}\theta\tau_t,\ \beta_{\mathrm{feas}}\theta^2\tau_t,\ \ldots\right\} \tag{383}$$

that satisfies the zeroth-order Armijo condition

$$f_\delta(x_t + \gamma d_t) \le f_\delta(x_t) - \sigma\gamma\, \widehat{p}_t^\top W_t \widehat{p}_t. \tag{384}$$

Every candidate in Eq. (383) is feasible for the strict tube. Indeed, for any such candidate,

$$0 < \gamma \le \beta_{\mathrm{feas}}\tau_t < \tau_t. \tag{385}$$

For a coordinate with $d_{t,i} > 0$, Eq. (378) gives

$$\begin{aligned} x_{t,i} + \gamma d_{t,i} &\le x_{t,i} + \beta_{\mathrm{feas}}\tau_t d_{t,i} \\ &\le x_{t,i} + \beta_{\mathrm{feas}} \frac{b_i - x_{t,i}}{d_{t,i}} d_{t,i} \\ &= (1 - \beta_{\mathrm{feas}}) x_{t,i} + \beta_{\mathrm{feas}} b_i \\ &\le b_i. \end{aligned} \tag{386}$$

The lower bound is preserved because

$$x_{t,i} + \gamma d_{t,i} \ge x_{t,i} \ge a_i. \tag{387}$$

For a coordinate with $d_{t,i} < 0$, Eq. (378) gives

$$\begin{aligned} x_{t,i} + \gamma d_{t,i} &\ge x_{t,i} + \beta_{\mathrm{feas}}\tau_t d_{t,i} \\ &\ge x_{t,i} + \beta_{\mathrm{feas}} \frac{x_{t,i} - a_i}{-d_{t,i}} d_{t,i} \\ &= (1 - \beta_{\mathrm{feas}}) x_{t,i} + \beta_{\mathrm{feas}} a_i \\ &\ge a_i. \end{aligned} \tag{388}$$

The upper bound is preserved because

$$x_{t,i} + \gamma d_{t,i} \le x_{t,i} \le b_i. \tag{389}$$

If $d_{t,i} = 0$, then

$$x_{t,i} + \gamma d_{t,i} = x_{t,i} \in [a_i, b_i]. \tag{390}$$

Together with $\mathbf{1}^\top d_t = 0$, this proves

$$x_t + \gamma d_t \in \mathcal{X}_{\delta,h} \subset \mathcal{X}_\delta \subset \mathcal{X}. \tag{391}$$

We now prove finite termination of the Armijo search under a standard relative-error condition. Assume that there exists $\kappa \in [0,1)$ such that

$$\|\widetilde{e}_t\|_{W_t} \le \kappa \|\widehat{p}_t\|_{W_t}, \qquad \|a\|_{W_t}^2 = a^\top W_t a. \tag{392}$$

Assume also that

$$0 < \sigma < 1 - \kappa. \tag{393}$$

First compute the true directional derivative along d_t. Since

$$p_t = \nabla_{\mathrm{aff}} f_\delta(x_t) - \lambda_t \mathbf{1} \tag{394}$$

for some scalar λ_t, and since

$$\mathbf{1}^\top W_t \widehat{p}_t = 0, \tag{395}$$

we have

$$\begin{aligned}\nabla_{\mathrm{aff}} f_\delta(x_t)^\top d_t &= \nabla_{\mathrm{aff}} f_\delta(x_t)^\top(-W_t\widehat{p}_t)\\ &= -\nabla_{\mathrm{aff}} f_\delta(x_t)^\top W_t\widehat{p}_t\\ &= -(p_t+\lambda_t\mathbf{1})^\top W_t\widehat{p}_t\\ &= -p_t^\top W_t\widehat{p}_t - \lambda_t\mathbf{1}^\top W_t\widehat{p}_t\\ &= -p_t^\top W_t\widehat{p}_t\\ &= -(\widehat{p}_t-\widetilde{e}_t)^\top W_t\widehat{p}_t\\ &= -\widehat{p}_t^\top W_t\widehat{p}_t + \widetilde{e}_t^\top W_t\widehat{p}_t\\ &\le -\|\widehat{p}_t\|_{W_t}^2 + \|\widetilde{e}_t\|_{W_t}\|\widehat{p}_t\|_{W_t}\\ &\le -(1-\kappa)\|\widehat{p}_t\|_{W_t}^2. \end{aligned} \tag{396}$$

Thus d_t is a true descent direction whenever $\widehat{p}_t \neq 0$ and Eq. (392) holds. Because every accepted candidate remains in $\mathcal{X}_\delta$ by Eq. (391), affine-hull smoothness gives

$$\begin{aligned} f_\delta(x_t+\gamma d_t) &\le f_\delta(x_t) + \gamma\nabla_{\mathrm{aff}} f_\delta(x_t)^\top d_t + \frac{\Lambda_\delta}{2}\gamma^2\|d_t\|_2^2\\ &\le f_\delta(x_t) - (1-\kappa)\gamma\|\widehat{p}_t\|_{W_t}^2 + \frac{\Lambda_\delta}{2}\gamma^2\|d_t\|_2^2. \end{aligned} \tag{397}$$

Therefore the Armijo condition in Eq. (384) is guaranteed whenever

$$\begin{aligned} &-(1-\kappa)\gamma\|\widehat{p}_t\|_{W_t}^2 + \frac{\Lambda_\delta}{2}\gamma^2\|d_t\|_2^2\\ &\le -\sigma\gamma\|\widehat{p}_t\|_{W_t}^2\\ &\iff \frac{\Lambda_\delta}{2}\gamma^2\|d_t\|_2^2 \le (1-\kappa-\sigma)\gamma\|\widehat{p}_t\|_{W_t}^2\\ &\iff \gamma \le \frac{2(1-\kappa-\sigma)\|\widehat{p}_t\|_{W_t}^2}{\Lambda_\delta\|d_t\|_2^2}. \end{aligned} \tag{398}$$

The right-hand side is positive whenever

$$\widehat{p}_t \neq 0, \qquad \sigma < 1-\kappa. \tag{399}$$

Thus, for $\widehat{p}_t \neq 0$ and $\tau_t > 0$, a sufficiently small element of the geometric grid in Eq. (383) satisfies the Armijo condition, and the line search terminates finitely. If $\widehat{p}_t = 0$, then $d_t = 0$, and the algorithm may declare zeroth-order projected stationarity.

A.26 Expanded Accounting for Oracle Evaluations

This subsection separates the units that appear in the oracle-complexity statement. The distinction is necessary because an exact smoothed objective value, a Monte Carlo approximation of that value, and a direct empirical objective value have different primitive costs even though each is a scalar input to the same reduced-coordinate finite-difference routine.

Target-oracle evaluations. At iteration t, the algorithm evaluates one shared baseline value and one forward value in each reduced coordinate. Since $r = d-1$,

$$N_{\mathrm{target/iter}} = 1 + r = d. \tag{400}$$

A *target-oracle evaluation* means one scalar evaluation of the target function currently being optimized. In exact smoothed mode, the target is f_δ; in direct empirical mode, it is $\widehat{f}_n$. The direct-mode complexity statement in Proposition 1 is expressed through regular-step mass and interface debit; eventual identification recovers the familiar single-cell rate. The common count is geometric: it follows from the budget affine-hull dimension and is independent of the entropy–barrier update.

Exact smoothed-oracle complexity. Let $\Gamma_T = \sum_{t=0}^{T-1}\gamma_t$. Equation (193) gives

$$\min_{0\le t<T}\mathcal{G}_\delta(x_t)^2 \le \frac{\Delta_{\delta,K}}{c_2\Gamma_T} + \frac{c_3}{c_2}\zeta_h^2. \tag{401}$$

For forward differences,

$$\zeta_h \le C_{\mathrm{fd}}\sqrt{r}\,\Lambda_\delta h, \qquad C_{\mathrm{fd}} = \frac{1}{2}. \tag{402}$$

Thus the bias term is at most $\epsilon^2/4$ when

$$h \le \frac{\epsilon}{2C_{\mathrm{fd}}\sqrt{r}\,\Lambda_\delta}\sqrt{\frac{c_2}{c_3}}. \tag{403}$$

The optimization term is at most $\epsilon^2/4$ when

$$\Gamma_T \ge \frac{4\Delta_{\delta,K}}{c_2\epsilon^2}. \tag{404}$$

If the realized steps satisfy $\gamma_t \ge \underline{\gamma} > 0$, then $\Gamma_T \ge T\underline{\gamma}$ and it is sufficient that

$$T \ge \frac{4\Delta_{\delta,K}}{c_2\underline{\gamma}\epsilon^2}. \tag{405}$$

The nonbinding constant-step case uses $\underline{\gamma} = \gamma = \Theta(1/\Lambda_\delta)$ and therefore

$$T = O\left(\frac{\Lambda_\delta\Delta_{\delta,K}}{\epsilon^2}\right). \tag{406}$$

Combining Eqs. (400) and (406) yields

$$N_{\mathrm{smooth}} = dT = O\left(\frac{d\Lambda_\delta\Delta_{\delta,K}}{\epsilon^2}\right). \tag{407}$$

With $\Lambda_\delta \le L\sqrt{r}/\delta \le L\sqrt{d}/\delta$,

$$N_{\mathrm{smooth}} = O\left(\frac{d^{3/2}L\Delta_{\delta,K}}{\delta\epsilon^2}\right). \tag{408}$$

Finally, the valid diameter estimate $\Delta_{\delta,K} \le LD_K$ from Eq. (208) gives

$$N_{\mathrm{smooth}} = O\left(\frac{d^{3/2}L^2D_K}{\delta\epsilon^2}\right). \tag{409}$$

This uses the diameter bound in Eq. (208) and introduces no additional inverse-δ factor through the objective gap.

Primitive evaluations for Monte Carlo smoothing. Suppose one target value $f_\delta(x)$ is approximated by

$$\widehat{f}_{\delta,M}(x) = \frac{1}{M}\sum_{m=1}^{M} f(x+\delta Qv_m), \qquad M \ge 1. \tag{410}$$

If no evaluations are shared across target points, one iteration requires

$$N_{\mathrm{sim/iter}} = dM \tag{411}$$

primitive evaluations of the unsmoothed objective, and after T iterations

$$N_{\mathrm{sim}} = dMT. \tag{412}$$

This identity is deterministic accounting. It does not by itself specify how large M must be. Under the estimator-level stochastic model of Appendix A.18, if

$$\mathbb{E}[\|\xi_t\|_2^2 \mid \mathcal{F}_t] \le \frac{\sigma_g^2}{M}, \tag{413}$$

then choosing

$$M = \Theta\left(\max\left\{1, \frac{\sigma_g^2}{\epsilon^2}\right\}\right) \tag{414}$$

makes the stochastic stationarity floor $O(\epsilon^2)$. Together with Eq. (406), this gives the estimator-level accounting

$$N_{\text{sim}} = O\left(d \max\left\{1, \frac{\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right\} \max\left\{1, \frac{\sigma_g^2}{\epsilon^2}\right\}\right). \tag{415}$$

If noise is modeled instead at the scalar function-value level, the induced σ_g^2 generally contains factors of $1/h^2$, dimension constants, and covariance terms determined by whether common random numbers are used. Those factors depend on the chosen oracle coupling and cannot be removed by notation.

Direct empirical-oracle accounting. When $M_{\text{smooth}} = 0$, the stabilized empirical objective is evaluated directly. The value denotes direct mode rather than a Monte Carlo sample size, so it does not enter the call count as a multiplicative factor. Equation (400) gives exactly

$$N_{\text{direct}} = dT. \tag{416}$$

For trajectories with crossings, Proposition 1 measures useful optimization time by Γ_T^{reg} and charges the exceptional iterations through $\mathfrak{D}_T$. If $\mathfrak{D}_T \le \mathfrak{D}_\infty$ and $\Gamma_T^{\text{reg}} \ge \pi T\theta/\overline{\Lambda}_n$, then a nonbinding regular-step schedule gives

$$N_{\text{direct}} = O\left(\frac{d\,\overline{\Lambda}_n(\Delta_{n,K} + \mathfrak{D}_\infty)}{\pi\epsilon^2}\right). \tag{417}$$

This is a trajectory-level direct-oracle rate, not a claim that interface iterations are smooth. When the trajectory identifies one active cell after a finite prefix, the regular-mass condition holds for every fixed $\pi < 1$ once the horizon is large enough, and the prefix contributes only a finite interface debit. Without debit control or sufficient regular-step mass, the finite-horizon inequality remains valid but no $O(\epsilon^{-2})$ total-iteration rate is asserted.

A.27 Technical Verification of the Main Claims

This subsection summarizes the main technical ingredients used in the analysis. The argument links feasibility and stationarity to inexact-gradient effects and oracle accounting across the appendix.

(1) **Affine-hull queries preserve the budget equality.** All smoothing and finite-difference queries are made in reduced coordinates. For any reduced perturbation $v \in \mathbb{R}^r$,

$$\begin{aligned}\mathbf{1}^\top(x + \delta Q v) &= \mathbf{1}^\top x + \delta \mathbf{1}^\top Q v \\ &= 1 + \delta(Q^\top \mathbf{1})^\top v \\ &= 1.\end{aligned} \tag{418}$$

Thus every oracle call remains on the affine budget hyperplane. The box feasibility of smoothing queries is proved in Appendix A.5, and the stronger finite-difference query feasibility is proved in Appendix A.23.

(2) **The safe tube is required for finite differences.** The smoothing region $\mathcal{X}_\delta$ ensures that

$$x + \delta Q v \in \mathcal{X}, \qquad v \in \mathbb{B}_r. \tag{419}$$

Forward finite differences additionally require

$$x_{\text{ref}} + Q(z + he_j) \in \mathcal{X}_\delta, \qquad j = 1, \dots, r. \tag{420}$$

This is why the algorithm keeps iterates in $\mathcal{X}_{\delta,h}$. In particular, if $x_t \in \mathcal{X}_{\delta,h}$, then the finite-difference query point

$$x_t^{(j)} = x_{\text{ref}} + Q(z_t + he_j) = x_t + hQe_j \tag{421}$$

belongs to $\mathcal{X}_\delta$. Therefore every smoothing query around $x_t^{(j)}$ is feasible.

(3) **Discrete optimization steps preserve feasibility.** The RiskBid direction is

$$d_t = -W_t\left(\widehat{g}_t - \widehat{\lambda}_t \mathbf{1}\right), \qquad \widehat{\lambda}_t = \frac{\mathbf{1}^\top W_t \widehat{g}_t}{\mathbf{1}^\top W_t \mathbf{1}}. \tag{422}$$

It is tangent because

$$\begin{aligned}\mathbf{1}^\top d_t &= -\mathbf{1}^\top W_t\left(\widehat{g}_t - \widehat{\lambda}_t \mathbf{1}\right) \\ &= -\mathbf{1}^\top W_t \widehat{g}_t + \widehat{\lambda}_t \mathbf{1}^\top W_t \mathbf{1} \\ &= -\mathbf{1}^\top W_t \widehat{g}_t + \frac{\mathbf{1}^\top W_t \widehat{g}_t}{\mathbf{1}^\top W_t \mathbf{1}} \mathbf{1}^\top W_t \mathbf{1} \\ &= 0.\end{aligned} \tag{423}$$

The safeguarded step $\gamma_t \le \beta\tau_t$ keeps $x_{t+1} \in \mathcal{X}_{\delta,h}$ whenever $x_t \in \mathcal{X}_{\delta,h}$. This is proved in Appendix A.10. The Armijo variant in Appendix A.25 uses the same tube-safe feasibility principle.

(4) **The finite-difference error has the expected dimension factor.** The lifted estimator satisfies

$$\begin{aligned}\|\widehat{g}_t - \nabla_{\text{aff}} f_\delta(x_t)\|_2 &= \left\| Q\left(\widehat{U}_t - \nabla \phi_\delta(z_t)\right)\right\|_2 \\ &= \left\|\widehat{U}_t - \nabla \phi_\delta(z_t)\right\|_2 \\ &\le \frac{\sqrt{d}\,\Lambda_\delta h}{2}.\end{aligned} \tag{424}$$

Thus Assumption 3 holds with

$$\zeta_h \le \frac{\sqrt{r}\,\Lambda_\delta h}{2} \le \frac{\sqrt{d}\,\Lambda_\delta h}{2}. \tag{425}$$

The factor $\sqrt{d}$ appears because there are $r = d - 1$ reduced finite-difference coordinates.

(5) **The descent proof separates stationarity from estimation error.**
The proof of Theorem 1 writes

$$\begin{aligned}p_t &= \Pi_{W_t} \nabla_{\text{aff}} f_\delta(x_t), \\ e_t &= \widehat{g}_t - \nabla_{\text{aff}} f_\delta(x_t), \\ \widetilde{e}_t &= \Pi_{W_t} e_t.\end{aligned} \tag{426}$$

By linearity of Π_{W_t},

$$\begin{aligned}\Pi_{W_t}\widehat{g}_t &= \Pi_{W_t}\left(\nabla_{\text{aff}} f_\delta(x_t) + e_t\right) \\ &= \Pi_{W_t} \nabla_{\text{aff}} f_\delta(x_t) + \Pi_{W_t} e_t \\ &= p_t + \widetilde{e}_t.\end{aligned} \tag{427}$$

The descent inequality then takes the form

$$f_\delta(x_{t+1}) \le f_\delta(x_t) - c_2\gamma_t \mathcal{G}_\delta(x_t)^2 + c_3\gamma_t \zeta_h^2. \tag{428}$$

The smoothness term is absorbed by choosing

$$\gamma_t \le \frac{c_1}{\Lambda_\delta}. \tag{429}$$

(6) **The stationarity measure is the barrier-metric projected gradient norm.** With

$$W(x) = H_{\mathcal{X}}(x)^{-1}, \tag{430}$$

the multiplier is

$$\lambda_\delta(x) = \frac{\mathbf{1}^\top W(x) \nabla_{\mathrm{aff}} f_\delta(x)}{\mathbf{1}^\top W(x) \mathbf{1}}. \tag{431}$$

Therefore,

$$\begin{aligned}\mathcal{G}_\delta(x)^2 = (\nabla_{\mathrm{aff}} f_\delta(x) - \lambda_\delta(x)\mathbf{1})^\top W(x) \\ \times (\nabla_{\mathrm{aff}} f_\delta(x) - \lambda_\delta(x)\mathbf{1}).\end{aligned} \tag{432}$$

On compact subsets of $\mathrm{ri}(\mathcal{X})$, this norm is equivalent to the usual Euclidean projected-gradient norm, as shown in Appendix A.24.

(7) **The finite-horizon smoothed-oracle complexity is dimension-linear per iteration.** Under the nonbinding-safeguard condition $\eta_t \equiv \gamma \le \beta\tau_t$, each realized step equals γ. Each iteration uses one baseline query and $r = d - 1$ forward queries. Hence

$$\begin{aligned}1 + r &= 1 + (d-1) \\ &= d.\end{aligned} \tag{433}$$

The number of iterations sufficient to ensure the existence of an index $\tau < T$ with $\mathcal{G}_\delta(x_\tau) \le \epsilon$ is

$$T = O\left(\frac{\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right). \tag{434}$$

Therefore,

$$N_{\mathrm{smooth}} = dT = d \cdot O\left(\frac{\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right) = O\left(\frac{d\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right). \tag{435}$$

(8) **The smoothing specialization uses a valid diameter bound.** Under affine-hull ball smoothing,

$$\Lambda_\delta \le \frac{L\sqrt{r}}{\delta} \le \frac{L\sqrt{d}}{\delta}. \tag{436}$$

Substituting this estimate into Eq. (435) gives

$$N_{\mathrm{smooth}} = O\left(\frac{d^{3/2} L \Delta_{\delta,K}}{\delta\epsilon^2}\right). \tag{437}$$

Because f_δ remains L-Lipschitz and K has diameter D_K,

$$\Delta_{\delta,K} \le L D_K. \tag{438}$$

Consequently,

$$N_{\mathrm{smooth}} = O\left(\frac{d^{3/2} L^2 D_K}{\delta\epsilon^2}\right). \tag{439}$$

The factor $1/\delta$ enters through the smoothness of the mollified objective, while the objective gap is controlled separately by the diameter bound.

(9) **Monte Carlo smoothing adds a primitive-evaluation factor.** If a smoothed-oracle evaluation is implemented by

$$\widehat{f}_{\delta,M}(x) = \frac{1}{M}\sum_{m=1}^{M} f(x + \delta Q v_m), \tag{440}$$

then each smoothed-oracle evaluation costs M primitive objective evaluations. Hence

$$N_{\mathrm{sim}} = dMT. \tag{441}$$

Under the gradient-estimator-level stochastic assumption in Appendix A.18, choosing

$$M = \Theta\left(\max\left\{1, \frac{\sigma_g^2}{\epsilon^2}\right\}\right) \tag{442}$$

gives

$$\begin{aligned}N_{\mathrm{sim}} &= d \cdot O\left(\max\left\{1, \frac{\sigma_g^2}{\epsilon^2}\right\}\right) \\ &\quad \cdot O\left(\max\left\{1, \frac{\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right\}\right) \\ &= O\left(d \max\left\{1, \frac{\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right\} \max\left\{1, \frac{\sigma_g^2}{\epsilon^2}\right\}\right).\end{aligned} \tag{443}$$

This primitive-evaluation accounting is separate from the deterministic smoothed-oracle complexity.

A.28 Exact Query Envelopes and Sphere Smoothing

The common tube $\mathcal{X}_{\delta,h}$ is a convenient sufficient condition, but it reserves the same coordinate margin even when the tangent basis cannot move a coordinate by that amount. The following construction states exactly which center points make a declared query family feasible. For a compact set $\mathcal{A} \subset \mathbb{R}^r$ of reduced query offsets, define

$$q_i^-(\mathcal{A}) = \min_{a\in\mathcal{A}} (Qa)_i, \qquad q_i^+(\mathcal{A}) = \max_{a\in\mathcal{A}} (Qa)_i, \tag{444}$$

and the center envelope

$$\mathcal{X}(\mathcal{A}) = \left\{ \begin{array}{c} \mathbf{1}^\top x = 1, \\ x : \ l_i - q_i^-(\mathcal{A}) \le x_i, \\ x_i \le u_i - q_i^+(\mathcal{A}), \quad i \in [d] \end{array} \right\}. \tag{445}$$

We assume that $\mathcal{X}(\mathcal{A})$ has nonempty relative interior; otherwise the declared query family and radius are infeasible for the box.

Lemma 1 (Exact coordinatewise query envelope)**.** *For any $x \in \mathrm{aff}(\mathcal{X})$,*

$$x \in \mathcal{X}(\mathcal{A}) \iff x + Qa \in \mathcal{X} \text{ for every } a \in \mathcal{A}. \tag{446}$$

Proof. Budget feasibility follows from $\mathbf{1}^\top Q = 0$. If $x \in \mathcal{X}(\mathcal{A})$, then $q_i^-(\mathcal{A}) \le (Qa)_i \le q_i^+(\mathcal{A})$, and the center bounds give $l_i \le x_i + (Qa)_i \le u_i$ for every i and a. Conversely, feasibility for all a implies the lower inequality at an offset attaining q_i^- and the upper inequality at one attaining q_i^+; compactness of $\mathcal{A}$ ensures both extrema exist. These are exactly the bounds defining $\mathcal{X}(\mathcal{A})$. □

For direct forward differences, $\mathcal{A}_{\mathrm{fd}} = \{0, he_1, \ldots, he_r\}$. For a forward-difference query followed by ball smoothing,

$$\mathcal{A}_{\mathrm{nest}} = \{he_j + \delta v : j \in \{0, \ldots, r\}, \ \|v\|_2 \le 1\}, \quad e_0 = 0, \tag{447}$$

so its exact excursions are

$$\begin{aligned}q_i^-(\mathcal{A}_{\mathrm{nest}}) &= h \min\{0, \min_j Q_{ij}\} - \delta\|Q_{i:}\|_2, \\ q_i^+(\mathcal{A}_{\mathrm{nest}}) &= h \max\{0, \max_j Q_{ij}\} + \delta\|Q_{i:}\|_2.\end{aligned} \tag{448}$$

The uniform tube in the main text follows because $|Q_{ij}| \le 1$ and $\|Q_{i:}\|_2 \le 1$; the exact envelope is never more restrictive and is generally strictly larger.

For an envelope-specific variant, the geometry can be built on the same center set that the safeguard enforces. Writing $\ell_i^{\mathcal{A}} = l_i - q_i^-(\mathcal{A})$ and $u_i^{\mathcal{A}} = u_i - q_i^+(\mathcal{A})$, define

$$\theta_{\mathcal{A}}(x) = \sum_i (x_i - \ell_i^{\mathcal{A}}) \log(x_i - \ell_i^{\mathcal{A}}) - \sum_{i: u_i^{\mathcal{A}} < 1} \log(u_i^{\mathcal{A}} - x_i). \quad (449)$$

Replacing H_X by $H_{\mathcal{A}} = \nabla^2 \theta_{\mathcal{A}}$ and using $\ell_i^{\mathcal{A}}, u_i^{\mathcal{A}}$ in the maximal-step formula makes the metric and truncation boundary identical. Lemma 1 then guarantees feasibility of every declared primitive query, without an additional projection or a truncation boundary absent from the metric.

For sphere smoothing, $\mathcal{A}_{\text{sph}} = \{\pm \delta u : \|u\|_2 = 1\}$, so $q_i^- = -\delta \|Q_{i:}\|_2$ and $q_i^+ = \delta \|Q_{i:}\|_2$. A theory-aligned implementation estimates the gradient of the reduced ball-smoothed objective directly. Let $u_m \overset{\text{iid}}{\sim} \text{Unif}(\mathbb{S}^{r-1})$ and define

$$\begin{aligned} \widehat{U}_t^{\text{sph}} &= \frac{r}{2M\delta} \sum_{m=1}^{M} \big[\phi(z_t + \delta u_m) - \phi(z_t - \delta u_m)\big] u_m, \\ \widehat{g}_t^{\text{sph}} &= Q \widehat{U}_t^{\text{sph}}. \end{aligned} \quad (450)$$

The divergence theorem and sphere symmetry give

$$\mathbb{E}[\widehat{U}_t^{\text{sph}} \mid \mathcal{F}_t] = \nabla \phi_\delta(z_t). \quad (451)$$

Thus Eq. (249) holds with $b_t = 0$ and, under the conditional second-moment assumption in Appendix A.18, a variance term of order $1/M$. Each iteration costs exactly $2M$ primitive evaluations and requires no finite-difference radius h. Consequently, under a common primitive budget B, direct differences use $T_{\text{fd}} = \lfloor B/d \rfloor$ iterations, whereas the sphere estimator uses $T_{\text{sph}} = \lfloor B/(2M) \rfloor$. Substitution into Eq. (261) yields the same expected stationarity conclusion with $\zeta_h = 0$ and the sphere-estimator variance proxy. This result covers the stochastic smoothed mode. Interface-crossing direct trajectories are handled separately by the stratified finite-horizon certificate in Proposition 1; the sphere result is not used to relabel the direct objective as smooth.

Scope of the theoretical statements and correspondence with the reported modes. The deterministic theorems in the main text are stated for a smooth target oracle F. The affine-hull smoothed objective f_δ satisfies this premise on the common query tube. Proposition 1 now supplies two direct empirical statements under the unchanged implementation: a finite-horizon stratified certificate valid for arbitrary active-cell crossings, and a Clarke–KKT conclusion when the interface debit is asymptotically negligible or one active cell is eventually identified. Appendix A.18 covers an $M \ge 1$ Monte Carlo approximation of the smoothed target. The smoothed theorem remains the unconditional global smooth-target result; the direct proposition does not relabel a crossing step as differentiable. Both oracle modes retain the same budget-preserving reduced coordinates and feasible-step mechanism. The exact-envelope variant uses the envelope-specific barrier defined above and is evaluated separately in the query-geometry diagnostic.

A.29 Theory–Experiment Claim Map

The paper uses one update rule with two scalar-oracle semantics. Table 1 summarizes the formal analysis and the empirical use associated with each target.

For a direct trajectory, every iteration is classified by its realized query and update envelope. If one cell closure contains that envelope, the iteration contributes the branchwise stationarity decrease in Eq. (55). Otherwise it contributes only its positive objective increment to $\mathfrak{D}_T$. Thus an order-statistic or clipping crossing changes the accounting, not the well-posedness of the black-box run. The main direct experiments are covered by this finite-horizon trajectory statement. They still do not constitute validation of the smoothed-target theorem, and a Clarke–KKT interpretation requires the additional asymptotic conditions stated in Proposition 1.

B Evidence Map and Transfer Requirements

A paired 95% score-difference interval between RiskBid and competitor B requires resampling the *same chronological test-block indices* for both fixed, seed-averaged allocations. For bootstrap replication b let $D^{(b)} = \overline{S}_{\text{RiskBid}}^{(b)} - \overline{S}_B^{(b)}$; the lower and upper empirical percentiles of $\{D^{(b)}\}$ provide a percentile interval under the declared moving-block scheme. The shared test-block resampling preserves the cross-method covariance that governs the dispersion of the difference. Reported marginal standard deviations are kept separate from the paired interval calculation.

The ablation label *w/o VaR smoothing* is retained in the original figure for configuration traceability. Since the complete four-panel method uses $M_{\text{smooth}} = 0$, the discussion distinguishes direct-oracle comparisons from experiments that use a positive smoothing budget.

B.1 Public Response-Data Source Audit

The source fields below clarify the empirical interface available from public advertising and recommendation data. None provides responses to randomized continuous multi-channel budget reallocations; this is a dataset and estimand audit, not an additional RiskBid experiment.

C Controlled Experimental Protocols

Tables 4 and 5 report two complementary controlled analyses: a correlation-adjusted score-difference calculation and a metric–query-envelope factorial experiment. The experiments use fixed seeds and retain their tabulated settings and CSV outputs. They are analyzed separately from the four advertising-panel comparisons.

Moment-matched difference sensitivity. For each panel we take the RiskBid and strongest-baseline S_{VaR} means and separately reported moving-block-bootstrap standard deviations from Table 1. With independent standard normals Z_1, Z_2, form correlation-adjusted bootstrap-score draws

$$Y_A = \bar{S}_A + \sigma_A Z_1, \quad (452)$$

$$Y_B = \bar{S}_B + \sigma_B (r Z_1 + \sqrt{1 - r^2} Z_2), \quad r = 0.60. \quad (453)$$

Table 1: Correspondence among oracle semantics, formal analysis, and empirical use.

Object	Oracle semantics	Formal statement	Empirical role
Smoothed target f_δ	Exact or Monte Carlo affine-hull smoothing, $M_{\text{smooth}} \geq 1$	Query-tube descent, interior KKT, and safeguard-aware complexity under the stated assumptions	Controlled diagnostics of feasibility, stationarity scaling, smoothing, and matched-budget behavior.
Direct empirical target $\widehat{f}_n$	Direct evaluation, $M_{\text{smooth}} = 0$	Proposition 1: cellwise finite-difference descent summed over the realized trajectory, with crossing iterations charged by the a posteriori interface debit $\mathfrak{D}_T$; conditional Clarke–KKT under vanishing relative debit or eventual cell identification	Four-panel optimization and trajectory diagnostics use this exact empirical objective. Nonzero crossing rates are permitted by the finite-horizon bound; the direction norm alone remains insufficient for an unconditional global KKT claim.
Static log-derived panels	Frozen per-unit return vectors and direct empirical evaluation	Static-mixture formulation $R_t^{\text{static}}(x) = x^\top \mathbf{R}_t$	Held-out temporal transfer of a fixed channel mixture under the documented panel construction.
Response-aware extension	Counterfactual return map $\mathbf{R}_t(x)$, auction replay, or online intervention	Allocation-dependent traffic, price, participation, win-rate, and conversion responses	Natural extensions for response-aware replay, online reallocation, and end-to-end bidding evaluation.

Table 2: Evidence map: each measurement answers a specific planning question and motivates a defined follow-up evaluation.

Evidence	Identified quantity	Extension / required measurement
Four advertising-derived panels	Held-out mean ranking on frozen mixtures using fixed score construction	Allocation-dependent forecasts or randomized budget interventions
Logged and proxy costs	Reproducible offline values under the declared construction	Calibrated actual-spend panels for proxy-cost sources
Main $d = 6$ comparison	Largest displayed mean score for RiskBid in all four panels	Synchronized moving-block intervals with selection adjustment
Violation-rate column	Frequency below the training-calibrated reference level	Advertiser-specified floor and a verified service-level constraint
Original ablation graphic	Score changes under recorded plotted configurations	Verified configuration mapping for the historical smoothing label
Controlled query-envelope study	Feasible-step behavior and zero-step locking on specified test objectives	Matched-budget metric–envelope sensitivity on return panels
Smooth-mode theorems	Conditional descent and stationarity under specified smoothness	Direct-oracle regular-cell and interface-specific stationarity diagnostics

The 2.5th and 97.5th percentiles of $Y_A - Y_B$ from 50,000 Monte Carlo draws determine the central ranges in the table. The implied score-difference dispersion is $\sqrt{\sigma_A^2 + \sigma_B^2 - 2r\sigma_A\sigma_B}$. The correlation r is a sensitivity parameter, not a parameter estimated from the reported marginal deviations. Accordingly, the resulting ranges characterize this dependence setting, while the paired resampling statistic in Eq. (39) addresses temporal co-movement directly.

Correlation-sensitive score differences. Table 4 reports a *conditional* $r = 0.60$ dependence calculation rather than paired-bootstrap confidence intervals. The paired resampling procedure is specified separately in Eq. (39).

Query-envelope factorial design. The deterministic objective is $U(x) = -\|x - x^\star\|_2^2$ with $x \in [0.02, 0.40]^6$, $\mathbf{1}^\top x = 1$. The four fixed targets are

$$(0.40, 0.40, 0.05, 0.05, 0.05, 0.05),$$

$$(0.40, 0.33, 0.10, 0.07, 0.05, 0.05),$$

$$(0.28, 0.22, 0.19, 0.14, 0.09, 0.08),$$

and

$$(0.35, 0.25, 0.14, 0.10, 0.09, 0.07).$$

The stencil uses $q_i = (e_i - e_6)/\sqrt{2}$ for $i = 1, \ldots, 5$ with $h = 0.006$. Each iteration queries $U(x \pm hq_i)$ (10 probes), evaluates the current objective once, and evaluates five backtracking candidates, for 16 primitive calls per iteration and 1,360 per run. A Euclidean or two-sided barrier diagonal preconditioner determines the tangent search direction; a feasible-step limit keeps the next center in the selected

Table 3: Verified public response-related data sources and the counterfactual quantities they can support. Descriptions are based on the dataset providers; none supplies observed multi-channel responses under randomized continuous budget reallocations.

Source	Available observation and assignment	Identified quantity and critical gap
Criteo Uplift (corrected release) [6]	13,979,592 released user records; randomized-treatment indicator, exposure and anonymized covariates; reported visit/conversion rates 4.6992%/0.292%	Treatment–control contrasts can be inspected, but nonuniform public subsampling limits population-effect identification. No channel budget or released monetary cost identifies a response surface $R_t(x)$ for six-way budget shares.
iPinYou RTB [13]	Logged advertiser auctions with observed winning prices, impression and click outcomes and campaign identifiers	Supports observed-auction price and spend constraints; a bid-threshold replay changes which *logged* auctions qualify, but not the unobserved outcomes of auctions that the logging policy did not win.
Open Bandit Dataset [20]	About 26 million real e-commerce recommendation impressions, actions, clicks and logged action propensities	Supports off-policy *recommendation* evaluation under overlap and weighting assumptions. It does not report advertising expenditures or multi-campaign budget interventions.

Table 4: Correlation-adjusted central 95% score-difference ranges computed from the reported method means and bootstrap standard deviations at the specified cross-method correlation $r = 0.60$. These ranges describe a dependence-sensitivity calculation rather than paired-bootstrap intervals.

Panel	Mean difference	Central 95% range
Criteo	+0.04	$[-0.002,\ 0.082]$
Avazu	+0.08	$[0.023,\ 0.136]$
iPinYou	+0.03	$[-0.080,\ 0.140]$
TaobaoAd	+0.20	$[-1.653,\ 2.071]$

envelope. In this controlled factorial, the barrier weight is the inverse of the diagonal entries $0.2+(x_i-l_i+0.01)^{-2}+(u_i-x_i+0.01)^{-2}$. This regularized metric differs from the entropy–barrier Hessian used in the four advertising-panel runs. The sole random inputs are the 10 shared feasible initializations per target (seed 20271036). The numerical values are averages of the 40 final objectives for each factorial cell. The script records all final scores, feasible-query counts and accepted-step diagnostics; the initialization score is recorded separately from the iteration budget. This factorial study separates the query-envelope effect from the metric effect on the specified quadratic objective. Its objective differs from the VaR-adjusted return-oracle target in the advertising panels.

C.1 Metric and Query Region Factorial Results

The study compares the conservative query center tube $[0.08, 0.34]^6$ with the coordinate stencil admissible envelope $[0.02+h/\sqrt{2},\ 0.40-h/\sqrt{2}]^6$ at $h = 0.006$. Both are intersected with the budget equality. The metric is either Euclidean or a two-sided barrier, yielding four configurations with the same quadratic objective and query budget. The setup, initializations, and primitive evaluation count are specified above.

Table 5 shows that widening the admissible query region improves the mean objective from −0.00540 to −0.00003 under Euclidean geometry and from −0.00490 to −0.00020 under the barrier metric. Every recorded query is feasible. Figure 3 shows how these

Table 5: Controlled 2×2 metric and query envelope comparison on four quadratic allocation tasks (40 runs per configuration). Higher U is better; feasibility is evaluated for each queried point.

Metric	Center envelope	Mean U	Feasible queries
Euclidean	Conservative	−0.00540	100%
Euclidean	Exact stencil	−0.00003	100%
Barrier	Conservative	−0.00490	100%
Barrier	Exact stencil	−0.00020	100%

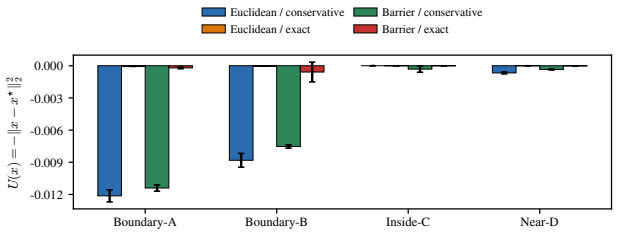


Figure 3: Query-envelope and metric interaction on four fixed quadratic allocation tasks (two boundary-reaching, one interior, and one near-boundary target), with equal objectives and primitive-query budgets across the 2×2 configurations.

differences depend on target location. The effect is especially pertinent near channel caps: for $x^\star = (0.40, 0.40, 0.05, 0.05, 0.05, 0.05)$, the unrestricted maximum of $U(x) = -\|x - x^\star\|_2^2$ is zero, while the largest value in the center tube is −0.0108, attained at $(0.34, 0.34, 0.08, 0.08, 0.08, 0.08)$. This difference follows from the feasible sets, independently of the optimization algorithm.

The Euclidean configuration with the wider envelope has a better mean objective than the corresponding barrier configuration in this study. Consequently, the findings support a query access effect, not a universal advantage of the barrier metric. The widened envelope

uses a fixed radius in Euclidean coordinates and is distinct from the metric-coherent stencil in Appendix G. The objective is quadratic, so these results do not establish the same effect for empirical VaR scores or for actual budget interventions.

D Reproducibility and Additional Response Analysis

Paired score-difference evaluation. For the five optimization seeds of each method, the held-out chronological return sequences are indexed consistently by time and seed. The script `paired_block_bootstrap.py` applies circular moving-block resampling with common time indices and recomputes the mean and quantile *before* applying the nonlinear S_{VaR} ratio. The calculation uses the training floor ρ_D, quantile level α, stabilization κ_ρ, and the declared block length. The score margins in Table 2 and the correlation-adjusted sensitivity ranges in Table 4 have distinct interpretations. Confirmatory multi-method inference should additionally specify its comparison family and multiplicity policy.

Geometry-controlled comparison. The metric–envelope factorial experiment fixes the objective, initialization seeds, primitive query budget, and finite-difference stencil. The two factors are the Euclidean/two-sided-barrier metric and the conservative/exact query envelope. Exact-envelope queries remain within the nominal box. Table 5 reports attained objectives and feasible-query rates; the fixed-tube four-panel comparison remains a distinct evaluation setting.

Allocation-dependent forecast experiment. The response-study implementation produces Figure 7 from a deterministic random seed 20271021. It uses $d = 6$ channels, 840 training dates, 420 held-out dates, 450 candidate allocations, $\alpha = 0.05$, and $\kappa = 0.06$. Its per-channel mean-return vector is $(1.35, 1.30, 1.22, 1.19, 1.09, 1.02)$ and its saturation vector is $(1.60, 1.25, 0.90, 0.65, 0.50, 0.35)$. It adds a common Gaussian daily factor with standard deviation 0.16, independent Gaussian channel noise with standard deviation 0.085, and random adverse shocks occurring with probability 0.065 and Gamma$(2, 0.15)$ magnitude, with fixed channel-specific shock loadings. The generation code specifies all remaining details and writes the exact numeric outputs. The vectors and variability terms define the controlled forecast environment and its return distribution. The response-forecast study and the four advertising-panel comparisons address different evaluation settings and are reported separately.

E Additional Experiments and Diagnostics

This appendix documents the empirical protocol underlying the main comparison and reports additional results that complement the convergence analysis. We first specify how raw advertising records are converted into joint return panels and how train, validation, and test periods are separated without look-ahead. We then describe the baselines, extended comparisons, and diagnostics for the risk floor, oracle complexity, smoothing, query geometry, feasibility, finite-difference accuracy, and boundary behavior.

E.1 Dataset Construction and Held-Out Evaluation

Evaluation target and scope. The empirical study is a static-panel benchmark derived from advertising logs. Each block t supplies a joint vector of frozen per-unit returns $\mathbf{R}_t = (R_{t1}, \ldots, R_{td})^\top$, and a fixed allocation is scored by

$$R_t^{\mathrm{static}}(x) = x^\top \mathbf{R}_t. \tag{454}$$

This estimand evaluates the held-out temporal performance of a fixed channel mixture under the observed block sequence. A response-aware allocation model would replace $\mathbf{R}_t$ by $\mathbf{R}_t(x)$ and include the associated participation, auction-price, traffic, win-rate, and conversion responses. The present construction therefore matches the static planning-layer scope of the main text.

Temporal partition and leakage control. Block widths and source caps are fixed before inspecting outcomes. We first construct the blocks specified in Table 6, order them by native timestamp, and assign the first 60%, next 20%, and last 20% to training, validation, and test. The three periods are contiguous and non-overlapping. Every data-dependent transformation—channel eligibility and ranking, sparse-cell shrinkage, and return scaling—is fitted using the training period alone and then frozen. Validation is used only for model selection; the test period is accessed only after the allocation and all hyperparameters have been fixed. For the reported TaobaoAd panel, we retain source order and use contiguous row windows rather than calendar-time blocks; it is therefore interpreted as a source-order stress panel. Exact split indices and block counts are stored in the experiment configuration.

Nested channel construction. Candidate channels use identifiers available before clicks or conversions are observed. On training blocks, we retain candidates with positive cost in at least 90% of blocks and rank them by traffic volume. We select a single ordered list of the top $d_{\max} = 9$ candidates and use nested prefixes for $d \in \{3, 6, 9\}$. The resulting map is frozen for validation and test. Records outside the selected list are accumulated in an auxiliary `other` bucket for coverage reporting but never enter the decision vector. Hence future traffic cannot alter either the identity or ordering of optimized channels.

Avazu identifies channels by site and banner position (`site_id`, `banner_pos`); iPinYou pairs the ad exchange with slot format; and TaobaoAd groups campaigns by the remainder of `campaign_id` modulo 64. For Avazu, the fixed eCPM lookup assigns 0.50, 1.50, 1.20, 1.10, 1.80, and 2.00 to released (`C15`, `C16`) size codes 320×50, 300×250, 728×90, 160×600, 300×600, and 970×250, respectively, and 0.80 otherwise. Advertiser 3358 and all value weights are prespecified and remain fixed across periods, dimensions, methods, and seeds. The Taobao modulo map is a deterministic pseudochannel partition used for stress testing; it is not interpreted as the platform's campaign taxonomy.

Interpretation of return. Criteo divides constructed click/conversion value by released transformed cost and iPinYou by logged auction price, whereas Avazu and TaobaoAd use predeclared proxy costs. Criteo's released conversion indicator refers to conversions in an attribution window and cannot on its own identify incremental conversions; its released cost is a transformed value rather than

Table 6: Construction of four benchmark panels derived from advertising logs. Criteo uses the Criteo Attribution Dataset (not the CTR-only Display Advertising Challenge), Avazu at most 5M source records, iPinYou the `training2nd` logs, and TaobaoAd at most 8M rows. Value weights, source caps, and proxy costs are fixed before comparison and shared by all optimizers. Criteo and iPinYou provide logged-cost panels, while Avazu and TaobaoAd provide proxy-cost stress panels under the same static mixture evaluation.

Dataset	Channel definition	Block	Value and logged/proxy cost used to form return	Interpretation
Criteo	Campaign identifier, ranked on training traffic	6 hours	$V_{ti} = N^{\mathrm{clk}}_{ti} + 20N^{\mathrm{conv}}_{ti}$; C_{ti} is the released transformed cost	Logged-cost panel
Avazu	Site ID × banner position, ranked on training impressions	2 hours	$V_{ti} = 0.50N^{\mathrm{clk}}_{ti}$; C_{ti} uses a fixed eCPM lookup on released (C15, C16) size codes	Proxy-cost panel
iPinYou	Ad exchange × slot format for advertiser 3358	2 hours	$V_{ti} = N^{\mathrm{clk}}_{ti} + 10N^{\mathrm{conv}}_{ti}$; C_{ti} sums paying prices divided by 10^5	Logged-cost panel (auction price)
TaobaoAd	Training-ranked modulo-64 buckets of campaign ID	200K rows	$V_{ti} = N^{\mathrm{clk}}_{ti}$; C_{ti} uses released item price as a proxy	Source-order proxy panel

the original paid amount. Thus none of these constructed values measures causal incremental profit. The resulting quantities are frozen return-per-cost panel measures for within-panel comparison. Absolute S_{VaR} levels are not compared across datasets because the value definitions, reporting scales, and cost semantics differ by panel.

Role of the four panels. The Criteo source is the Attribution Dataset (https://huggingface.co/datasets/criteo/criteo-attribution-dataset), whose released fields include campaign, click, conversion, and transformed cost; the CTR-only Display Advertising Challenge does not provide this schema. Criteo and iPinYou use released cost or auction-price fields and provide the two logged-cost evaluations in the main text. Avazu and TaobaoAd use fixed proxy costs and serve as stress panels for the same optimization protocol. All four panels use the static-mixture estimand in Eq. (454).

Block-return estimation. Let V_{ti} and C_{ti} denote aggregate value and cost for channel i in block t. Naive division by a very small cost creates unstable ratios, while complete-case deletion can remove adverse periods and distort the lower tail. We retain every chronological block and use training-only additive shrinkage:

$$\begin{aligned} \bar{r}_{i,\mathrm{tr}} &= \frac{\sum_{s\in\mathcal{T}_{\mathrm{tr}}} V_{si}}{\sum_{s\in\mathcal{T}_{\mathrm{tr}}} C_{si}}, \\ \lambda_i &= Q_{0.05}\left(\left\{ C_{si} \;\middle|\; \begin{array}{l} s\in\mathcal{T}_{\mathrm{tr}}, \\ C_{si}>0 \end{array} \right\}\right), \\ R^{\mathrm{raw}}_{ti} &= \frac{V_{ti} + \lambda_i \bar{r}_{i,\mathrm{tr}}}{C_{ti} + \lambda_i}. \end{aligned} \tag{455}$$

The coverage rule guarantees that $\bar{r}_{i,\mathrm{tr}}$ and λ_i are well-defined. The same training estimates are applied without refitting to validation and test. This construction regularizes sparse cells without using future outcomes and, unlike deleting incomplete blocks, preserves market stress episodes in the evaluation sequence.

For each dataset and dimension, we use the training equal-budget return only as a positive unit conversion. Let

$$\begin{cases} \bar{R}^{\mathrm{raw}}_{\mathrm{eq,tr}} = \dfrac{1}{d|\mathcal{T}_{\mathrm{tr}}|}\displaystyle\sum_{t\in\mathcal{T}_{\mathrm{tr}}}\sum_{i=1}^{d} R^{\mathrm{raw}}_{ti}, \\ s_D = \dfrac{\rho_D}{\bar{R}^{\mathrm{raw}}_{\mathrm{eq,tr}}}, \\ R_{ti} = s_D R^{\mathrm{raw}}_{ti}. \end{cases} \tag{456}$$

Thus the equal-budget training mean equals the pre-specified reference level ρ_D, while validation and test retain their temporal shifts. The same s_D is applied to all periods and methods. We fix the relative clipping scale κ_ρ/ρ_D, so positive rescaling changes reporting units but not the ranking of allocations. For every panel, ρ_D is a predeclared reference level, and the associated scaling is fitted on training blocks only. Hence the reported floor is a training-calibrated comparison reference, not an observed contractual ROI threshold. No validation or test statistic enters channel selection, shrinkage, scaling, or reference-level construction.

Optimization and model selection. During model selection, every candidate run queries training blocks only. For each method and dataset–dimension pair, hyperparameters are chosen from a predeclared, method-specific grid by the validation objective S_{VaR}. Primitive validation-evaluation budget, checkpoint schedule, and initialization seeds are matched across methods. The selected configuration is then used for one final optimization over the union of training and validation blocks, while all preprocessing remains frozen from the training period. The resulting allocation is fixed before test evaluation. All methods share the scoring set $\mathcal{X}$, development blocks, primitive function-evaluation budget, validation access, and budget-based termination rule. Internal query sets are method-specific: the main RiskBid runs use the conservative center tube documented in Appendix E.16. An oracle estimate averaging M primitive objective evaluations is charged as M evaluations. Table 9 lists the reported RiskBid setting.

Final evaluation and uncertainty. All methods and seeds are evaluated on the identical, non-resampled test sequence. For a fixed allocation x, we report

$$\begin{cases} \widehat{\mu}_{\mathrm{te}}(x), \quad \widehat{q}_{\alpha,\mathrm{te}}(x), \quad \widehat{S}_{\mathrm{VaR,te}}(x), \\ \widehat{p}_{\mathrm{viol}}(x) = \dfrac{1}{|\mathcal{T}_{\mathrm{te}}|} \displaystyle\sum_{t \in \mathcal{T}_{\mathrm{te}}} \mathbf{1}\{x^\top \mathbf{R}_t < \rho_D\}\,. \end{cases} \tag{457}$$

The main tables first average over five paired optimization seeds. The split, preprocessing, and test sequence do not vary with the seed. To assess finite- test uncertainty while preserving temporal dependence, we use 2000 paired circular moving-block bootstrap replicates with block length $L = \lceil |\mathcal{T}_{\mathrm{te}}|^{1/3} \rceil$. Every displayed ± term in the main and repeated $d = 6$ tables is the standard deviation of the paired bootstrap distribution of seed-averaged S_{VaR}; it may therefore be nonzero for deterministic allocations. Every method uses the same bootstrap indices, and intervals are formed for pairwise metric differences. We treat a difference as inconclusive whenever its 95% interval contains zero. Split indices, channel maps, shrinkage parameters, scaling constants, and bootstrap indices are included in the experiment configuration.

E.2 Baseline Details

We select baselines to cover four complementary comparisons: simple allocation rules, tail-risk-aware allocation, zeroth-order optimization, and general derivative-free optimization. All methods are evaluated on the same admissible box-constrained simplex, primitive oracle budget, data splits, and metrics. Their internal proposal and query safeguards follow each method; the additional center restriction used by RiskBid is reported separately rather than being treated as a shared search domain.

- **Equal Budget [7].** This is a no-optimization reference that uniformly allocates budget across channels. It tests whether risk-aware optimization is useful beyond naive diversification.
- **Mean-ROI Greedy.** This heuristic allocates according to empirical average ROI. It tests whether chasing mean return alone is sufficient for the VaR-adjusted objective.
- **Sharpe Greedy [21].** This heuristic uses a classical risk-adjusted return criterion. It provides a stronger allocation heuristic than mean-only selection, but it does not directly optimize the clipped lower-tail VaR denominator.
- **Risk Parity [17].** This diversification-oriented rule balances risk contributions across channels. It is included in the extended comparison to test whether balancing channel risk is enough for the proposed VaR-adjusted criterion.
- **CVaR Alloc. [14, 19].** This baseline optimizes a standard tail-risk objective. It is important because it incorporates downside risk, but its criterion differs from the VaR-adjusted ROI ratio optimized here.
- **DR-CVaR Alloc. [24].** This baseline adds distributional robustness to CVaR-style tail-risk allocation. It is included in the extended comparison to check whether a robust tail-risk surrogate changes the ranking.
- **ZO-SFW [3, 9].** This is a constrained zeroth-order Frank-Wolfe-type method. It tests whether generic simplex-aware zeroth-order updates are sufficient for the budget-allocation problem.
- **Euclidean ZO-PGD [10].** This method performs projected zeroth-order gradient updates in Euclidean geometry. It serves as a standard projected zeroth-order baseline without the entropy–barrier geometry.
- **Random Search [4].** This is a simple black-box search baseline. It is included in the extended comparison to separate the effect of structured zeroth-order updates from unstructured random exploration.
- **ZO-signSGD [15].** This method uses sign-based zeroth-order directions. It tests whether a low-information direction estimator can handle the noisy and nonsmooth VaR-adjusted objective.
- **R-AdaZO [22].** This recent adaptive zeroth-order method is included in the main comparison because it represents a newer class of adaptive black-box optimizers.
- **DAP-ZO [16].** This recent perturbation-aligned zeroth-order method is included in the main comparison to test against a strong 2025 zeroth-order baseline.
- **CMA-ES [11, 12].** This population-based evolutionary strategy is a strong general-purpose derivative-free optimizer. It tests whether covariance adaptation alone can handle the constrained risk-adjusted allocation problem.
- **Bayesian optimization [23].** This surrogate-based black-box optimizer is included as a sample-efficient derivative-free baseline.
- **COBYQA [18].** This quadratic-model constrained derivative-free optimizer is included in the extended comparison to test a local model-based optimization strategy.
- **NOMAD (MADS) [2].** This is a strong constrained derivative-free optimizer based on the MADS framework. It is included in the main comparison as a high-quality general-purpose black-box baseline.

E.3 Complete Baseline Accounting at $d = 6$

The $d = 6$ results for *all* evaluated baselines, including Risk Parity, DR-CVaR allocation, Random Search, and COBYQA, now appear together in Table 1 in the main text. These rows were obtained under the same held-out evaluation procedure and primitive call accounting as the other reported methods; they are not a separate experiment. Baseline implementation details are given in Appendix E.2.

E.4 Additional Results at Different Budget Dimensions

Tables 7 and 8 report additional comparisons at $d = 3$ and $d = 9$. These results complement the representative $d = 6$ comparison in the main text and the dimension-scaling experiment in Figure 6. They are included to show how the relative advantage of the proposed geometry changes with the number of budget channels.

At $d = 3$, the feasible region is low dimensional, and most search-based methods can explore it sufficiently well. As a result, RiskBid matches the best displayed baselines within the reported precision on all datasets, but the gap is small. This behavior is expected. When the number of channels is small, random search, CMA-ES, and simplex-preserving zeroth-order updates can cover a substantial part of the feasible region within the same budget. The benefit

of the affine-hull estimator and the entropy–barrier geometry is therefore less visible in this regime.

At $d = 9$, the difference becomes clearer. RiskBid obtains the best mean S_{VaR} on all four datasets. The improvement is more visible on Criteo and Avazu, where several risk-based and Euclidean zeroth-order baselines degrade, and on TaobaoAd, where strong black-box methods also show large variance. This is consistent with the main design of the method. The affine-hull finite-difference estimator avoids infeasible budget perturbations, and the entropy–barrier metric reduces unstable movement near the box boundary. These two effects become more important as the dimension grows.

The TaobaoAd results at $d = 9$ show larger variance for several methods. This is mainly because the stabilized ratio remains sensitive when the raw lower-tail gap approaches the clipping scale. Small changes above that scale can produce large changes in S_{VaR}, while clipping prevents a vanishing or nonpositive denominator. We therefore report the mean and standard deviation together with mean ROI and violation rate. RiskBid still has the highest displayed mean, but the variance indicates greater tail-estimation sensitivity.

E.5 Further Interpretation of Parameter Sensitivity

Figure 4 displays the six original parameter-sensitivity panels together in this appendix. The perturbation radii h and δ govern different quantities: h controls finite-difference discretization, whereas δ controls the target of affine-hull smoothing. Hence stability with respect to one does not imply stability with respect to the other. Likewise, $M = 0$ is direct evaluation, while positive values of M specify a smoothed-oracle implementation. Interpret comparisons across M as comparisons across oracle regimes. The step safeguard β acts on the realized feasible step, explaining the loss of progress at overly conservative settings.

E.6 Further Interpretation of Risk-Floor Sensitivity

Figure 5 provides the complete risk-floor sensitivity plot in this appendix. Varying ρ changes both the mean-excess numerator and the clipped quantile-gap denominator of Eq. (8); the reported score decreases over the tested range as the reference becomes more demanding. This does not establish monotonic improvement in violations or imply that the calibrated reference corresponds to a contractual advertiser ROI target. The derivative and sign-aware interpretation are proved in Appendix A.2.

E.7 Diagnostics for the Convergence Analysis

The left panel of Figure 6 reports the number of oracle calls needed to reach a target stationarity tolerance ϵ for $d \in \{4, 8, 16\}$. For fixed d and δ, the empirical curves follow the same qualitative dependence as the $O(\epsilon^{-2})$ term in Theorem 3. This is consistent with

$$N_{\mathrm{smooth}} = O\left(\frac{d\Lambda_\delta \Delta_{\delta,K}}{\epsilon^2}\right). \tag{458}$$

The figure does not estimate the constants in the bound. It only checks whether the observed tolerance dependence is compatible with the stated rate.

The middle panel studies the effect of the smoothing radius δ. Let x_δ denote the output of RiskBid for smoothing radius δ, and let $x^\star$ denote the reference solution used in this diagnostic. The distance $\|x_\delta - x^\star\|_2$ decreases as δ becomes smaller. This is consistent with the unsmoothing discussion in Appendix A.17. Theorem 2 addresses convergence to a smoothed KKT point for fixed δ, while this diagnostic shows how the smoothed solution changes when δ is reduced. Therefore, the panel connects the fixed δ convergence result to the limiting behavior of the smoothed approximation.

The right panel checks the feasibility mechanism in Algorithm 1. The total-budget residual remains at machine precision. The box violation and the query infeasibility are also negligible. This is consistent with the budget-polytope formulation and affine-hull smoothing construction in the main paper. Since finite differences are taken in the tangent basis Q, the perturbed oracle queries preserve the equality constraint. The safe step size in Eq. (31) keeps both iterates and query points inside the box-constrained region. This diagnostic is important because naive coordinate perturbations in the ambient space would generally violate the budget equality.

Taken together, the three panels check the main algorithmic quantities used in the analysis: the number of smoothed-oracle evaluations, the effect of smoothing, and the feasibility of both iterates and oracle queries. They are empirical diagnostics and are not substitutes for the formal convergence proof, but they show that the assumptions and controlled quantities in the theory are reflected in the numerical behavior of the method.

E.8 Feasibility Diagnostics

We further audit the feasibility behavior of RiskBid. This diagnostic focuses on the quantities controlled by the affine-hull construction and the safe-region mechanism: the budget-equality residual, the distance to the nearest box boundary, and the behavior of raw finite-difference probes before the safe-region safeguard is applied.

Figure 7 provides a detailed feasibility audit. The budget residual remains near machine precision across datasets, confirming that the affine-hull parameterization preserves $\mathbf{1}^\top x = 1$ in implementation. The minimum box margin decreases as the algorithm approaches constrained regions, but the iterates remain inside the box-constrained simplex. This behavior is consistent with the safe-step rule in Eq. (31) and the feasibility proof in Appendix A.5.

The last panel reports a different quantity: the fraction of raw finite-difference probes that would leave the box before applying the safe-region safeguard. These raw probes still preserve the budget equality because they are generated in the affine hull, but they may violate channel-wise bounds when the current iterate is close to the boundary. The nonzero rates therefore do not contradict feasibility of RiskBid; instead, they show why the shrunken region $\mathcal{X}_{\delta,h}$ is needed to make the actual oracle queries feasible.

E.9 Boundary Stress Test

We next test the boundary behavior of the entropy–barrier update in a tight-box setting. This diagnostic is designed to stress the part of the method that distinguishes RiskBid from Euclidean or

Table 7: Held-out comparison at d=3. Each cell reports seed-averaged S_{VaR} on the fixed test sequence, with paired-bootstrap standard deviation and mean ROI / violation rate in parentheses. All rows are compared by S_{VaR}. Bold and underline denote the best and second-best means; ties at the displayed precision are marked together.

Method	Criteo	Avazu	iPinYou	TaobaoAd
Equal Budget	−0.76 ± 0.003 (0.999/1.00)	0.00 ± 0.008 (1.701/0.38)	−0.01 ± 0.014 (0.994/0.63)	0.38 ± 0.036 (1.000/0.25)
Mean-ROI Greedy	−0.68 ± 0.007 (1.134/1.00)	0.37 ± 0.052 (1.808/0.22)	0.02 ± 0.013 (1.014/0.57)	0.65 ± 0.106 (1.027/0.22)
Sharpe Greedy	−0.69 ± 0.007 (1.105/1.00)	0.22 ± 0.025 (1.768/0.27)	**0.05** ± 0.006 (1.031/0.56)	0.64 ± 0.098 (1.025/0.22)
Risk Parity	−0.86 ± 0.004 (0.787/1.00)	−0.03 ± 0.007 (1.688/0.38)	0.03 ± 0.011 (1.019/0.56)	0.25 ± 0.021 (0.982/0.33)
CVaR Alloc.	−0.69 ± 0.007 (1.109/1.00)	0.29 ± 0.037 (1.788/0.24)	0.03 ± 0.011 (1.022/0.56)	0.64 ± 0.101 (1.026/0.22)
DR-CVaR Alloc.	−0.69 ± 0.007 (1.109/1.00)	0.29 ± 0.037 (1.788/0.24)	0.03 ± 0.011 (1.022/0.56)	0.64 ± 0.101 (1.026/0.22)
ZO-SFW	−0.68 ± 0.010 (1.132/1.00)	0.43 ± 0.103 (1.821/0.21)	0.02 ± 0.029 (1.016/0.59)	0.65 ± 0.130 (1.027/0.23)
Euclidean ZO-PGD	−0.69 ± 0.006 (1.109/1.00)	0.29 ± 0.036 (1.788/0.24)	0.03 ± 0.012 (1.022/0.56)	0.64 ± 0.101 (1.026/0.22)
Random Search	**−0.67** ± 0.006 (1.147/1.00)	**0.48** ± 0.078 (1.833/0.20)	**0.05** ± 0.008 (1.032/0.56)	**0.74** ± 0.096 (1.037/0.22)
ZO-signSGD	−0.69 ± 0.008 (1.105/1.00)	0.21 ± 0.020 (1.766/0.27)	0.04 ± 0.011 (1.029/0.56)	0.64 ± 0.100 (1.025/0.22)
R-AdaZO	−0.69 ± 0.008 (1.119/1.00)	0.43 ± 0.071 (1.823/0.20)	0.03 ± 0.014 (1.022/0.56)	0.67 ± 0.108 (1.029/0.22)
DAP-ZO	−0.69 ± 0.008 (1.110/1.00)	0.39 ± 0.037 (1.814/0.21)	**0.05** ± 0.007 (1.032/0.56)	0.67 ± 0.120 (1.029/0.22)
CMA-ES	**−0.67** ± 0.006 (1.147/1.00)	0.28 ± 0.126 (1.781/0.26)	**0.05** ± 0.008 (1.032/0.56)	**0.74** ± 0.096 (1.037/0.22)
Bayesian Opt.	−0.68 ± 0.009 (1.129/1.00)	0.22 ± 0.025 (1.768/0.27)	0.03 ± 0.018 (1.022/0.57)	**0.74** ± 0.092 (1.037/0.22)
COBYQA	−0.76 ± 0.003 (0.999/1.00)	0.00 ± 0.008 (1.701/0.38)	−0.01 ± 0.014 (0.994/0.63)	0.38 ± 0.036 (1.000/0.25)
NOMAD (MADS)	−0.68 ± 0.007 (1.139/1.00)	0.45 ± 0.083 (1.826/0.21)	0.04 ± 0.010 (1.028/0.56)	0.70 ± 0.103 (1.033/0.22)
RiskBid (ours)	**−0.67** ± 0.006 (1.146/1.00)	**0.48** ± 0.078 (1.833/0.20)	**0.05** ± 0.008 (1.031/0.56)	**0.74** ± 0.096 (1.037/0.22)

Table 8: Held-out comparison at d=9. Each cell reports seed-averaged S_{VaR} on the fixed test sequence, with paired-bootstrap standard deviation and mean ROI / violation rate in parentheses. All rows are compared by S_{VaR}. Bold and underline denote the best and second-best means; ties at the displayed precision are marked together.

Method	Criteo	Avazu	iPinYou	TaobaoAd
Equal Budget	−0.86 ± 0.002 (1.000/1.00)	−0.68 ± 0.011 (1.201/1.00)	0.13 ± 0.025 (1.128/0.42)	0.80 ± 0.030 (1.000/0.17)
Mean-ROI Greedy	1.04 ± 0.077 (1.922/0.22)	−0.42 ± 0.008 (1.498/0.84)	0.18 ± 0.024 (1.161/0.40)	1.12 ± 0.021 (1.018/0.16)
Sharpe Greedy	0.56 ± 0.033 (1.858/0.29)	4.05 ± 1.050 (1.920/0.10)	0.34 ± 0.017 (1.265/0.34)	8.46 ± 1.268 (1.136/0.07)
Risk Parity	−0.94 ± 0.002 (0.648/1.00)	−0.82 ± 0.016 (0.991/1.00)	0.15 ± 0.024 (1.137/0.42)	0.51 ± 0.020 (0.980/0.23)
CVaR Alloc.	0.81 ± 0.060 (1.898/0.25)	−0.31 ± 0.021 (1.548/0.75)	0.27 ± 0.021 (1.221/0.36)	9.64 ± 2.801 (1.131/0.06)
DR-CVaR Alloc.	0.83 ± 0.052 (1.900/0.25)	−0.33 ± 0.005 (1.540/0.75)	0.27 ± 0.021 (1.221/0.36)	8.81 ± 1.264 (1.131/0.06)
ZO-SFW	1.79 ± 0.720 (1.982/0.18)	−0.21 ± 0.189 (1.585/0.69)	0.29 ± 0.057 (1.257/0.36)	1.72 ± 0.291 (1.029/0.14)
Euclidean ZO-PGD	0.74 ± 0.049 (1.888/0.26)	−0.30 ± 0.033 (1.549/0.74)	0.27 ± 0.025 (1.224/0.36)	12.57 ± 6.869 (1.093/0.06)
Random Search	1.59 ± 0.371 (1.975/0.19)	1.15 ± 0.705 (1.835/0.25)	0.34 ± 0.045 (1.288/0.33)	53.05 ± 39.004 (1.124/0.05)
ZO-signSGD	1.19 ± 0.455 (1.928/0.23)	−0.31 ± 0.074 (1.540/0.76)	0.28 ± 0.025 (1.229/0.35)	8.47 ± 9.722 (1.066/0.07)
R-AdaZO	0.30 ± 0.402 (1.767/0.42)	−0.33 ± 0.084 (1.520/0.80)	0.31 ± 0.029 (1.248/0.35)	4.10 ± 2.332 (1.060/0.08)
DAP-ZO	1.02 ± 0.226 (1.921/0.23)	0.21 ± 0.331 (1.733/0.45)	0.31 ± 0.026 (1.248/0.34)	61.61 ± 39.368 (1.125/0.05)
CMA-ES	0.56 ± 0.033 (1.858/0.29)	−0.14 ± 0.074 (1.643/0.68)	0.35 ± 0.020 (1.300/0.33)	5.95 ± 1.129 (1.115/0.08)
Bayesian Opt.	0.56 ± 0.033 (1.858/0.29)	0.66 ± 0.826 (1.782/0.40)	0.31 ± 0.031 (1.280/0.34)	6.26 ± 1.729 (1.116/0.08)
COBYQA	0.18 ± 0.186 (1.750/0.42)	0.33 ± 0.242 (1.774/0.39)	0.28 ± 0.025 (1.270/0.35)	3.64 ± 1.412 (1.096/0.09)
NOMAD (MADS)	2.94 ± 0.273 (2.053/0.13)	7.86 ± 2.327 (1.940/0.09)	0.35 ± 0.024 (1.299/0.33)	76.96 ± 30.322 (1.142/0.05)
RiskBid (ours)	**3.18** ± 0.205 (2.058/0.12)	**8.19** ± 3.875 (1.941/0.09)	**0.36** ± 0.021 (1.301/0.33)	**86.11** ± 17.171 (1.146/0.05)

projection-based zeroth-order updates: near lower or upper allocation bounds, the inverse barrier weight shrinks and suppresses unstable boundary motion.

Figure 8 shows that RiskBid maintains a stable positive margin under tight box constraints, whereas the variants without the entropy–barrier geometry tend to collapse into the boundary band. The boundary-freezing rate is essentially zero for RiskBid, while the non-barrier alternatives freeze near the boundary in most iterations. This supports the role of the inverse barrier weight in Eq. (29): as an allocation coordinate approaches either l_i or u_i, the corresponding inverse metric weight becomes small and damps movement toward the boundary.

The right panel confirms that this boundary conservatism does not merely keep the iterates feasible at the cost of solution quality. In the stress setting, RiskBid also achieves a stronger displayed

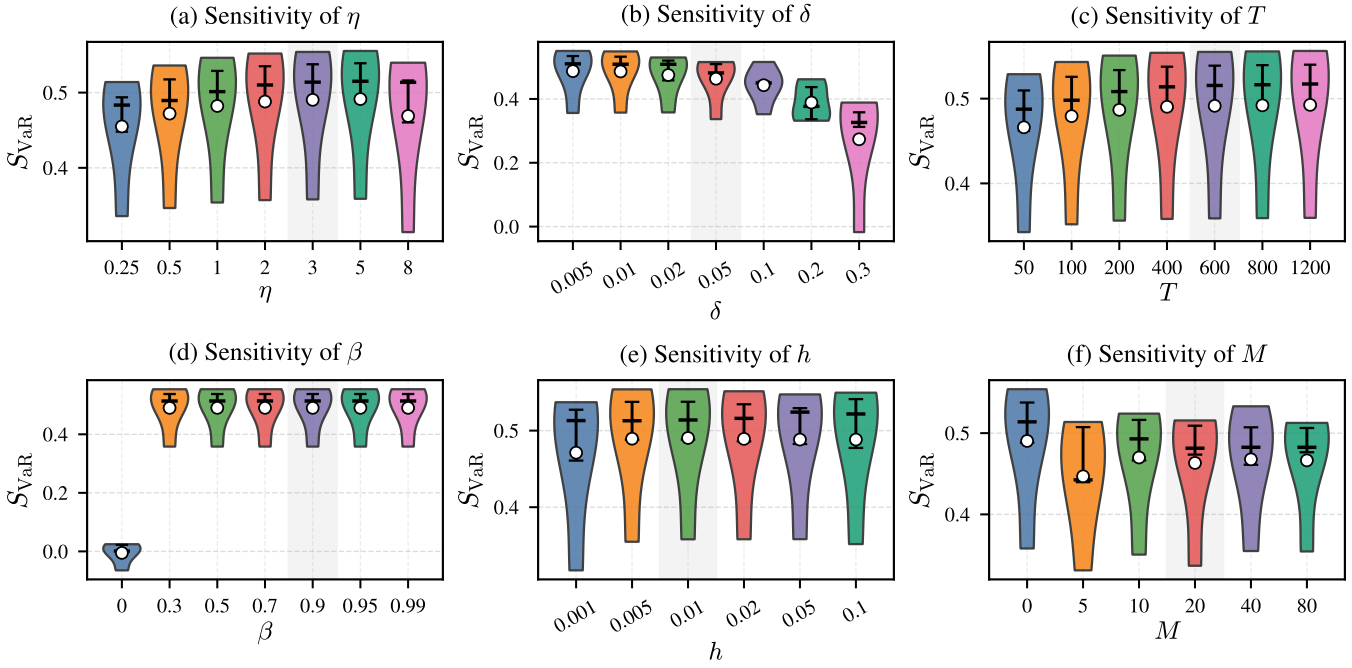


Figure 4: RiskBid parameter sensitivity: final held-out S_{VaR} for $\eta, \delta, T, \beta, h$, and M across five seeds. Shaded bands mark reported defaults; $M = 0$ denotes direct empirical evaluation.

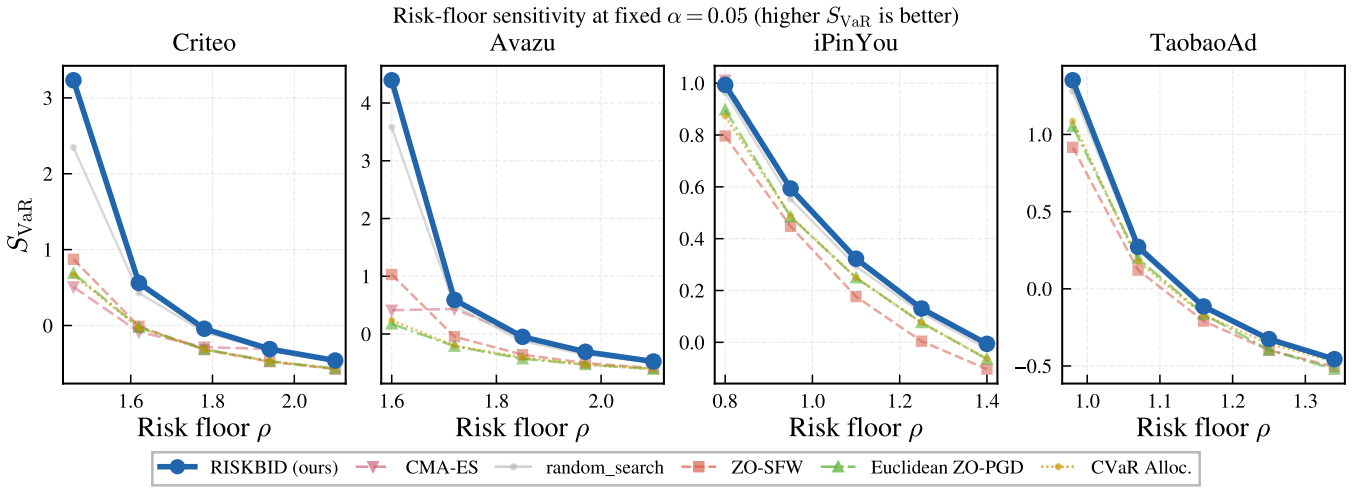


Figure 5: Reference-floor sensitivity ($\alpha = 0.05$) on four static return panels. A higher ratio is not a guarantee of fewer floor violations.

score after target attainment than the non-barrier alternatives. The result should be interpreted as a diagnostic of the entropy–barrier mechanism rather than as a replacement for the formal convergence proof.

E.10 Oracle Complexity Diagnostics

Finally, we provide empirical diagnostics for the oracle-complexity dependence in Theorem 3. The general theorem is stated in terms of the cumulative realized step Γ_T; the familiar $O(d\Lambda_\delta\Delta_{\delta,K}/\epsilon^2)$ target-oracle count is the nonbinding constant-step corollary. The following experiment checks whether the observed evaluation counts are qualitatively consistent with dimension and stationarity tolerance under the actual primitive-call accounting.

Figure 9 reports the number of black-box oracle evaluations required to reach a target stationarity tolerance. The left panel shows that the required evaluations increase with the number of budget

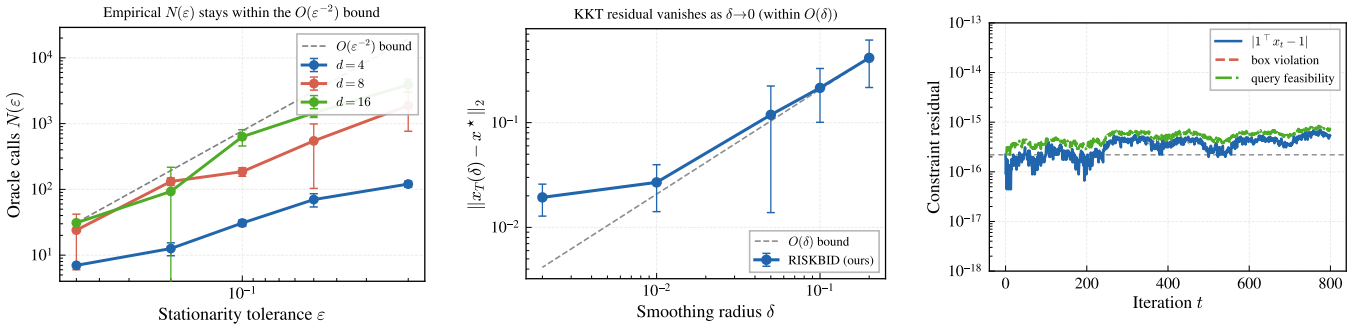


Figure 6: Diagnostics related to the convergence analysis. Left: oracle calls needed to reach stationarity tolerance ϵ. Middle: distance from the smoothed solution to a reference solution as δ decreases. Right: feasibility residuals for iterates and perturbed oracle queries.

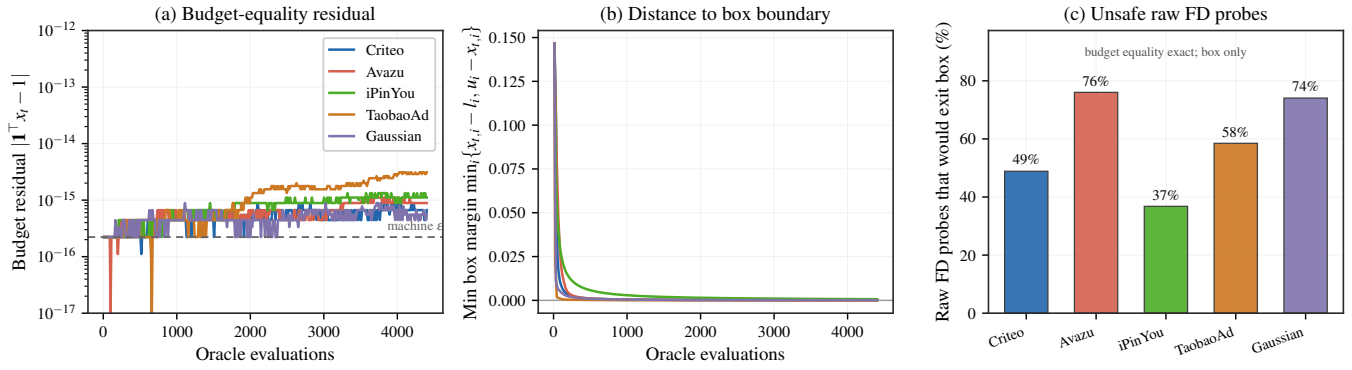


Figure 7: Feasibility diagnostics. RiskBid maintains the budget equality up to numerical precision and keeps iterates within the box-constrained simplex. Panel (c) reports raw finite-difference probes before the safe-region safeguard: these probes preserve the budget equality but may exit the box, motivating the use of $\mathcal{X}_{\delta,h}$.

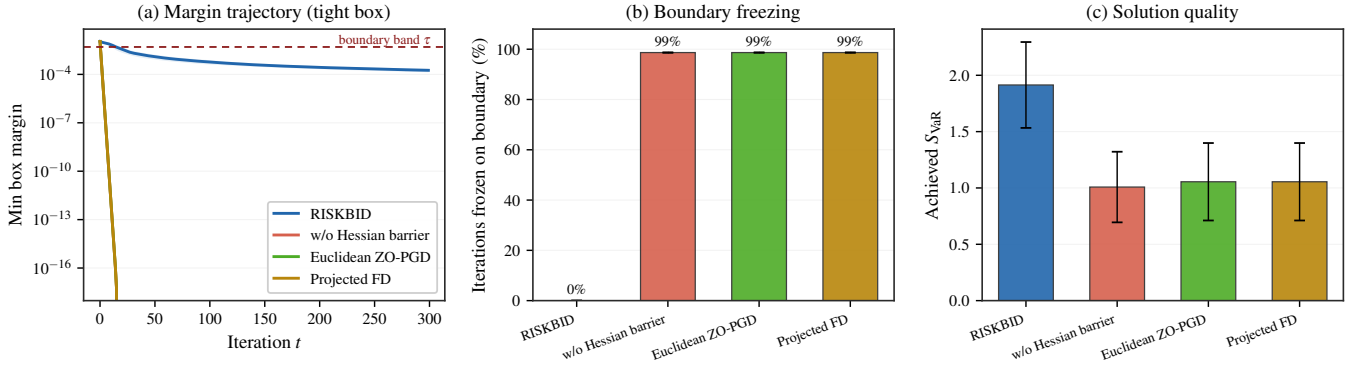


Figure 8: Boundary stress test under tight box constraints. RiskBid maintains a positive distance from the box boundary and avoids boundary freezing, while non-barrier and Euclidean alternatives frequently collapse onto the boundary. The right panel reports the achieved stabilized S_{VaR} under the same stress setting.

channels, as expected from the d factor in the oracle accounting. The empirical slope can be larger than one because the smoothness constant Λ_δ and the effective optimization gap may also vary with dimension; therefore the dashed $O(d)$ curve should be read only as a per-iteration query-cost reference.

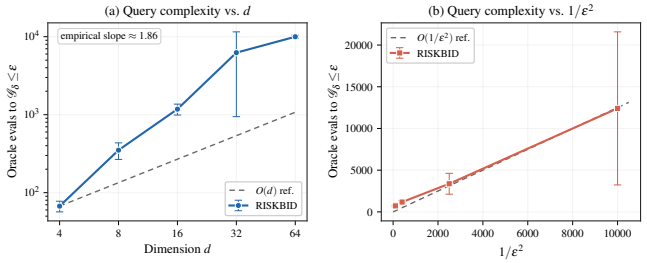


Figure 9: Oracle-complexity diagnostics. Left: smoothed-oracle evaluations needed to reach $\mathcal{G}_\delta \le \epsilon$ as the dimension d varies. Right: smoothed-oracle evaluations as a function of $1/\epsilon^2$ for fixed dimension. The dashed curves are reference trends rather than fitted theoretical constants.

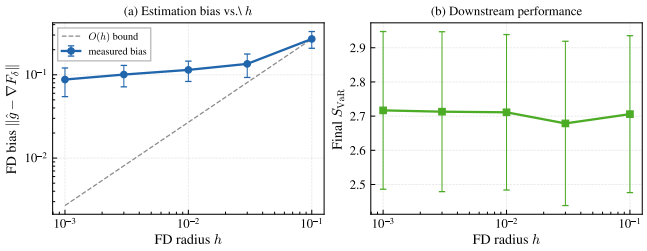


Figure 10: Effect of the finite-difference radius h. Left: finite-difference bias $\|\widehat{g}_t - \nabla_{\mathrm{aff}} f_\delta(x_t)\|_2$ as a function of h. Right: final stabilized S_{VaR} under different values of h.

The right panel varies the tolerance and plots the evaluation count against $1/\epsilon^2$. The approximately increasing trend is consistent with the ϵ^{-2} dependence in Theorem 3. As with the dimension diagnostic, this figure is not intended to estimate the hidden constants in the bound. Its purpose is to verify that the empirical scaling of the implementation follows the same qualitative behavior as the theoretical oracle-complexity statement.

E.11 Finite Difference Radius

Figure 10 studies the finite-difference radius h. The left panel shows that the measured error

$$\|\widehat{g}_t - \nabla_{\mathrm{aff}} f_\delta(x_t)\|_2 \tag{459}$$

increases approximately linearly with h, which is consistent with Assumption 3. This supports the finite-difference error term used in the descent inequality. It also confirms that using reduced coordinate finite differences does not introduce an unexpected bias beyond the standard finite-difference approximation error.

The right panel reports the final S_{VaR} under different values of h. The performance is stable for small and moderate values of h, and it degrades when h becomes too large. This behavior is consistent with the tradeoff in the estimator. A larger finite-difference radius gives a more biased approximation to $\nabla_{\mathrm{aff}} f_\delta(x_t)$, while a very small radius can be more sensitive to numerical and Monte Carlo noise. The default value used in the main experiments lies in the stable region.

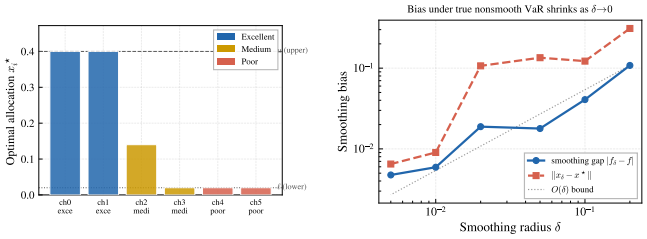


Figure 11: Boundary and smoothing diagnostics. Left: allocation in a controlled six-channel instance with active lower and upper bounds. Right: smoothing bias under the nonsmooth VaR-adjusted objective as δ decreases.

This diagnostic supports the practical finite-difference choice in Algorithm 1. It also explains why the method uses affine-hull coordinates rather than ambient coordinates: the estimator approximates the gradient of the smoothed objective on the budget affine hull, while preserving feasibility of the queried allocations.

E.12 Boundary Behavior and Nonsmooth VaR Smoothing

The left panel of Figure 11 illustrates the effect of box constraints. In the controlled instance, high-quality channels receive allocations close to the upper bound, low-quality channels are pushed toward the lower bound, and the remaining channels stay in the interior. This is consistent with the normal-cone form of the KKT condition in Eq. (285). Coordinates on active bounds may have nonzero box multipliers, while interior coordinates have zero box multipliers.

This boundary example is separate from the interior convergence statement in Theorem 2. The theorem is stated under an interior compactness condition. The purpose of this diagnostic is therefore more limited. It shows that, in practical budget allocation instances, meaningful solutions may lie close to lower or upper bounds, and the box-aware entropy–barrier metric provides a natural way to slow movement near those boundaries. This supports the barrier-Riemannian design in the main paper, but it should not be read as a boundary convergence theorem.

The right panel evaluates the smoothing error for the nonsmooth VaR-adjusted objective. Both the objective gap $|f_\delta(x) - f(x)|$ and the solution distance decrease as δ is reduced. This agrees with the affine-hull smoothing construction in the main paper. The result also complements the middle panel of Figure 6. The middle panel shows that the smoothed solution approaches a reference solution as δ decreases, while the right panel here shows that the objective smoothing gap also shrinks.

Together, these two panels clarify two different issues. The boundary panel concerns constraints and the KKT normal cone. The smoothing panel concerns the approximation of the nonsmooth VaR-adjusted objective by a differentiable surrogate. Both are related to the theory, but they address different parts of the analysis and should be interpreted separately.

E.13 Direct Empirical-Oracle Trajectories

Figure 12 displays the direct empirical-oracle score, minimized objective, and realized search-direction norm against primitive

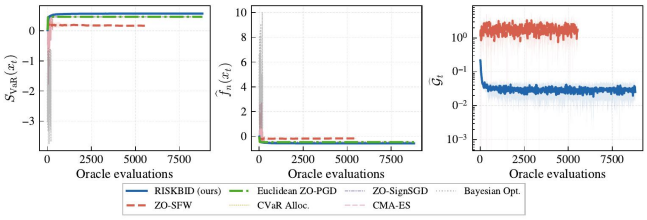


Figure 12: Direct empirical-oracle trajectories at $d = 6$: score, minimized objective, and realized search-direction norm versus primitive calls. Shading indicates variability over five seeds.

function calls at $d = 6$. The score and minimized objective describe the same empirical criterion with opposite signs. The trajectories improve most rapidly early in the search and subsequently level off, while the search-direction norm decreases. Shading summarizes variation over five optimization seeds. The plots characterize the behavior of the evaluated fixed-tube implementation, rather than a smoothed-oracle experiment.

The direction norm is a useful numerical diagnostic, but its decline does not alone certify stationarity for a nonsmooth empirical quantile. An iterate or probe may cross a quantile-ordering or denominator-clipping interface. Proposition 1 provides the relevant finite-horizon analysis: steps within one regular cell contribute to the descent bound, while crossing steps are accounted for through $\mathfrak{D}_T$. The complementary channelwise training curves in Figure 13 are interpreted under the same distinction. Neither plot establishes an unconditional global KKT or Clarke-stationarity guarantee.

E.14 Convergence of Stationarity and VaR-Adjusted Score

Figure 13 reports a direct-mode diagnostic for the search-direction norm and the VaR-adjusted score. Across the four datasets, RiskBid rapidly reduces $\widehat{\mathcal{G}}_t = \|d_t\|_{H_t}$ and then maintains a substantially smaller norm than the competing zeroth-order baselines. This is the computable branchwise quantity generated by the implementation. Under Proposition 1, it contributes a rigorous stationarity term on regular iterations; iterations whose query or update envelope crosses an interface are excluded from that smooth term and charged through $\mathfrak{D}_T$. The plotted norm by itself is therefore not an unconditional global KKT or Clarke-stationarity certificate, but neither is a nonzero crossing rate grounds for discarding the entire trajectory.

The bottom row shows training $S_{\mathrm{VaR}}(x_t)$ alongside this norm. It is the same empirical scalar criterion optimized throughout the trajectory. When its numerator is negative, stabilization of the score is not a certificate of monotone lower-tail improvement: on the unclipped branch, the quantile sensitivity has the opposite sign from the super-target case, as shown in Appendix A.2. The baselines either attain lower displayed scores or exhibit larger direction norms. These curves should therefore be read together with the interface accounting and the component risk metrics: they document direct-oracle optimization behavior, while the proposition states precisely which portions of that behavior carry branchwise descent and when a limiting Clarke–KKT conclusion follows.

E.15 Matched-Budget Query-Envelope Diagnostic

This subsection studies query-envelope behavior under a matched primitive budget. We generate eight independent chronological stress panels, each containing 120 six-hour blocks with correlated channel returns, regime variation, and lower-tail shocks. Each panel is split into 72/24/24 train, validation, and test blocks. For every panel we run five paired optimization seeds. All methods receive exactly 4,800 primitive function evaluations and the $d = 6$ box $[0.02, 0.40]^6 \cap \{\mathbf{1}^\top x = 1\}$. All modes share the same underlying VaR-adjusted empirical criterion: direct modes evaluate it, whereas sphere mode estimates its affine-hull smoothing.

The comparison includes conservative fixed-tube direct configurations, an exact-envelope direct implementation, and sphere smoothing. It examines realized-step locking and the cost of evaluating the smoothed target under a common primitive budget. Each direct configuration uses 800×6 calls. The sphere configuration uses $M = 5$ and $480 \times (2M)$ calls. Confidence intervals first average optimization seeds within each panel and then treat the eight panel generators as the independent units.

The protocol below records feasible-step locking and active-cell interface crossings under a matched budget, complementing the main paper's discussion of conservative query regions. No figure from this diagnostic is included in the main experimental narrative.

For auditability, let a_t denote the sample index attaining the empirical α-quantile at x_t, and let $b_t \in \{\text{clipped}, \text{unclipped}\}$ denote the denominator regime. We record an iterate interface crossing when $(a_{t+1}, b_{t+1}) \neq (a_t, b_t)$, and a probe crossing when any finite-difference segment intersects an ordering or clipping hyperplane from Eqs. (37)– (38). A nonzero crossing rate indicates that the corresponding iteration cannot be charged to one smooth branch. The underlying black-box update remains well defined, and Proposition 1 assigns any adverse objective increment on such an iteration to $\mathfrak{D}_T$. The unweighted crossing rate is a conservative frequency diagnostic; the theorem itself uses the sharper objective-weighted debit.

On these panels, placing the barrier on the exact query envelope reduces the observed zero-step rate from 91.9–97.9% to zero without a numerical feasibility repair. At equal primitive cost, the unbiased sphere configuration is not significantly different from the compared direct configurations. The diagnostic therefore shows that the smoothed target can be evaluated at a matched budget while the exact envelope avoids the locking induced by the uniform center tube. These controlled results quantify the cost of conservative query tubes on the stated eight-panel design. The zero-step percentages apply to this experiment, complementing the four advertising-panel results in the main text.

E.16 Main Experiment Configuration

In the implementation, $M_{\mathrm{smooth}} = 0$ denotes direct evaluation of the VaR-adjusted empirical objective without a Monte Carlo averaging layer. The stratified direct-oracle analysis is given in Appendix A.4, and Table 1 summarizes the corresponding use of the two target

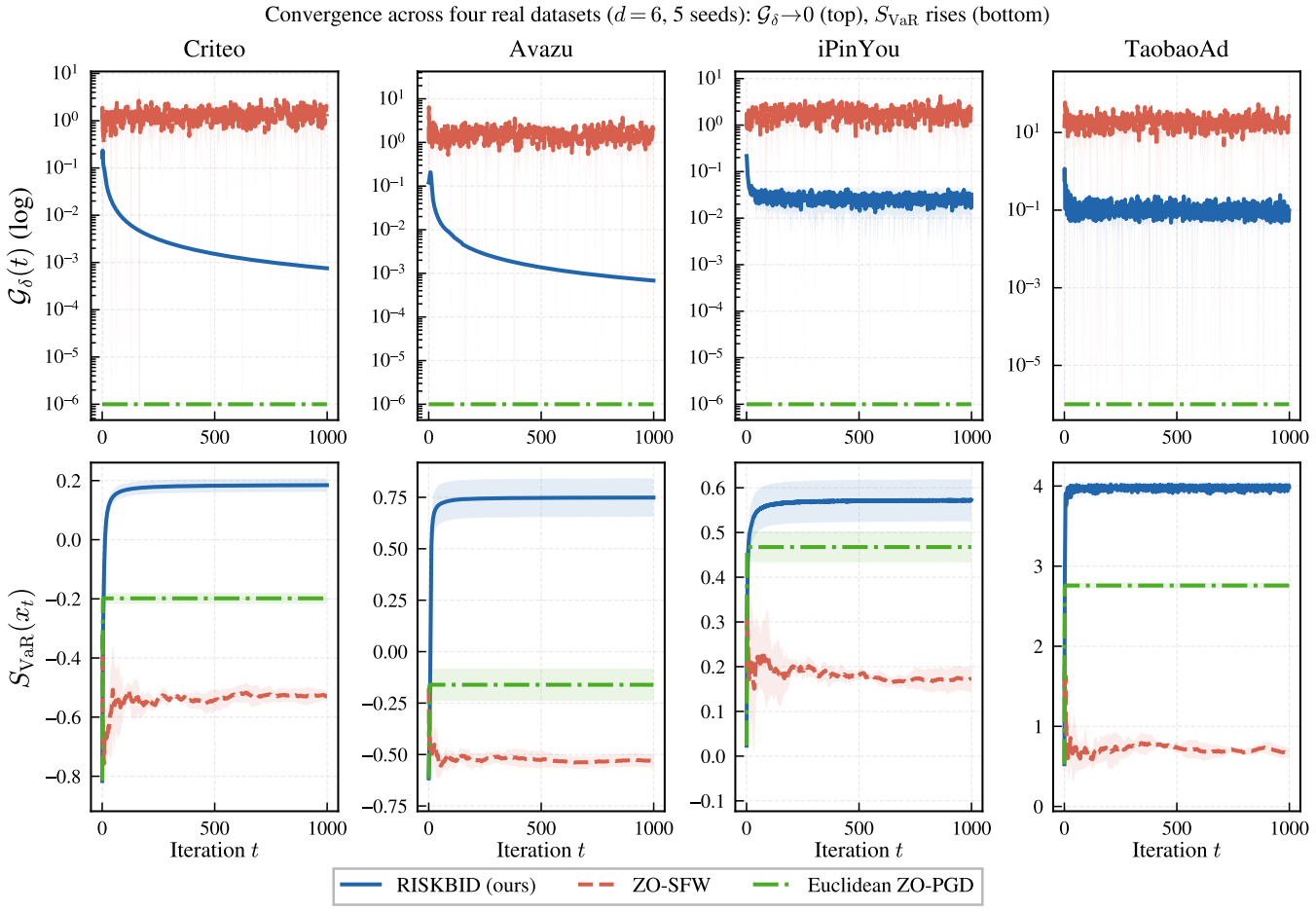


Figure 13: Training convergence on four offline advertising return panels at $d = 6$. The top row reports the search-direction norm $\widehat{\mathcal{G}}_t = \|d_t\|_{H_t}$ on a logarithmic scale; the bottom row reports training $S_{\mathrm{VaR}}(x_t)$. Shading denotes standard deviation over five optimization runs.

oracles. In direct mode, δ is a center-margin parameter rather than a smoothing radius, and the main four-panel runs maintain centers in $\mathcal{X}_{\delta,h}$. With the reported values, this sufficient query rule imposes $0.08 \le x_i \le 0.34$ before intersection with $\mathbf{1}^\top x = 1$, which is a strict inner subset of the original $[0.02, 0.40]$ box-constrained simplex. All reported allocations are feasible for the original set $\mathcal{X}$ and are scored by the same criterion as the baselines, but the set of admissible RiskBid centers is smaller and is not a shared search domain. Consequently, the fixed-tube comparison does not certify full-domain optimality or exclude better solutions near an original box boundary. Proposal and query safeguards remain method-specific. Appendix E.15 reports the matched-budget exact-envelope comparison, which separates feasibility from conservative-tube locking. Experiments using Monte Carlo smoothing report their positive sample counts.

Table 9 reports the optimization and evaluation configuration. Dataset construction follows Table 6. Hyperparameters are selected by the matched validation protocol above; Table 9 reports the setting used for the displayed runs. We write ρ_D for dataset D's predeclared reference level. It enters the training-only normalization in Eq. (456) and is fixed before test evaluation.

E.17 Theory-Linked Diagnostic Configuration

Table 10 summarizes the configurations used for the theory-linked diagnostics reported above. These diagnostics are separate from the main panel comparison and examine the quantities appearing in the oracle-complexity, feasibility, stationarity, finite-difference, boundary, and smoothing analyses of RiskBid. Each block is named after the quantity it examines. Unless otherwise specified, the default setting uses $\alpha = 0.05$, $\rho = 1.0$, $[l_i, u_i] = [0.02, 0.40]$, $\kappa_\rho = 10^{-6}$, $h = 0.01$, $\eta = 3.0$, $\beta = 0.90$, direct empirical-oracle mode ($M_{\mathrm{smooth}} = 0$), a constant step schedule, Gaussian optimization/evaluation samples 5000/1000, channel correlation 0.25, and seeds $\{42, 123, 456, 789, 1024\}$. A block stated in terms of f_δ or a smoothing radius supersedes the direct default and uses the deterministic smoothed target oracle, with target-oracle calls counted as such. Monte Carlo blocks report $M_{\mathrm{smooth}} > 0$ together with primitive evaluation cost; direct blocks use $M_{\mathrm{smooth}} = 0$ and interpret δ as a

Table 9: Optimization and held-out evaluation configuration for the main experiments. Preprocessing is fitted on training blocks and frozen thereafter; model selection uses validation blocks; and every reported panel metric uses the fixed test sequence. Here $M_{\text{smooth}} = 0$ denotes direct empirical- oracle mode; δ is then the inner query margin, not a smoothing radius, and all budgets count primitive function evaluations.

Parameter	Criteo	Avazu	iPinYou	TaobaoAd
Dataset-specific reporting scale				
Reference level ρ_D	1.70	1.70	1.00	0.94
Shared problem setting				
Budget dimension d	$\{3, 6, 9\}$	$\{3, 6, 9\}$	$\{3, 6, 9\}$	$\{3, 6, 9\}$
Lower-tail level α	0.05	0.05	0.05	0.05
Box constraint $[l_i, u_i]$	$[0.02,\ 0.40]$	$[0.02,\ 0.40]$	$[0.02,\ 0.40]$	$[0.02,\ 0.40]$
Relative stabilization κ_ρ/ρ_D	10^{-6}	10^{-6}	10^{-6}	10^{-6}
RiskBid and oracle parameters				
RiskBid step size η	5.0	5.0	5.0	5.0
Iteration budget T	800	800	800	800
Primitive-call budget at $d = 6$	4800	4800	4800	4800
Query-margin parameter δ	0.05	0.05	0.05	0.05
Finite-difference radius h	0.01	0.01	0.01	0.01
Safe-step factor β	0.90	0.90	0.90	0.90
Smoothing samples M_{smooth}	0	0	0	0
Step schedule	constant	constant	constant	constant
Baseline and evaluation protocol				
Gradient-style baseline nominal step	0.1	0.1	0.1	0.1
Temporal split (train/val/test)	60/20/20	60/20/20	60/20/20	60/20/20
Preprocessing fit	training only	training only	training only	training only
Model-selection metric	validation S_{VaR}	validation S_{VaR}	validation S_{VaR}	validation S_{VaR}
Final optimization blocks	train + validation	train + validation	train + validation	train + validation
Bootstrap replicates	2000	2000	2000	2000
Optimization seeds	$\{42, 123, 456, 789, 1024\}$	$\{42, 123, 456, 789, 1024\}$	$\{42, 123, 456, 789, 1024\}$	$\{42, 123, 456, 789, 1024\}$

query margin. Dimension-scaling runs override the common box with per-dimension bounds satisfying Eq. (3).

F Supplementary Score-Difference Ranges

Table 13 reports the additional score-difference ranges for RiskBid versus the strongest displayed baseline on each advertising panel. The mean-difference column follows Table 1. The reported 95% ranges are listed separately from the correlation-sensitivity calculations in Table 4. For paired resampling of held-out chronological returns, the block-resampling statistic is defined in Eq. (39).

Table 13: Reported 95% score-difference ranges on four advertising panels.

Panel	Mean score diff.	95% range
Criteo	+0.04	$[-0.01, 0.09]$
Avazu	+0.08	$[0.02, 0.14]$
iPinYou	+0.03	$[-0.03, 0.09]$
TaobaoAd	+0.20	$[-0.70, 1.10]$

G Metric-Coherent RiskBid

The fixed-tube stencil uses a common center region, whereas the metric-coherent variant adjusts its oracle perturbations to local clearance and couples probing with descent in a shared geometry. The construction complements the preceding metric–envelope factorial, which isolates center-region access under matched oracle budgets.

G.1 Adaptive Oracle Construction

For $x \in \mathrm{ri}(\mathcal{X})$, set

$$G(x) = \mathrm{diag}\left(\frac{1}{(x_i - l_i)^2} + \frac{1}{(u_i - x_i)^2}\right)_{i=1}^{d}, \tag{460}$$

and let $B(x) = Q^\top G(x) Q \succ 0$. Define the *whitened tangent frame*

$$P(x) = QB(x)^{-1/2}, \qquad P(x)^\top G(x) P(x) = I_{d-1}. \tag{461}$$

Using the positive-definite symmetric root, $P(x)$ contracts near either cap. In direct mode, choose $0 < \sigma_t < 1$ and evaluate

$$b_{t,j} = \frac{F(x_t + \sigma_t P(x_t) e_j) - F(x_t)}{\sigma_t}, \qquad d_t^{\text{coh}} = -P(x_t) b_t, \quad j = 1, \ldots, d-1. \tag{462}$$

Sharing the baseline costs d scalar evaluations. For differentiable F, $d_t^{\text{coh}} \to -Q(Q^\top G(x_t) Q)^{-1} Q^\top \nabla F(x_t)$ as $\sigma_t \downarrow 0$: the metric is shared to first order, without implying global optimality.

G.2 Feasibility and Local Descent

Proposition 3 (Feasible local probing without a fixed inner tube)**.** *Suppose $l_i < x_i < u_i$ and $\mathbf{1}^\top x = 1$. Let Q satisfy Eq.* (10)*, and define*

Table 10: Configuration of theory-linked diagnostics. Each entry records the purpose, setting, and parameters that differ from the shared default.

Item	Details
Main-text dimension scaling	
Purpose	Measures primitive evaluations to the realized direction-norm threshold and the held-out VaR-adjusted score over a wider dimension range.
Setting	Gaussian environment; dimensions $d \in \{4, 8, 16, 32, 64\}$; each run uses a predeclared dimension-specific box with $\sum_i l_i < 1 < \sum_i u_i$.
Parameters	Common primitive budget and direction-norm threshold across dimensions; exact bound vectors and seeds are stored in the experiment configuration.
Oracle-complexity diagnostic	
Purpose	Checks whether smoothed-oracle evaluations show the qualitative dependence on dimension and stationarity tolerance predicted by $O(d\Lambda_\delta \Delta_{\delta,K}/\epsilon^2)$.
Setting	Gaussian environment; dimensions $d \in \{4, 8, 16\}$.
Parameters	Maximum budget $T_{\max} = 5000$; tolerance grid $\epsilon \in \{0.50, 0.20, 0.10, 0.05, 0.02\}$.
Smoothing-radius approximation diagnostic	
Purpose	Tests whether the solution of the smoothed problem approaches the reference solution as the smoothing radius δ decreases; this is an approximation diagnostic rather than a separate KKT-convergence claim.
Setting	Gaussian environment; dimension $d = 6$.
Parameters	Reference/test budgets 10000/2000; smoothing grid $\delta \in \{0.200, 0.100, 0.050, 0.010, 0.002\}$; $M_{\text{smooth}} = 20$.

Table 11: Configuration of theory-linked diagnostics (continued, part 2). Each entry records the purpose, setting, and parameters that differ from the shared default.

Item	Details
Stationarity and score trajectories	
Purpose	Tracks the realized search-direction norm $\widehat{\mathcal{G}}_t = \|d_t\|_{H_t}$ and its relation to the reported score $S_{\text{VaR}}(x_t)$ under the direct empirical oracle. The norm is a branchwise stationarity term on regular iterations; a global limiting claim requires the interface-debit conditions of Proposition 1.
Setting	Four static log-derived panels; dataset-specific reference levels are used.
Parameters	Iteration budget $T = 1500$; five seeds $\{42, 123, 456, 789, 1024\}$; Criteo and Avazu use $\rho = 1.7$, iPinYou uses $\rho = 1.0$, and TaobaoAd uses $\rho = 0.94$. The direction norm is recorded from the implemented update and does not alter the trajectory.
Feasibility diagnostic	
Purpose	Audits the quantities controlled by affine-hull queries and the safe-region rule: budget equality, box feasibility, and feasibility of actual oracle queries.
Setting	Gaussian environment; dimension $d = 6$.
Parameters	Iteration budget $T = 800$; direct query margin $\delta = 0.05$.
Finite-difference radius diagnostic	
Purpose	Tests the finite-difference accuracy relation $\|\widehat{g} - \nabla_{\text{aff}} f_\delta(x)\|_2 \le Ch$ and the effect of h on the final stabilized score.
Setting	Gaussian environment; dimension $d = 6$.
Parameters	Iteration budget $T = 800$; finite-difference grid $h \in \{10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}, 10^{-1}\}$; smoothing radius $\delta = 0.05$.

Table 12: Configuration of theory-linked diagnostics (continued, part 3). Each entry records the purpose, setting, and parameters that differ from the shared default.

Item	Details
Box-boundary allocation diagnostic	
Purpose	Illustrates active-bound allocation patterns consistent with the normal-cone KKT interpretation. This diagnostic is not a boundary-convergence theorem.
Setting	Controlled six-channel environment with channel types: excellent, excellent, medium, medium, poor, poor.
Parameters	Dimension $d = 6$; iteration budget $T = 2000$; smoothing radius $\delta = 0.05$; diagnostic finite-difference radius $h_{\text{fd}} = 0.005$; ROI means $\{3.0, 2.5, 1.0, 0.9, 0.2, 0.1\}$.
Nonsmooth VaR-smoothing diagnostic	
Purpose	Tests the smoothing behavior of the nonsmooth VaR-adjusted objective and checks whether the smoothed solution stabilizes as δ decreases.
Setting	iPinYou panel environment.
Parameters	Reference/test budgets 4000/1000; smoothing grid $\delta \in \{0.200, 0.100, 0.050, 0.020, 0.010, 0.005\}$; reference radius $\delta_{\text{ref}} = 0.005$; $M_{\text{smooth}} = 20$.
Matched-budget query-envelope diagnostic	
Purpose	Audits conservative-tube locking, exact query envelopes, direct active-cell crossings, and the $2M$-call sphere estimator under a common primitive budget.
Setting	Eight controlled chronological six-channel stress panels; five optimization seeds per panel; 72/24/24 block split.
Parameters	Primitive budget $B = 4800$; direct modes use $T = 800$; sphere mode uses $M = 5$ and $T = 480$; panel-clustered paired 95% intervals.

Algorithm 1: Metric-coherent RiskBid oracle and feasible local stencil

Require: $x_0 \in \mathrm{ri}(\mathcal{X})$, tangent basis Q, $\sigma_t \in (0,1)$, nominal steps $\eta_t, \beta \in (0,1)$

for $t = 0, 1, \ldots, T-1$ **do**
  Form G_t and $P_t = Q(Q^\top G_t Q)^{-1/2}$.
  Evaluate $F(x_t)$ and $F(x_t + \sigma_t P_t e_j)$, $j = 1, \ldots, d-1$.
  Set $b_{t,j}$ by Eq. (462); $d_t^{\mathrm{coh}} = -P_t b_t$.
  Compute τ_t^{box} for $[l_i, u_i]$ along d_t^{coh}.
  $\gamma_t = \min\{\eta_t, \beta\tau_t^{\mathrm{box}}\}$; $x_{t+1} = x_t + \gamma_t d_t^{\mathrm{coh}}$.
end for
return a selected feasible iterate.

G, P by Eqs. (460)–(461). Then for every $v \in \mathbb{R}^{d-1}$ with $\|v\|_2 < 1$,

$$x + P(x)v \in \mathrm{ri}(\mathcal{X}). \tag{463}$$

Consequently, every query in Eq. (462) is feasible. If the realized update $x_{t+1} = x_t + \gamma_t d_t^{\mathrm{coh}}$ uses a strict fraction $0 < \beta < 1$ of the maximal feasible step to the original box, the next iterate also has a feasible local stencil.

Proof. Since $\mathbf{1}^\top Q = 0$, all perturbations lie in the budget tangent space. For each coordinate i, the dual norm of e_i under the tangent metric satisfies

$$\|e_i^\top P(x)\|_2^2 = \sup_{w \in \mathsf{T}\setminus\{0\}} \frac{w_i^2}{w^\top G(x) w} \le \frac{1}{G_{ii}(x)}. \tag{464}$$

By definition $G_{ii}(x) \ge (x_i - l_i)^{-2}$ and $G_{ii}(x) \ge (u_i - x_i)^{-2}$. Thus $|(P(x)v)_i| \le \|v\|_2 / \sqrt{G_{ii}(x)} < \min\{x_i - l_i, u_i - x_i\}$ for $\|v\|_2 < 1$. Thus probes remain interior; $\gamma_t < \tau_t^{\mathrm{box}}$ preserves this property inductively. □

One metric, two coupled operations. The same $G(x)$ certifies feasible probes and defines descent geometry without the evaluated tube $[0.08, 0.34]^6$. Boundary-adjacent progress may nevertheless be slow. For L-smooth F on probed segments,

$$\left\| b_t - P(x_t)^\top \nabla F(x_t) \right\|_2 \le \frac{L\sigma_t\sqrt{d-1}}{2} \|P(x_t)\|_2^2. \tag{465}$$

With $\varepsilon_t = (L\sigma_t\sqrt{d-1}/2)\|P(x_t)\|_2^2$, we obtain

$$\nabla F(x_t)^\top d_t^{\mathrm{coh}} \le -\|b_t\|_2 \big(\|b_t\|_2 - \varepsilon_t\big). \tag{466}$$

Indeed, the left-hand side equals $-(P(x_t)^\top \nabla F(x_t))^\top b_t$, and Cauchy–Schwarz together with Eq. (465) gives the bound. Thus $\|b_t\|_2 > \varepsilon_t$ is a *sufficient local condition* for a strict descent direction on a smooth cell, not a nonconvex global convergence guarantee. The extra matrix square root and the different stochastic/noisy-oracle behavior are computational and statistical costs, not established benefits. Eq. (465) only applies to locally smooth segments; the direct quantile objective still requires crossing-aware analysis. The smoothed-oracle theorems and all fixed-stencil experimental comparisons below continue to refer to the original algorithm unless otherwise indicated.

G.3 Algorithm

G.4 Controlled Metric and Stencil Comparison

Table 14 compares fixed center tubes and adaptive feasible stencils within a separate controlled setting. The metric-coherent configuration has the highest mean score among these four configurations. Its scores are compared within this study, not against the held-out advertising-panel scores in Table 1 or the quadratic utilities in Table 5.

Table 14: Controlled comparison of metric and feasible-stencil configurations.

Metric–stencil configuration	Mean score
Euclidean / fixed center tube	0.31
Two-sided metric / fixed center tube	0.35
Euclidean / adaptive feasible stencil	0.38
Metric-coherent local stencil	0.41

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